



MASTER'S THESIS

Effects of Spurious Modes on Superconducting Resonators

A thesis submitted in partial fulfilment of the requirements
for the degree of *Quantum Technologies and Engineering (QuantTEEM)*

by

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Abstract

On-chip superconducting resonators are read out through a feedline they are coupled to, and the resonator is characterized by two loss rates: the internal loss rate κ_i and the external coupling rate κ_c . At Alice & Bob, the κ_c predicted by simulation of a chip's design and the κ_c extracted from its measurement disagree, with no known correction resolving the gap. This thesis explores the origin of this so-called κ_c discrepancy. We first benchmark the fitting algorithm using synthetic data with noise, cable delay, impedance-mismatch phase, and a realistic background. We then model the physical mechanisms that could be degrading the resonator's external coupling rate (κ_c): spurious electromagnetic modes of the chip package, and the wirebonds connecting the chip to the printed circuit board (PCB), both of which couple parasitically into the resonator. We derive their effect on κ_c from first principles using the quantum input-output (SLH) formalism, and validate the result against an independent circuit solver, giving a predictive model for how such spurious modes corrupt superconducting resonator readout.

Acknowledgments

This thesis is the product of several months of work that I could not have carried out alone, and I am glad to have the chance here to thank the people who made it possible.

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धैर्यं सर्वस्य साधनम्

Perseverance is the means to accomplish everything.

Chapter 1

Introduction

1.1 Host Company: Alice & Bob

This master's thesis was carried out at Alice & Bob, a French quantum-computing startup building superconducting quantum processors around the *cat qubit* [1]. A cat qubit encodes a bit of quantum information in the two-fold degenerate coherent states of a nonlinear superconducting oscillator, stabilized by an engineered two-photon dissipation process that exponentially suppresses bit-flip errors [2]. This comes at the cost of a phase-flip error that is much easier to correct, meaning fewer physical qubits are needed per logical qubit.

Physically, a cat qubit is a Josephson-junction-based nonlinear oscillator coupled to buffer and readout modes, and is read out through a feedline. This readout signal typically lies in the microwave-photon regime, where a wirebond-induced impedance mismatch or a parasitic package resonance can each shift the qubit's frequency or coupling rate. Engineering this microwave environment reliably, and accurately predicting these parameters, is the role of the Electronic Design Automation (EDA) team hosting this internship.

1.2 The EDA Team

The EDA team builds and benchmarks the tools used to take a circuit design (a resonator, a coupler, a full processor layout) to a fabricated, characterized device. Among the tools it maintains is an internally developed simulator called SPICE: it takes the scattering matrices of smaller sub-circuits and simulates their combination, so a full chip is assembled from many smaller pieces rather than solved as one large problem. A second tool, the measurement-side fitting algorithm that extracts a resonator's physical parameters from its measured transmission, is developed and maintained by the experimental groups; this thesis critically reviews that algorithm rather than building it. This thesis sits at the junction between the two; Section 1.3 lays out the discrepancy between them that motivates this work.

1.3 Background and Motivation: The Simulation-versus-Measurement κ_c Discrepancy

The measurement chain is as follows: a vector network analyzer drives a feedline coupled to an on-chip resonator in the hanger (notch) geometry and records the complex transmission $S_{21}(f)$; a fitting algorithm is then applied to extract the resonance frequency f_0 and the internal and external loss rates κ_i, κ_c (equivalently quality factors Q_i, Q_c). This pipeline is exercised systematically on NTC (Nanofab Test Coupon) chips, dedicated test structures whose linear resonators are used purely for characterization of the developed tools and fabrication methods.

It is on these chips that the discrepancy motivating this work first appears: κ_c predicted by chip-level SPICE simulation and the κ_c extracted by the fitting algorithm (external coupling loss) disagree inconsistently, in a way that no single correction can explain. This is what this thesis refers to simply as the κ_c discrepancy (Figure 1.1). Such a gap can only have two origins: error in the fitting algorithm itself, benchmarked later using synthetic S_{21} data with known, controllable non-idealities, or a genuinely incomplete SPICE circuit model. This thesis therefore performs a critical analysis of both tools independently, in an attempt to understand the physics behind this discrepancy.

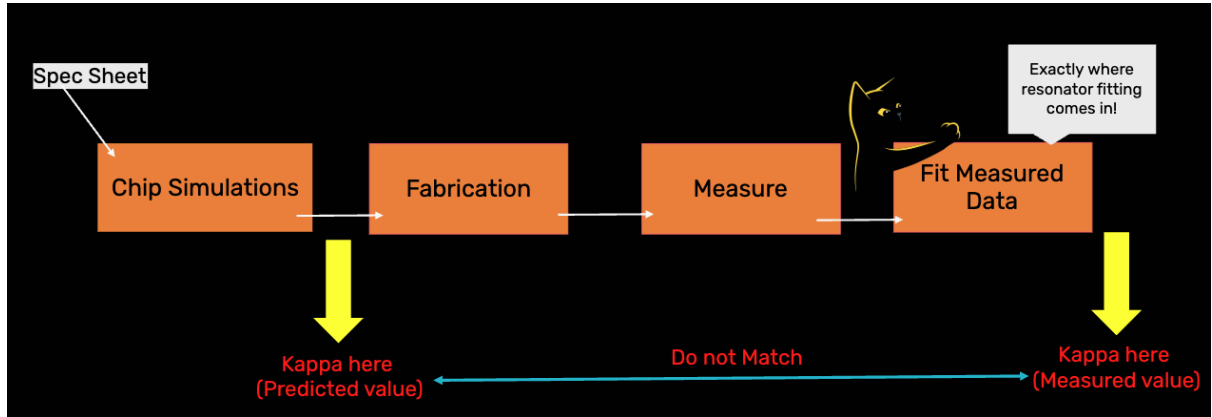


Figure 1.1: The chip design workflow, and the discrepancy in the process.

Figures 1.2 and 1.3 quantify this experimental motivation. Across the six resonators and measured devices, the simulated and measured coupling rates differ by values ranging from a few kilohertz to a few tens of kilohertz. The characteristic scale of the discrepancy is approximately 10–30 kHz, although its sign and magnitude fluctuate between resonators and devices.

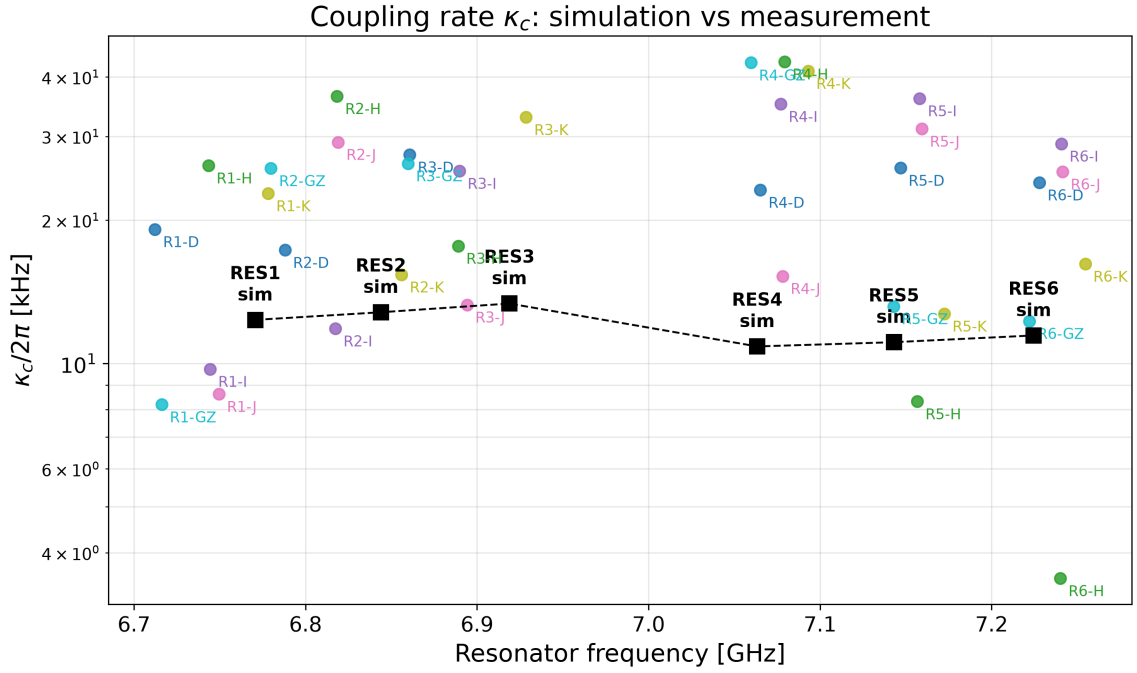


Figure 1.2: Comparison of the simulated and measured coupling rates $\kappa_c/2\pi$ for six resonators across the measured devices. The measurements exhibit substantial resonator-to-resonator and device-to-device variation around the simulated values.

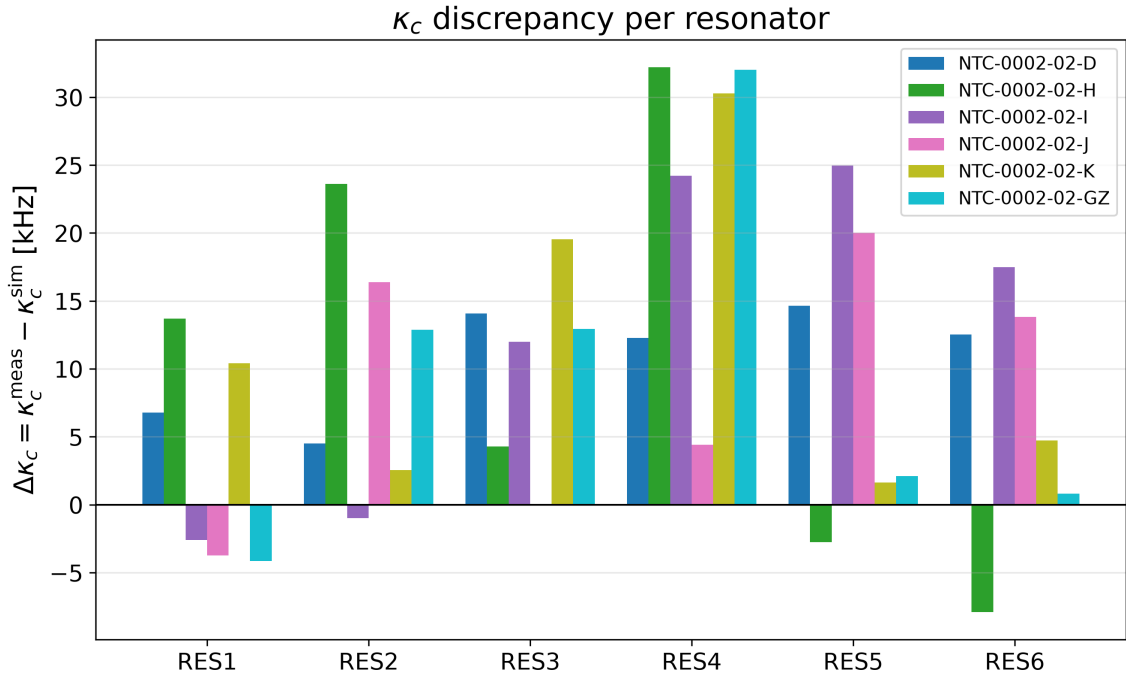


Figure 1.3: Coupling-rate discrepancy $\Delta\kappa_c/2\pi = (\kappa_c^{\text{meas}} - \kappa_c^{\text{sim}})/2\pi$ for each resonator and device. Most discrepancies are of order a few to a few tens of kilohertz, with the largest values approaching 30 kHz.

1.4 Research Objectives

This work pursues three linked objectives:

- Benchmark the fitting algorithm’s robustness to noise, cable delay, mismatch phase, and a realistic background, to bound how much of the κ_c discrepancy the extraction step alone could explain.
- Build analytical and SPICE circuit models quantifying how transmission-line segments, wirebond inductance, and parasitic package resonances affect κ_c and f_0 .
- Develop physical models that explain the behavior observed in the measurement results, and turn the findings into usable, well-documented, and maintainable code.

1.5 Thesis Structure

Chapters 2 and 3 lay the electromagnetic and microwave-network foundations (transmission lines, impedance mismatch, ABCD matrices, Q factors, and the hanger model) that the rest of the thesis builds on. Chapter 4 then benchmarks the resonator fitting algorithm against non-idealities in synthetic data. Chapter 5 develops the SPICE physical model of the resonator’s environment accounting for the wirebonds connecting the chip to its package, and the package’s own box modes. Further deriving the box mode’s effect on the resonator’s loss rates from first principles through the quantum Langevin equation and the SLH input-output framework, then validating that prediction against an independent nodal-admittance solver and a fitting routine built on the same model. The SLH-based fitting routine and its sweep plots are collected in Appendix G. Appendices A and B document the background-subtraction and experimental-SNR methodology used to prepare Chapter 4’s benchmark data. Appendices C - F collect the supporting derivations and numerical methods, the hanger transmission formula, the quantum Langevin equation, the nodal solver, and the energy-participation-ratio mode identifier that Chapter 5 simulations relies on.

Chapter 2

Theory I: Electromagnetic Foundations

2.1 Maxwell's Equations and Plane Wave Solutions

This chapter develops the electromagnetic and transmission-line foundations used throughout the thesis, following Pozar [3, Ch. 1–2].

In differential form, Maxwell's equations relate the electric field intensity $\mathbf{E}(\mathbf{r}, t)$, the magnetic field intensity $\mathbf{H}(\mathbf{r}, t)$, the electric flux density $\mathbf{D}(\mathbf{r}, t)$, and the magnetic flux density $\mathbf{B}(\mathbf{r}, t)$ as

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}, \quad (2.1)$$

$$\nabla \cdot \mathbf{D} = \rho, \quad \nabla \cdot \mathbf{B} = 0, \quad (2.2)$$

with $\mathbf{J}(\mathbf{r}, t)$ the current density and $\rho(\mathbf{r}, t)$ the charge density.

Throughout this thesis, the medium is assumed *linear* (the constitutive relations $\mathbf{D} = \epsilon \mathbf{E}$, $\mathbf{B} = \mu \mathbf{H}$ hold with ϵ, μ independent of field strength), *isotropic* (ϵ and μ are scalars, so the material responds the same way in every direction), *source-free* (no free charge or impressed current, $\rho = 0$), and the fields are *time-harmonic* (every field oscillates at a single angular frequency ω , so it can be written as $\text{Re}[\text{phasor} \cdot e^{j\omega t}]$, with the phasor a complex number or complex function of position only). Under these assumptions Equations (2.1) and (2.2) reduce to the time-harmonic (phasor) Maxwell equations

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}, \quad \nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E}, \quad (2.3)$$

$$\nabla \cdot \mathbf{D} = 0, \quad \nabla \cdot \mathbf{B} = 0, \quad (2.4)$$

where every field is now a phasor (a function of position, not time), ϵ [F/m] is the permittivity, μ [H/m] the permeability.

Taking the curl of the first equation in Equation (2.3) and substituting the second, together with $\nabla \cdot \mathbf{E} = 0$, decouples $\mathbf{E}$ and $\mathbf{H}$ into the vector Helmholtz equation

$$\nabla^2 \mathbf{E} + \omega^2 \mu \epsilon \mathbf{E} = 0, \quad (2.5)$$

with an identical equation for $\mathbf{H}$. Equation (2.5) defines the wavenumber $k = \omega\sqrt{\mu\epsilon}$, complex in general for a lossy medium. Restricting now to a *lossless* medium ($\sigma = 0$, real ϵ, μ), and taking a wave polarized along $\hat{x}$ and uniform in x, y , Equation (2.5) reduces to the one-dimensional wave equation whose forward-traveling solution is

$$E_x(z) = E^+ e^{-jkz}, \quad H_y(z) = \frac{E^+}{\eta} e^{-jkz}, \quad \eta = \sqrt{\frac{\mu}{\epsilon}}, \quad (2.6)$$

with k now real and η the (real) intrinsic impedance of the medium. Equation (2.6) is the phasor representation of a *uniform plane wave*: surfaces of constant phase are infinite planes perpendicular to the direction of propagation, and E_x, H_y are uniform (independent of x, y) on each such plane. Figure 2.1 sketches this solution: $\mathbf{E}$ and $\mathbf{H}$ oscillate in phase, perpendicular to each other and to the propagation direction $\mathbf{k}$.

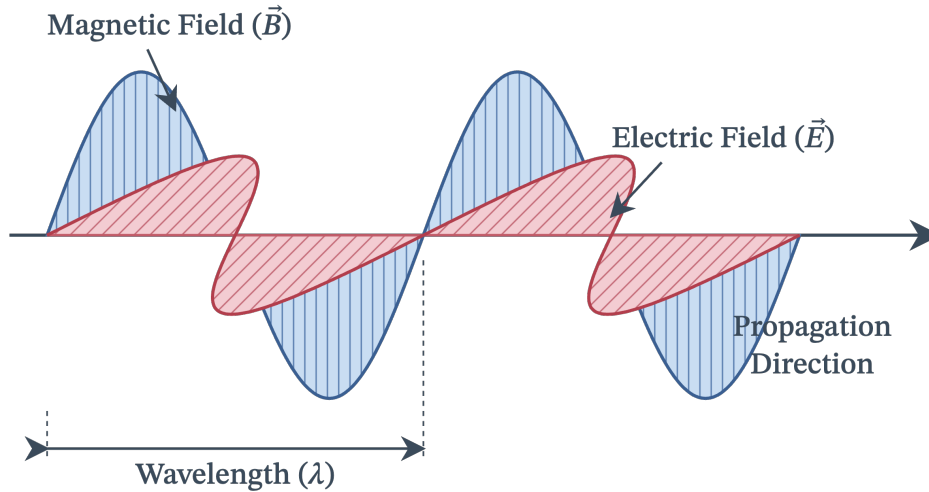


Figure 2.1: A uniform plane wave propagating along z

The phase velocity $v_p = \omega/k = 1/\sqrt{\mu\epsilon}$ and wavelength $\lambda = v_p/f$ defined here reappear throughout the thesis as the quantities that set the electrical length $\beta\ell$ of every transmission-line segment considered in later chapters.

2.2 Lumped-Element vs. Distributed Representations

Ordinary circuit theory treats a component's voltage and current as single numbers; it implicitly assumes the wave in Equation (2.6) barely changes phase across the component [3, §1.1]. Over a physical length ℓ the wave accumulates phase $k\ell$, so this lumped-element approximation only holds when

$$k\ell = \frac{2\pi\ell}{\lambda} \ll 1, \quad (2.7)$$

i.e. when ℓ is a small fraction of a wavelength (conventionally $\ell \lesssim \lambda/10$), as illustrated in Figure 2.2. Beyond that point the component must instead be treated as *distributed*: its voltage and current depend on position along its length.

In this thesis, every circuit element ends up represented in lumped form, though for two different reasons. A wirebond is lumped because it satisfies the criterion above directly: it is short enough that the fields across it are essentially uniform. A CPW resonator is lumped for a different reason: it is built to resonate, so it is not electrically short and its internal fields are just as distributed as the feedline's, but near its resonant frequency, its input impedance as seen from its two terminals happens to take exactly

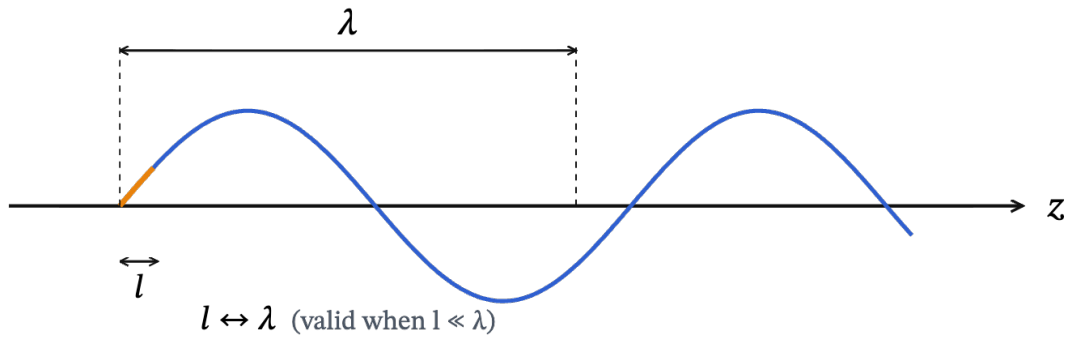


Figure 2.2: The lumped-element approximation (a component of physical length ℓ that is a small fraction of the wavelength λ)

the form of a lumped parallel LC circuit, an equivalence derived in Section 3.3.3. It is this lumped-element representation that we use extensively when setting up our simulations in Chapter 5.

2.3 Transmission Lines and the Telegrapher's Equations

2.3.1 The Telegrapher's Equations

A transmission line is modeled as a cascade of infinitesimal sections of length Δz , each with series impedance $(R + j\omega L)\Delta z$ and shunt admittance $(G + j\omega C)\Delta z$, where R, L, G, C are the per-unit-length resistance, inductance, conductance, and capacitance [3, §2.1]. L and C store energy in the magnetic and electric fields around the line, and are what let a wave propagate along it at all; R and G are the two places that energy leaks away as the wave travels: R in the conductors, G in the dielectric. Figure 2.3 shows one such section.

Applying Kirchhoff's voltage and current laws across one section and letting $\Delta z \rightarrow 0$ gives the telegrapher's equations; in phasor form,

$$\frac{dV}{dz} = -(R + j\omega L)I, \quad \frac{dI}{dz} = -(G + j\omega C)V. \quad (2.8)$$

Physically, the voltage drops along the series impedance, and the current leaks away through the shunt admittance, at a rate set by how much impedance/admittance sits in each infinitesimal length Δz .

Decoupling Equation (2.8) into a single equation for V alone (and identically for I) gives a wave equation, so a signal on the line travels with propagation constant γ . The ratio of V to I in that traveling-wave solution is fixed regardless of position, and defines the characteristic impedance Z_0

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}, \quad Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}. \quad (2.9)$$

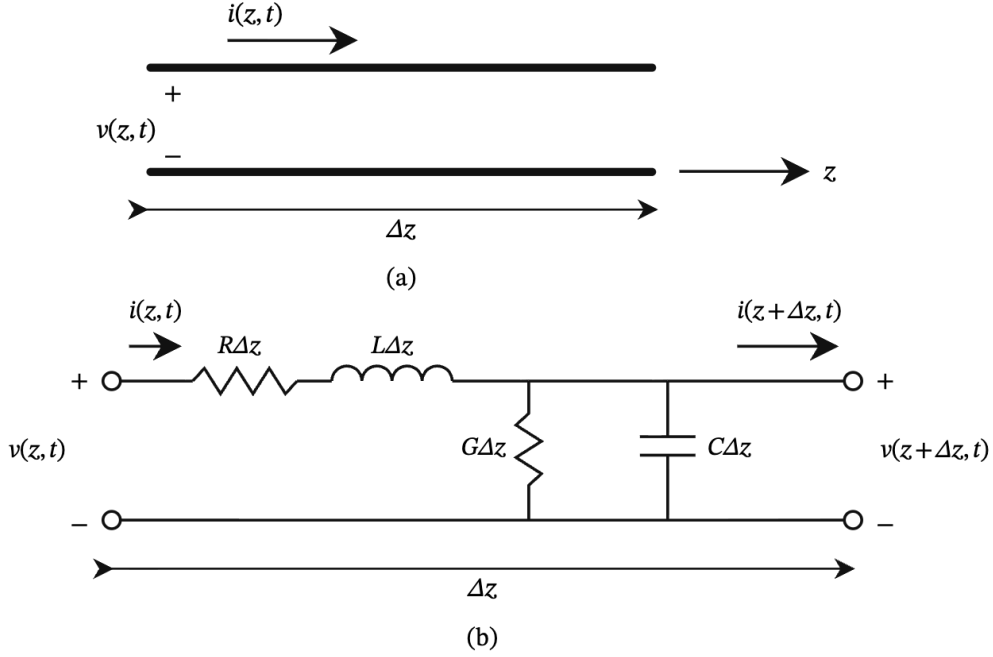


Figure 2.3: (a) A short section Δz of a two-conductor transmission line, carrying voltage $v(z, t)$ and current $i(z, t)$; (b) its lumped-element equivalent circuit.

For the low-loss lines relevant to superconducting circuits, both loss mechanisms are physically negligible: R is the resistance of the conductors themselves, essentially absent in a superconductor, and G is dielectric loss in the substrate, kept small by choosing a low-loss material (sapphire, high-resistivity silicon). Setting $R, G \rightarrow 0$ in Equation (2.9) gives the lossless case $\gamma = j\beta = j\omega\sqrt{LC}$: a purely imaginary propagation constant, meaning zero attenuation ($\alpha = 0$) the wave only picks up phase as it travels, never losing amplitude. The characteristic impedance likewise becomes purely real, $Z_0 = \sqrt{L/C}$, fixed only by the line's geometry. Here $\beta = k$ is the same wavenumber from Section 2.1, evaluated with the line's *effective* permittivity $\epsilon_{r,\text{eff}}$.

Writing the line voltage as a superposition of forward- and backward-traveling waves,

$$V(z) = V_0^+ e^{-j\beta z} + V_0^- e^{j\beta z}, \quad I(z) = \frac{V_0^+}{Z_0} e^{-j\beta z} - \frac{V_0^-}{Z_0} e^{j\beta z}, \quad (2.10)$$

is the *steady-state* solution of Equation (2.8). $V_0^+ e^{-j\beta z}$ travels in the $+z$ direction, from source to load; this is the *incident* wave, launched by the source. $V_0^- e^{j\beta z}$ travels in the $-z$ direction, back toward the source; this is the *reflected* wave, which exists only because whatever terminates the line generally does not absorb all the power it receives. How much comes back, and the one condition under which none does, is the subject of the next section.

2.3.2 The Terminated Line and the General Impedance Transformation

Enforcing the boundary condition at a load Z_L placed at $z = 0$ defines the (load) reflection coefficient $\Gamma \equiv V_0^-/V_0^+$. Substituting Equation (2.10) into $Z_L = V(0)/I(0)$ and solving for Γ gives

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}, \quad (2.11)$$

which is carried back along the line as $\Gamma(\ell) = \Gamma(0) e^{-2j\beta\ell}$ [3, §2.3]. Forming the input impedance $Z_{in}(\ell) = V(-\ell)/I(-\ell)$ from Equation (2.10) and substituting Equation (2.11) gives, after collapsing the exponentials with Euler's identity, the general transmission-line impedance transformation

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan(\beta\ell)}{Z_0 + jZ_L \tan(\beta\ell)}. \quad (2.12)$$

Equation (2.12) is an important formula in this thesis: it is the exact expression evaluated numerically for every feedline segment in the SPICE circuit models of Chapter 5, and it specializes to every case considered there (an ideal termination, a matched line, a mismatched segment of arbitrary length) simply by choosing Z_0 , Z_L , and $\beta\ell$ appropriately. Section 2.4 derives it from first principles and examines what it says physically.

2.4 Impedance Mismatch and Reflection

2.4.1 Deriving the Input Impedance

This section works through the derivation behind Equations (2.11)–(2.12) in full, since every feedline calculation in this thesis reduces to it. Figure 2.4 sets up the problem: a line of characteristic impedance Z_0 and phase constant β , terminated in a load Z_L at $z = 0$, with the input impedance sought looking in from $z = -\ell$.

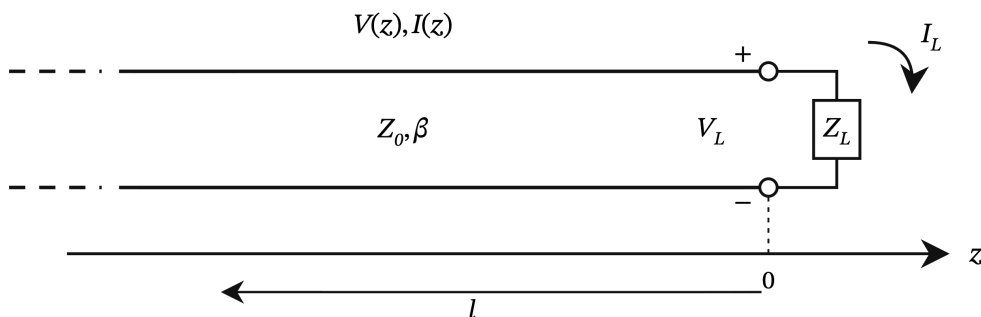


Figure 2.4: A transmission line of characteristic impedance Z_0 and phase constant β , terminated in a load Z_L at $z = 0$.

At the load ($z = 0$), Equation (2.10) gives $Z_L = V(0)/I(0) = Z_0 (V_0^+ + V_0^-)/(V_0^+ - V_0^-)$. Solving for the ratio V_0^-/V_0^+ gives the reflection coefficient Equation (2.11) directly: Z_L and Z_0 agreeing exactly ($\Gamma = 0$) is the special case in which no wave is reflected at all, and every other load reflects some fraction Γ of the incident wave back toward the source.

To find the impedance seen further back along the line, evaluate Equation (2.10) at $z = -\ell$ and divide:

$$Z_{\text{in}} = \frac{V(-\ell)}{I(-\ell)} = Z_0 \frac{e^{j\beta\ell} + \Gamma e^{-j\beta\ell}}{e^{j\beta\ell} - \Gamma e^{-j\beta\ell}}. \quad (2.13)$$

Substituting $\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$ into Equation (2.13) and multiplying numerator and denominator by $(Z_L + Z_0)$ groups the exponentials by Z_L and Z_0 :

$$Z_{\text{in}} = Z_0 \frac{Z_L(e^{j\beta\ell} + e^{-j\beta\ell}) + Z_0(e^{j\beta\ell} - e^{-j\beta\ell})}{Z_0(e^{j\beta\ell} + e^{-j\beta\ell}) + Z_L(e^{j\beta\ell} - e^{-j\beta\ell})}. \quad (2.14)$$

Euler's identity turns each bracket into a sine or cosine, $e^{j\theta} + e^{-j\theta} = 2\cos\theta$ and $e^{j\theta} - e^{-j\theta} = 2j\sin\theta$, so Equation (2.14) becomes

$$Z_{\text{in}} = Z_0 \frac{Z_L \cos\beta\ell + jZ_0 \sin\beta\ell}{Z_0 \cos\beta\ell + jZ_L \sin\beta\ell}, \quad (2.15)$$

and dividing through by $\cos\beta\ell$ gives exactly the general impedance transformation already stated as Equation (2.12). Two features of Equation (2.12) are worth noting immediately: it depends on the load only through Z_L (not on V_0^+ or V_0^- separately), and it depends on position only through $\tan(\beta\ell)$, i.e. only through the electrical length of the segment between the observation point and the load, consistent with Equation (2.7)'s criterion for when that length can be ignored altogether.

2.4.2 Standing Waves and the Matched Condition

Away from a perfectly matched load ($\Gamma \neq 0$), the incident and reflected waves of Equation (2.10) interfere, and the voltage magnitude along the line is no longer constant but oscillates with position

$$|V(-\ell)| = |V_0^+| |1 + \Gamma e^{-2j\beta\ell}|, \quad (2.16)$$

which is exactly the *standing wave* this section's derivation predicts once a reflection is present. Equation (2.16) reaches a maximum $V_{\text{max}} = |V_0^+|(1 + |\Gamma|)$ wherever the two waves add in phase, and a minimum $V_{\text{min}} = |V_0^+|(1 - |\Gamma|)$ wherever they add out of phase. Two standard quantities are built from these: their ratio is the voltage standing wave ratio, and the return loss is given by,

$$\text{VSWR} = \frac{V_{\text{max}}}{V_{\text{min}}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}, \quad \text{RL} = -20 \log_{10} |\Gamma| \text{ dB}. \quad (2.17)$$

Physically, VSWR (or return loss) is what a measurement actually sees: it is the steady-state signature, on the line itself, of exactly the mismatch Γ derived above. Return loss

in particular is a power ratio: the incident and reflected waves carry power proportional to $|V_0^+|^2$ and $|V_0^-|^2 = |\Gamma|^2 |V_0^+|^2$, so $RL = -20 \log_{10} |\Gamma| = -10 \log_{10} |\Gamma|^2$ is, in decibels, how much smaller the reflected power is than the incident power – a large RL means almost nothing comes back. Only in the matched limit $\Gamma \rightarrow 0$ does the standing wave collapse to a constant-amplitude traveling wave, $VSWR \rightarrow 1$, $RL \rightarrow \infty$, and $Z_{in} = Z_L$.

Chapter 3

Theory II: Microwave Networks, Resonators, and Coupling

3.1 Microwave Network Analysis

We saw in Section 2.2 that every element considered in this thesis a coupling capacitor, a feedline, or a resonator reduces to a lumped-element representation. What that does not yet provide is a way to combine these elements into a full circuit: solving Maxwell's equations directly gives the electric and magnetic fields at every point in space, which is computationally heavy, whereas in practice only a handful of terminal quantities (a voltage, a current, a power flow) are ever needed. Network analysis instead characterizes each element, from the outside, only by how the voltages and currents at its ports relate to one another. This port-level description can be combined with others and re-analyzed without re-solving the fields of each element in combination with its neighbors. This section introduces the two representations used throughout the rest of the thesis: the impedance/admittance matrix that the thesis's own analytical solver is built on, and the ABCD (transmission) matrix used for cascading circuit elements [3, Ch. 4].

3.1.1 Impedance and Admittance Matrices

The impedance concept originates with Heaviside's definition of the complex ratio

$$Z \equiv \frac{V}{I} \quad (3.1)$$

for AC circuits of resistors, inductors, and capacitors, and was later extended to electromagnetic fields more generally: impedance is characteristic of the type of field as much as of the medium, forming the link between field theory and the circuit theory [3, §4.1]. At microwave frequencies, however, a single number $Z = V/I$ is not enough: a real circuit element or discontinuity generally has more than one terminal pair, or *port*, and its behavior is only captured once the relationship between the voltages and currents at *every* port is known simultaneously.

For an N -port network with port voltages and currents V_n, I_n (Figure 3.1), the impedance matrix generalizes Equation (3.1) to

$$[V] = [Z][I], \quad (3.2)$$

with $[Y] \equiv [Z]^{-1}$ giving the equivalent admittance-matrix relation $[I] = [Y][V]$. Each entry is measured by driving port j and open-circuiting (Z) or short-circuiting (Y) every other port,

$$Z_{ij} = \left. \frac{V_i}{I_j} \right|_{I_k=0, k \neq j}, \quad Y_{ij} = \left. \frac{I_i}{V_j} \right|_{V_k=0, k \neq j} : \quad (3.3)$$

Z_{ii} is the input impedance seen looking into port i with every other port open, and Z_{ij} the transfer impedance between ports i and j under the same condition (dually for Y_{ii} , Y_{ij} with every other port shorted) [3, §4.2]. For a reciprocal network (no active devices), the reciprocity theorem forces $[Z]$ and $[Y]$ to be symmetric; for a lossless network, zero net real power flow forces every entry to be purely imaginary. Both properties are useful consistency checks on a fitted or simulated circuit model.

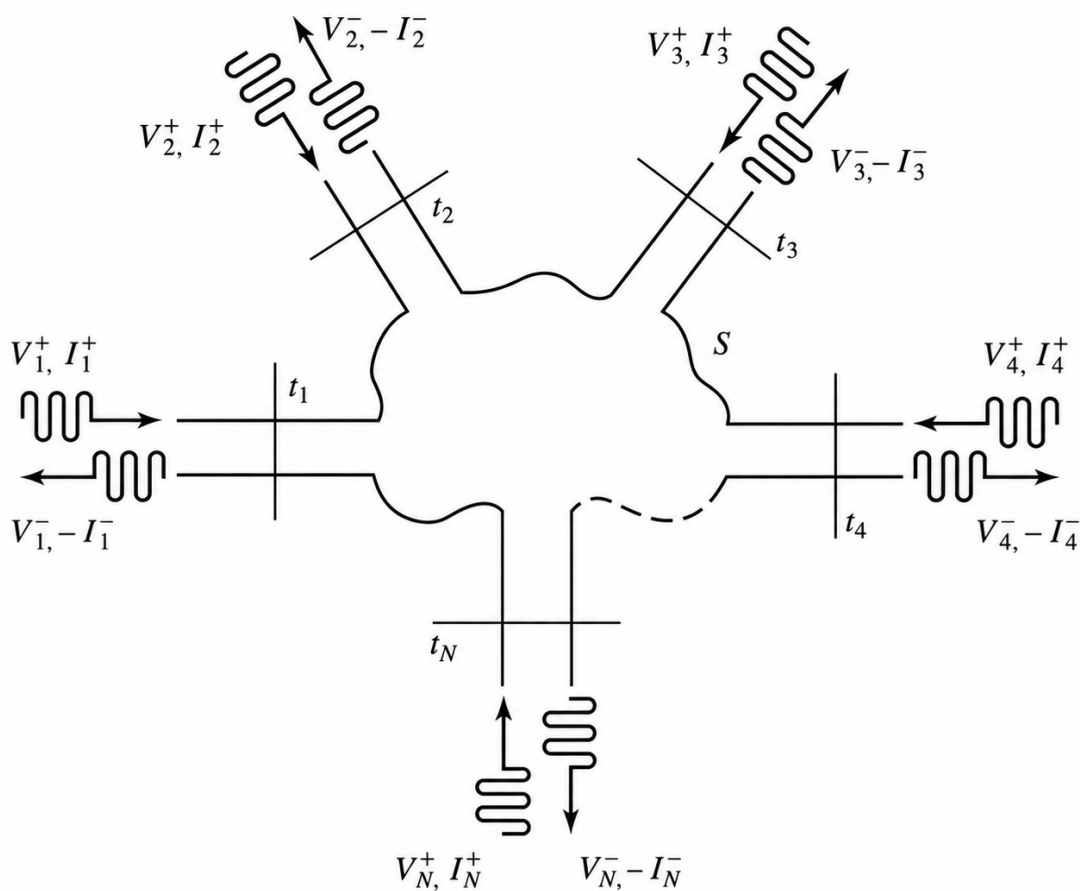


Figure 3.1: An arbitrary N -port microwave network

Appendix D builds this admittance matrix directly from a circuit netlist (provided via KiCAD) rather than from a field solution, using it as the basis for nodal, the thesis's own analytical solver: it extracts each mode's resonance frequency and loss rates as the complex roots of $\det[Y(\omega)] = 0$.

3.1.2 The ABCD Matrix and Cascading

The ABCD matrix relates the voltage and current at port 1 to those at port 2,

$$\begin{pmatrix} V_1 \\ I_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} V_2 \\ I_2 \end{pmatrix}, \quad (3.4)$$

with the sign convention that I_2 flows *out* of port 2. This convention exists for exactly one reason: it makes cascading two-ports a matrix product. If a network is built by connecting two two-ports in series, the current leaving the first equals the current entering the second, so

$$\begin{pmatrix} V_1 \\ I_1 \end{pmatrix} = \begin{pmatrix} A_1 & B_1 \\ C_1 & D_1 \end{pmatrix} \begin{pmatrix} A_2 & B_2 \\ C_2 & D_2 \end{pmatrix} \begin{pmatrix} V_3 \\ I_3 \end{pmatrix}, \quad (3.5)$$

and the ABCD matrix of the cascade is simply the (order-dependent) product of the individual matrices [3, §4.4].

The idea is simple: every basic building block of a circuit a length of line, a series element, or a shunt element has its own small, fixed ABCD matrix, worked out once and then simply reused. For example, a line of length ℓ gives

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}_{\text{line}} = \begin{pmatrix} \cos \beta \ell & jZ_0 \sin \beta \ell \\ jY_0 \sin \beta \ell & \cos \beta \ell \end{pmatrix}, \quad (3.6)$$

while a series impedance Z and a shunt admittance Y give

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}_{\text{series}} = \begin{pmatrix} 1 & Z \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} A & B \\ C & D \end{pmatrix}_{\text{shunt}} = \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix}. \quad (3.7)$$

To build a full feedline-to-resonator chain, each piece is assigned the matrix that matches it, and the matrices are simply multiplied together in the same order the pieces are physically connected. Once the chain's overall ABCD matrix is known, it converts back into an impedance (to compare against a load) with

$$A = \frac{Z_{11}}{Z_{21}}, \quad D = \frac{Z_{22}}{Z_{21}}, \quad C = \frac{1}{Z_{21}}, \quad (3.8)$$

and as a quick sanity check, reciprocity simply requires

$$AD - BC = 1. \quad (3.9)$$

3.2 Scattering Matrix

The impedance and admittance matrices of the previous section are, at microwave frequencies, somewhat of an abstraction: they require open- or short-circuit conditions that are impossible to realize in practice, and even where a voltage or current is well defined, a direct measurement only ever gives the magnitude and phase of a wave

traveling in some direction, or of a standing wave it is never V or I themselves. Scattering parameters instead reference every port to a matched termination Z_0 and relate incident and reflected waves directly, which is why they are what a vector network analyzer actually measures. Writing the port voltage as a sum of incident and reflected waves, $[V^-] = [S][V^+]$, each entry

$$S_{ij} = \left. \frac{V_i^-}{V_j^+} \right|_{V_k^+ = 0, k \neq j} \quad (3.10)$$

is measured by driving port j with all other ports matched-terminated (so $V_k^+ = 0$) and reading the wave reflected out of port i . As with Z and Y , reciprocity forces $[S] = [S]^T$ and losslessness forces $[S]$ to be unitary, $[S]^\dagger[S] = [U]$ i.e. the row/column vectors of a lossless scattering matrix are orthonormal. Closed-form conversions between S , Z , Y , and ABCD parameters exist in both directions [3, Table 4.2]; for a two-port referenced to Z_0 ,

$$S_{21} = \frac{2}{A + B/Z_0 + CZ_0 + D}. \quad (3.11)$$

With the impedance, admittance, and scattering descriptions of a general network now in hand, the rest of this chapter specializes to one particular network: a resonator coupled to a feedline. Its terminal behavior how strongly it couples power in and how quickly it loses it, is captured by a single number, the quality factor Q , introduced next.

3.3 Resonators and Quality Factors

A microwave resonator's behavior is very similar to that of an ordinary lumped-element resonant circuit: at frequencies near resonance it can always be modeled by a simple series or parallel RLC circuit [3, Ch. 6], a picture developed in detail for superconducting microwave resonators specifically in [4]. For the parallel case, the input admittance is

$$Y_{\text{in}} = \frac{1}{R} + j\omega C + \frac{1}{j\omega L}, \quad (3.12)$$

and resonance, $\omega_0 = 1/\sqrt{LC}$, occurs exactly when the average energy stored in the capacitor equals that stored in the inductor [3, §6.1]. Using the same reduction established in Section 2.2, near its own resonance a real resonator reduces to specific values of R , L , C , derived explicitly for a CPW resonator in Section 3.3.3.

3.3.1 Q , Loaded Q , and External Q

The quality factor of any resonant circuit is defined as 2π times the ratio of energy stored to energy dissipated per cycle, or equivalently

$$Q = \omega_0 \frac{\text{energy stored}}{\text{power dissipated}} \quad (3.13)$$

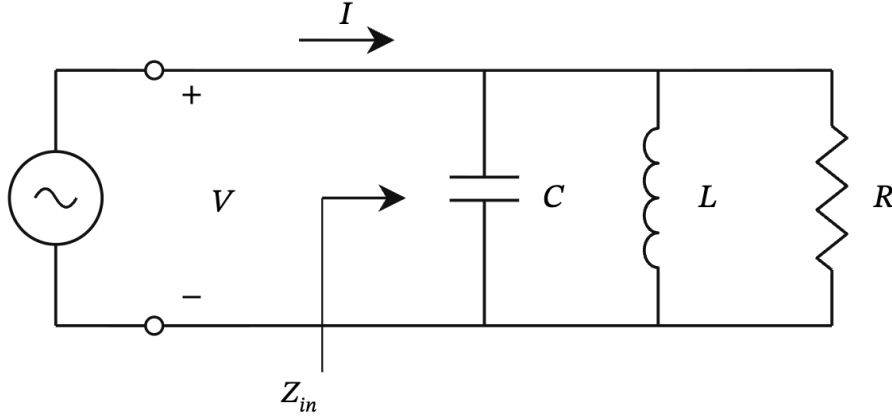


Figure 3.2: A parallel RLC circuit at resonance frequency $\omega_0 = 1/\sqrt{LC}$

Evaluated directly on Equation (3.12), this gives the internal quality factor $Q_i = \omega_0 RC = R/(\omega_0 L)$. Coupling the resonator to an external load R_L (for a parallel resonator, an additional shunt resistance) defines an external quality factor $Q_c = \omega_0 R_L C$, and the two loss channels combine reciprocally into the loaded quality factor,

$$\frac{1}{Q_l} = \frac{1}{Q_i} + \frac{1}{Q_c}. \quad (3.14)$$

3.3.2 Energy Decay and the Coupling Rate κ

Expanding Equation (3.12) to first order around resonance, $\omega = \omega_0 + \Delta\omega$, gives $Y_{in} \approx 1/R + 2j\Delta\omega C$, which vanishes, the circuit rings freely with no external drive but not at the real frequency ω_0 but at the complex frequency $\omega_0 + j/(2RC)$. Loss therefore shifts a resonator's natural frequency into the complex plane, $\omega_0 \rightarrow \omega_0(1 + \frac{j}{2Q_0})$, so that a freely decaying signal $\propto e^{j\omega_0 t}$ acquires an envelope $e^{-\omega_0 t/2Q_0}$ and an energy that decays as $e^{-\omega_0 t/Q_0}$. This decay rate is the coupling rate $\kappa \equiv \omega_0/Q$, used interchangeably with Q throughout this thesis, and combined with $Q_c = \omega_0 R_L C$ above gives the admittance-domain form used directly in the SPICE derivations of Chapter 5:

$$\kappa_c \equiv \frac{\omega_0}{Q_e} = \frac{1}{R_L C} = \frac{\text{Re}[Y_{\text{ext}}]}{C_{\text{total}}}. \quad (3.15)$$

Equation (3.15) is the bridge equation of the thesis: the left side is textbook Q -theory, the right side is the quantity computed directly from an analytical solver (check appendix D). This complex-frequency picture generalizes from the single RLC branch treated here to an arbitrary multi-node network, and is exactly the tool this thesis's own solver uses to separate internal from external loss; that generalization is given in Appendix D.

3.3.3 Coplanar Waveguide Superconducting Resonators as Lumped LC Circuits

The resonators modeled in this thesis are physically distributed CPW sections, yet Chapter 5 represents each one in a KiCAD netlist as a single capacitor C_1 and inductor L_1 . This substitution is derived, not assumed: near its own resonance, a distributed line's input impedance takes on exactly the same mathematical form as the lumped RLC circuit of Figure 3.2.

An open-circuited line of length ℓ has exact input impedance

$$Z_{\text{in}} = Z_0 \coth[(\alpha + j\beta)\ell], \quad (3.16)$$

a transcendental function of frequency with no obvious relation to any simple circuit. Its first resonance occurs at $\ell = \lambda/2$, the shortest length at which Equation (3.16) is purely real; call this frequency ω_0 . Writing $\omega = \omega_0 + \Delta\omega$ gives $\tan \beta\ell \approx \pi\Delta\omega/\omega_0$, and for a low-loss line ($\alpha\ell \ll 1$) also $\tanh \alpha\ell \approx \alpha\ell$; substituting both into Equation (3.16) and dropping the resulting doubly-small cross term leaves

$$Z_{\text{in}} \approx \frac{Z_0}{\alpha\ell + j\pi\Delta\omega/\omega_0} = \left(\frac{1}{R} + 2j\Delta\omega C \right)^{-1}, \quad (3.17)$$

which is exactly the near-resonance impedance of a parallel RLC circuit (Section 3.3.2). Matching term by term identifies the equivalent lumped components:

$$R = \frac{Z_0}{\alpha\ell}, \quad C = \frac{\pi}{2\omega_0 Z_0}, \quad L = \frac{1}{\omega_0^2 C}, \quad (3.18)$$

with internal quality factor $Q_i = \omega_0 RC = \beta/(2\alpha)$: C is fixed by the line's impedance and resonant frequency, L by $\omega_0 = 1/\sqrt{LC}$, and R grows as the line's own loss shrinks.

Equation (3.18) is therefore the justification for representing a CPW resonator, whose Z_0 , β , and α are fixed by its physical geometry, as the lumped pair C_1 , L_1 , near resonance the two are mathematically identical and since a hanger measurement only ever probes that neighborhood, nothing is lost by simulating C_1 , L_1 instead of the full distributed line.

3.4 Coupling to a Feedline and the Hanger (Notch) Geometry

3.4.1 The Coupling Coefficient and Critical Coupling

The ratio of internal to external quality factor, (internal loss dominates), $Q_i/Q_c = 1$ is critical coupling (the resonator absorbs the maximum possible power from the feedline), and $Q_i/Q_c > 1$ is overcoupled (coupling loss dominates).

This coupling-regime picture is exactly what determines how well the fitting routine of Chapter 4 performs on a real, fabricated chip. The hanger geometry is routinely fabricated for precisely this reason: it is simple enough to realize repeatably, yet realistic enough to characterize fabrication techniques and benchmark the tools developed.

3.4.2 The Hanger (Notch) Transmission Model

A resonator capacitively coupled to the side of a through feedline the “hanger” or “notch” geometry used throughout this thesis (Figure 3.3). It produces a transmission dip rather than the transmission peak of an ideal series-coupled resonator. Following Khalil et al. [5] and Probst et al. [6], its measured transmission is modeled as



Figure 3.3: Detail of the NTC-0002-02 chip showing resonator RES2 in the hanger geometry. The resonator runs parallel to the through feedline and couples to it capacitively across the narrow gap at the open end.

$$S_{21}(f) = a e^{i\alpha} e^{-i2\pi f\tau} \left(1 - \frac{Q_l}{Q_c} e^{i\phi} \frac{1}{1 + 2iQ_l \frac{f - f_r}{f_r}} \right), \quad (3.19)$$

where $ae^{i\alpha}$ is an off-resonance background amplitude and phase, τ is the cable delay accumulated between the resonator and the measurement plane, and ϕ is an impedance-mismatch phase that rotates the coupling quality factor into the complex plane, $Q_c \rightarrow |Q_c|e^{-i\phi}$. Only the real part of $1/Q_c$ contributes to the linewidth, $1/Q_l = 1/Q_i + \text{Re}\{1/Q_c\}$, so a mismatch that is purely reactive ($\phi \rightarrow \pm\pi/2$) shifts the resonance frequency but contributes no additional linewidth at all. An in-depth derivation of this model formula is provided in Appendix C. The two non-idealities τ and ϕ that remain are exactly what Chapter 4 benchmarks the fitting algorithm against, and ϕ specifically is the fitting-algorithm-level signature of the physical feedline-mismatch mechanism modeled from first principles in Chapter 5.

Chapter 4

Resonator Fitting Algorithm: Benchmark and Verification

4.1 Introduction

Section 1.3 identified the two possible origins of the κ_c discrepancy that motivates this thesis: the fitting algorithm's own extraction error, and an incomplete physical circuit model on the SPICE side. This chapter addresses the first of the two, benchmarking the resonator fitting algorithm against non-idealities such as the impedance-mismatch phase ϕ and cable delay τ , together with measurement noise and a realistic frequency-dependent background, using synthetic S_{21} data so that the algorithm's behavior can be characterized independently of any particular measurement (see [7] for a more detailed discussion of the fitting algorithm itself).

The chapter is organized as a sequence of five verification studies, each switching on one additional non-ideality at a time: Verification I establishes a numerical baseline with none of them present; II and III introduce the cable delay τ ; IV introduces the impedance-mismatch phase ϕ ; and V evaluates the complete model, including a realistic background, from real chip data. Together they bound how much of the simulation-versus-measurement κ_c discrepancy the extraction step alone could plausibly account for.

4.2 Model

The model function fitted throughout this chapter is the hanger (notch) transmission model introduced in Section 3.4, Equation (3.19); it is restated here in the form used by the fitting algorithm,

$$S_{\text{model},21}(f) = e^{-i2\pi f\tau} \cdot z(f) = e^{-i2\pi f\tau} \cdot ae^{i\phi} \left(1 - \frac{Q_l}{Q_c} e^{i\phi} \frac{1}{1 + 2iQ_l \frac{f-f_r}{f_r}} \right) \quad (4.1)$$

Table 4.1: Description of the variables

Symbol	Meaning
f	Frequency of the probe signal
f_r	Resonance frequency of the resonator
τ	Cable delay (accounts for signal propagation time through cables)
ϕ	Mismatch phase (accounts for the impedance mismatch)
$z(f)$	Resonator's complex response without cable delay
$ae^{i\alpha}$	Off-resonance complex transmission amplitude and phase (signal background)
Q_c	Coupling quality factor (describes energy loss through coupling to the feedline)
Q_l	Loaded quality factor (includes both internal and external losses)
Q_i	Internal quality factor (describes intrinsic losses of the resonator)

The fitting algorithm itself does not fit Equation (4.1) directly; it works instead from the equivalent circle-fit form $S_{21} = (a + b\omega)/(c + d\omega)$. As the probe frequency crosses the resonance, the complex transmission traces a circle in the complex plane. The routine first determines the circle's center and radius r , and then uses the frequency-dependent phase around the circle to extract f_r and Q_l . Finally, the radius and mismatch phase determine Q_c , from which Q_i follows through $Q_l^{-1} = Q_i^{-1} + Q_c^{-1}$. The rational representation provides a numerically convenient form for performing these steps.

The algorithm is benchmarked on synthetic data generated from this model. The synthetic-signal generator (see [here](#)) simulates S_{21} data and can additionally incorporate the effects under study: noise, electrical delay, and impedance mismatch. Throughout this chapter the signal-to-noise ratio (SNR) is defined as,

$$\text{SNR} = \frac{r_0}{\sigma_r} \quad (4.2)$$

where r_0 is the radius of the resonance circle (i.e., the signal strength) and σ_r is the spread in that radius due to noise [6].

4.3 Default Input Parameters

Unless stated otherwise, every simulation in this chapter uses the default parameters listed below; each verification section then varies only the quantities relevant to the non-ideality under study.

Table 4.2: Default input parameters

Name	Parameter Value	Units	Comments
f_0	5	GHz	Resonance frequency
Span (Δf_0)	100	MHz	n_{freq} points linearly spaced over $[f_0 - \Delta f_0/2, f_0 + \Delta f_0/2]$
n_{freq}	501	-	Taken from experimental measurements
N_{avg}	1	-	Number of averages per Q_i/Q_c value
Signal-to-noise (SNR) Ratio	21 points log-spaced over $[1, 1000]$	-	
Q_i/Q_c	21 points log-spaced over $[10^{-5}, 10^5]$	-	
Q_l	100	-	
Electrical delay (τ)	0	ns	
Impedance mismatch phase (ϕ)	0	rad	

4.4 Verification - I : S_{21} Parameter Prediction

4.4.1 Motivation

The natural starting point is the idealized limit identified in Section 3.4: with no added noise, no electrical delay ($\tau = 0$), and no impedance mismatch phase ($\phi = 0$), the hanger-model Equation (4.1) reduces to a purely real, symmetric Lorentzian, and any residual error in the fit can only be numerical in origin. Synthetic S_{21} data are therefore first generated under these ideal conditions, and the fitting procedure is applied to evaluate how accurately it reconstructs the original S_{21} signal. This establishes whether the errors in the fitted S_{21} are limited purely by numerical precision; that is, whether the algorithm effectively reaches the numerical noise floor in absolute and relative error before the non-idealities τ and ϕ are introduced one at a time in the sections that follow.

4.4.2 Parameters

These were the parameters used for the simulations and the analysis of the results.

Table 4.3: Verification I: simulation parameters

	Parameter Type	Value	Remarks
N_{avg}	Input	1	Number of averages per Q_i/Q_c value
Noise Type	Input	None	-
SNR	Input	No Noise	-
Statistical Measure	Output	Absolute Error, Root Mean Square Error (RMSE)	Used only for further data analysis

4.4.3 Plots

The absolute error and RMSE in the prediction of the S_{21} parameter are shown in Figures 4.1 and 4.2, followed by the absolute and relative errors in the extracted Q_i and Q_c , which are shown in Figures 4.3–4.6.

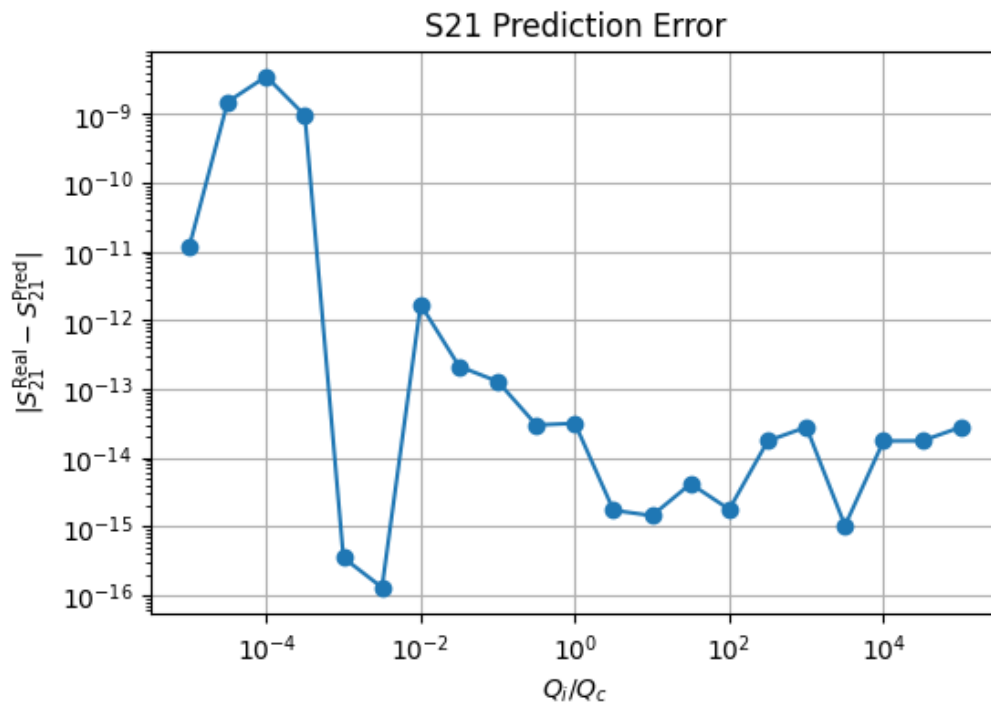


Figure 4.1: Absolute difference between the fitted and simulated S_{21} responses as a function of Q_i/Q_c .

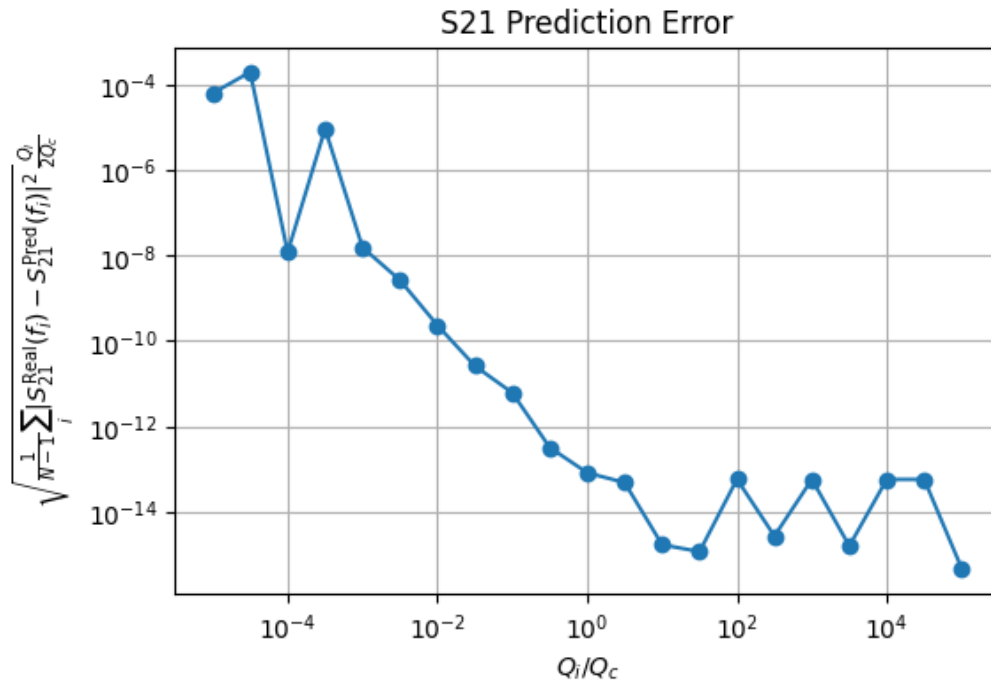


Figure 4.2: Root-mean-square error between the fitted and simulated S_{21} responses as a function of Q_i/Q_c .

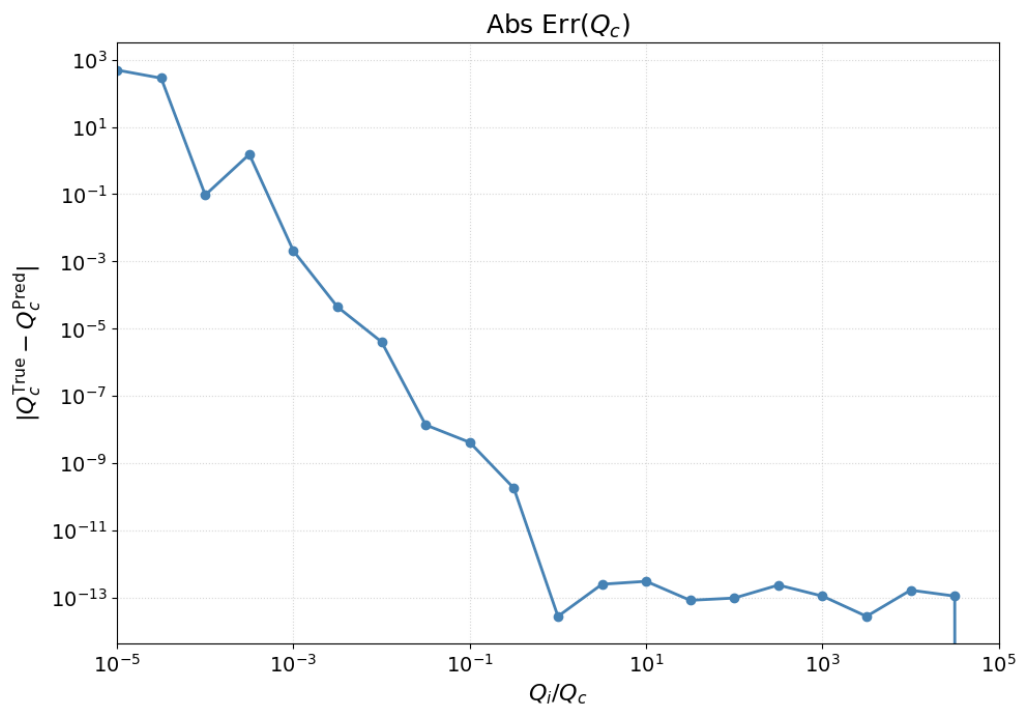


Figure 4.3: Absolute difference between the fitted and simulated Q_c values as a function of Q_i/Q_c .

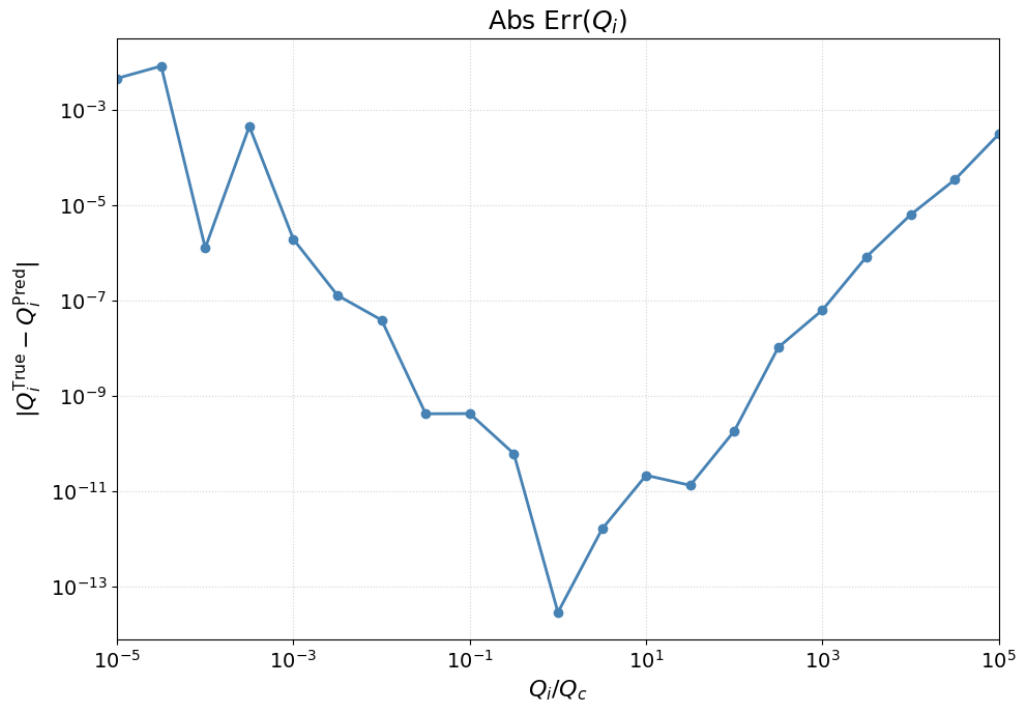


Figure 4.4: Absolute difference between the fitted and simulated Q_i values as a function of Q_i/Q_c .

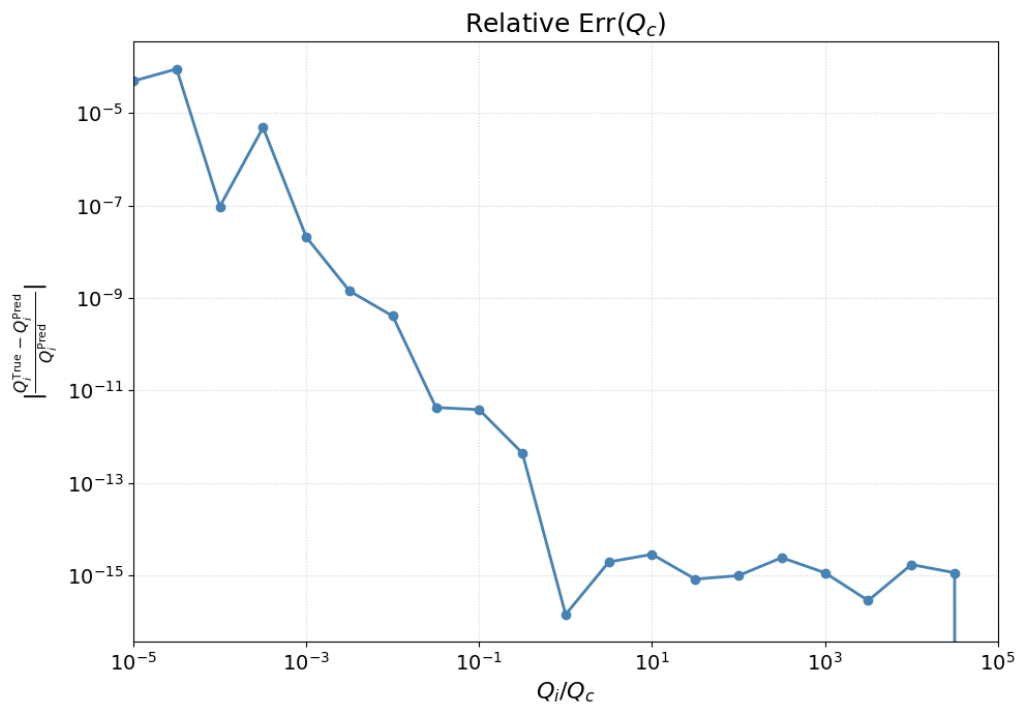


Figure 4.5: Relative error in the fitted Q_c as a function of Q_i/Q_c .

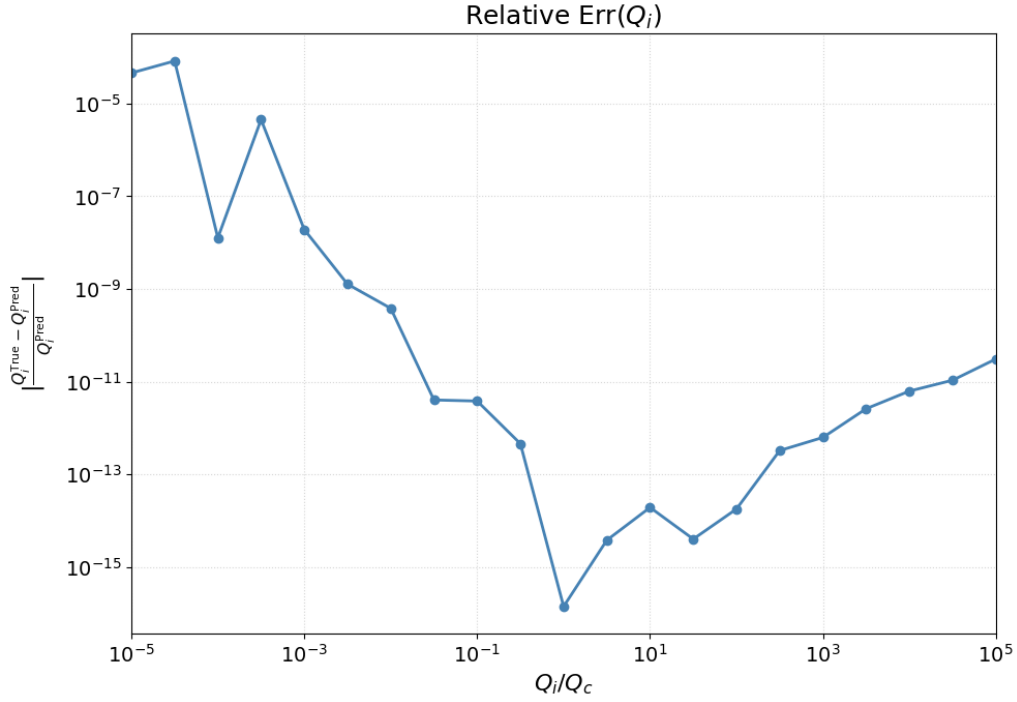


Figure 4.6: Relative error in the fitted Q_i as a function of Q_i/Q_c .

4.4.3.1 Some Observations from the plots

The results demonstrate that, in the absence of noise, electrical delay, and impedance mismatch phase, the fitting algorithm is capable of reproducing the S_{21} response with very high accuracy (Figure 4.1, 4.2). Double-precision floating-point arithmetic has a machine epsilon of the order of 10^{-16} ; the observed absolute error floor in the S_{21} prediction is $\sim 10^{-15}$ (Figure 4.1). The higher absolute errors seen in the undercoupled regime ($Q_i \ll Q_c$) arise from increased sensitivity in estimating the radius of the resonance circle: as $Q_i/Q_c \rightarrow 0$ the radius $r = \frac{Q_i}{2Q_c}$ shrinks, which degrades the prediction of the a, b, c, d parameters [7] and thereby also the extraction of the resonator parameters in this regime. The RMSE reaches values in the order of 10^{-14} (Figure 4.2), indicating that the algorithm operates close to the floating-point precision limit; that the residual lies slightly above the machine epsilon points to accumulated rounding effects, such as subtraction of nearly equal quantities or division by small numbers.

The trends in the absolute and relative errors for the prediction of Q_i and Q_c reveal how strongly the sensitivity of the extraction depends on the coupling regime (Figure 4.3–4.6). For Q_c (Figure 4.3, 4.5), in the regime $Q_i \ll Q_c$, since $Q_c = \frac{Q_l}{2r}$ [6], one has $\frac{dQ_c}{dr} = \frac{-Q_l}{2r^2}$; for small $r \rightarrow 0$, small errors in the prediction of r therefore produce large errors in the prediction of Q_c , making it very sensitive in this regime. In the opposite regime $Q_i \gg Q_c$, as $r \rightarrow 0.5$, $\frac{dQ_c}{dr} \approx -2Q_l$, which is a finite value, making the extraction of Q_c less sensitive and more stable. For Q_i (Figure 4.4, 4.6), the pattern reverses: for $Q_i \ll Q_c$, as $Q_i = Q_l \left(\frac{1}{1-2r} \right)$ with $r \rightarrow 0$, $(1-2r) \approx 1 \Rightarrow \frac{dQ_i}{dr} = \frac{2Q_l}{(1-2r)^2} \approx 2Q_l$, where Q_l is a constant value for these simulations; for $Q_i \gg Q_c$, as $r \rightarrow 0.5$, $(1-2r) \rightarrow 0 \Rightarrow \frac{dQ_i}{dr} \rightarrow \infty$,

causing small errors in r to produce large absolute errors in the prediction of Q_i , making it very sensitive in this regime.

This sensitivity has a direct physical origin, understood by considering how energy is lost in each coupling regime. In the undercoupled regime ($Q_i \ll Q_c$, i.e. $\kappa_i \gg \kappa_c$), internal losses dominate: energy stored in the resonator is dissipated primarily through intrinsic mechanisms such as two-level systems (TLS), rather than being coupled out to the measurement line, so the resonator feature becomes weak and less dominant in the measured S_{21} , making it difficult to accurately extract resonator parameters. In the overcoupled regime ($Q_i \gg Q_c$, i.e. $\kappa_i \ll \kappa_c$), coupling to the transmission line dominates instead: the resonator exchanges energy very strongly with the external environment, which broadens the resonance and reduces sensitivity to internal properties, so the extracted parameters become less reliable. Only near the critically coupled regime ($Q_i \approx Q_c$) does the balance between internal dissipation and external coupling give a well-defined resonance and the most accurate parameter extraction.

4.4.4 Conclusion

The plots above show that the fitting algorithm achieves its highest accuracy in the regime $Q_i \approx Q_c$. At this critical coupling condition the same regime singled out in Section 3.4; the resonance circle is well-defined ($r \approx 0.25$), resulting in successful and accurate extraction of the resonator parameters [Figure 4.3, 4.4]. Consequently, both absolute and relative errors are minimized, making this regime the most reliable for parameter estimation. With this numerical baseline established, the non-idealities of the measurement chain can now be introduced one at a time, beginning with the electrical delay τ .

4.5 Verification - II : Electrical Delay (τ)

4.5.1 Motivation

Having established the numerical baseline, the first non-ideality of the hanger model, Equation (3.19), to be switched on is the electrical delay (τ), which enters as the phase factor $e^{-i2\pi f\tau}$. This delay introduces an additional phase shift due to the finite propagation time of the signal through the measurement setup. It typically arises from long cables within the measurement chain both inside and outside the cryostat, as well as from components such as connectors, transmission lines, and other intermediate elements between the source and the device under test. To assess its impact, synthetic data is generated with electrical delay values of $\tau \in \{10, 20, 30, 40, 50\}$ ns, both in the absence and presence of Gaussian noise. The performance of the fitting algorithm is then evaluated at a fixed SNR by analyzing the absolute error in the extracted values of Q_c [Figure 4.7] and Q_i [Figure 4.9] in the absence of noise, and the standard deviation of Q_c [Figure 4.8] and Q_i [Figure 4.10] in the presence of noise.

4.5.2 Parameters

The following table lists the parameters used for this verification's simulations and analysis.

Table 4.4: Verification II: simulation parameters

	Parameter Type	Noise Simulations	No Noise Simulations	Units
Electrical delay	Input	(10, 20, 30, 40, 50)	(10, 20, 30, 40, 50)	ns
Q_l	Input	100	100	-
Q_i/Q_c	Input	($1e^{-5}$, $1e^5$)	($1e^{-5}$, $1e^5$)	-
N_{avg}	Input	80	1	-
Noise Type	Input	Gaussian	None	-
Signal-to-noise Ratio	Input	65	N/A	-
Statistical Measure	Output	Standard Deviation	Absolute Error	-

4.5.3 Plots

The two non-idealities are evaluated with different statistics because one is deterministic and the other is not: without noise the simulation is deterministic, so a single run already gives a well-defined absolute error, whereas with noise each run is one random realization, so a single run's error is not representative and N_{avg} noisy runs are instead characterized by the spread (standard deviation) of the extracted values around their mean. The absolute error [Figure 4.7, 4.9] and standard deviation [Figure 4.8, 4.10] of the extracted quality factors are shown below, followed by a discussion of the observed trends.

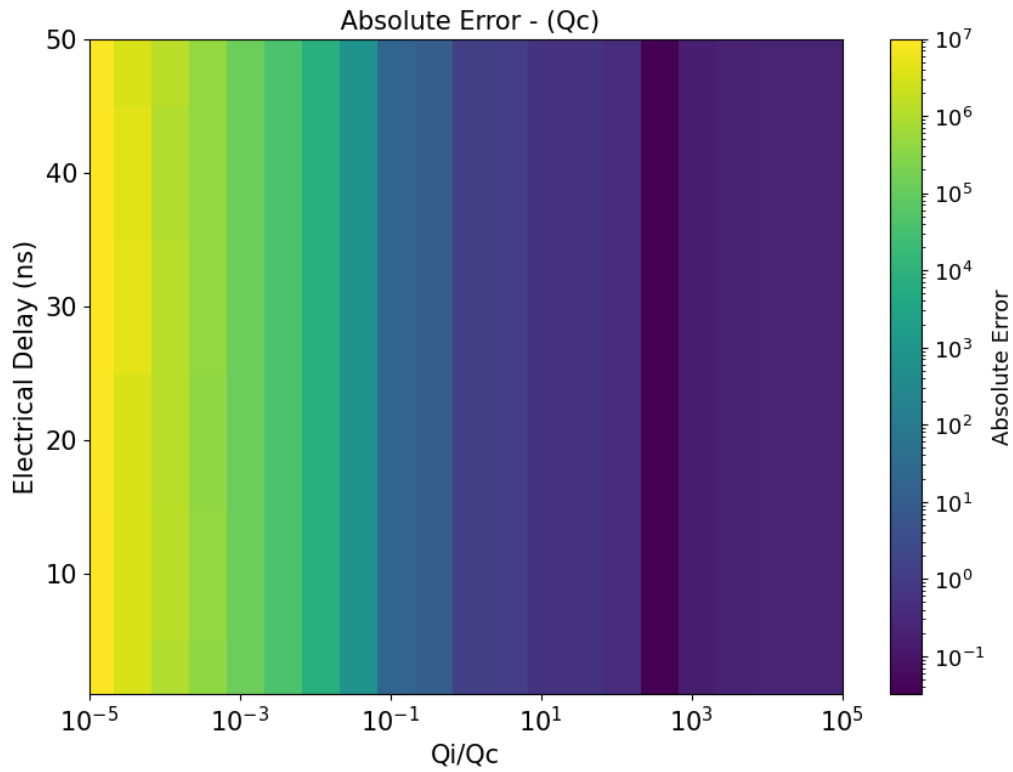


Figure 4.7: Absolute error in the fitted Q_c as a function of Q_i/Q_c and electrical delay, without noise.

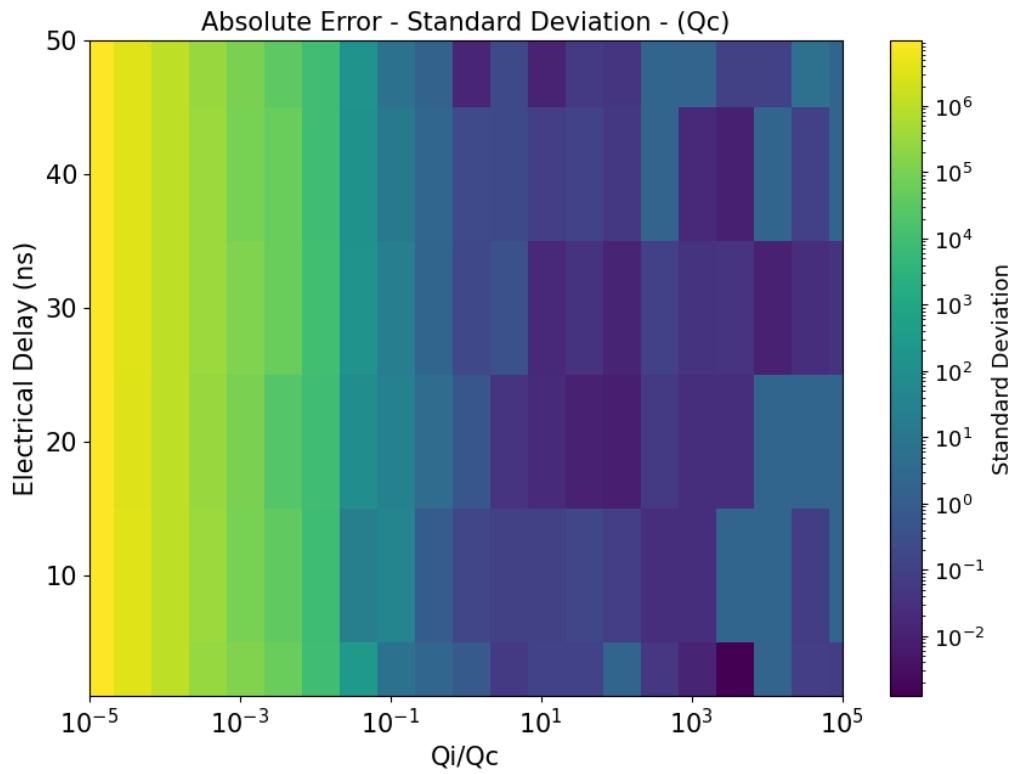


Figure 4.8: Standard deviation of the fitted Q_c as a function of Q_i/Q_c and electrical delay, with noise and 80 averages.

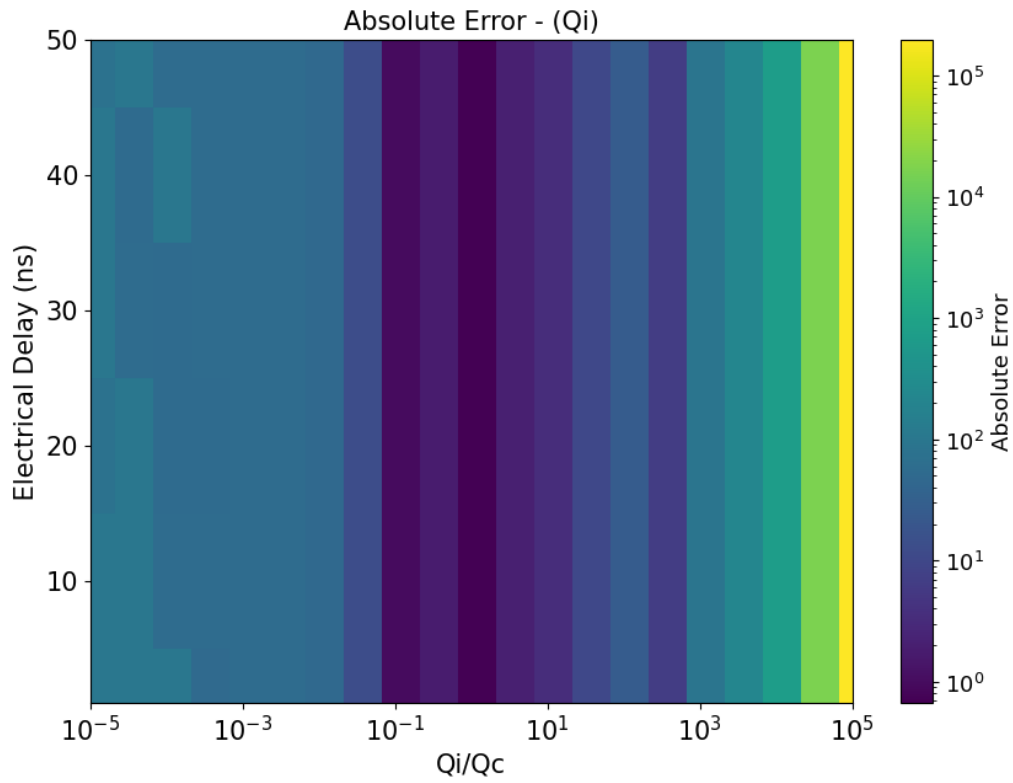


Figure 4.9: Absolute error in the fitted Q_i as a function of Q_i/Q_c and electrical delay, without noise.

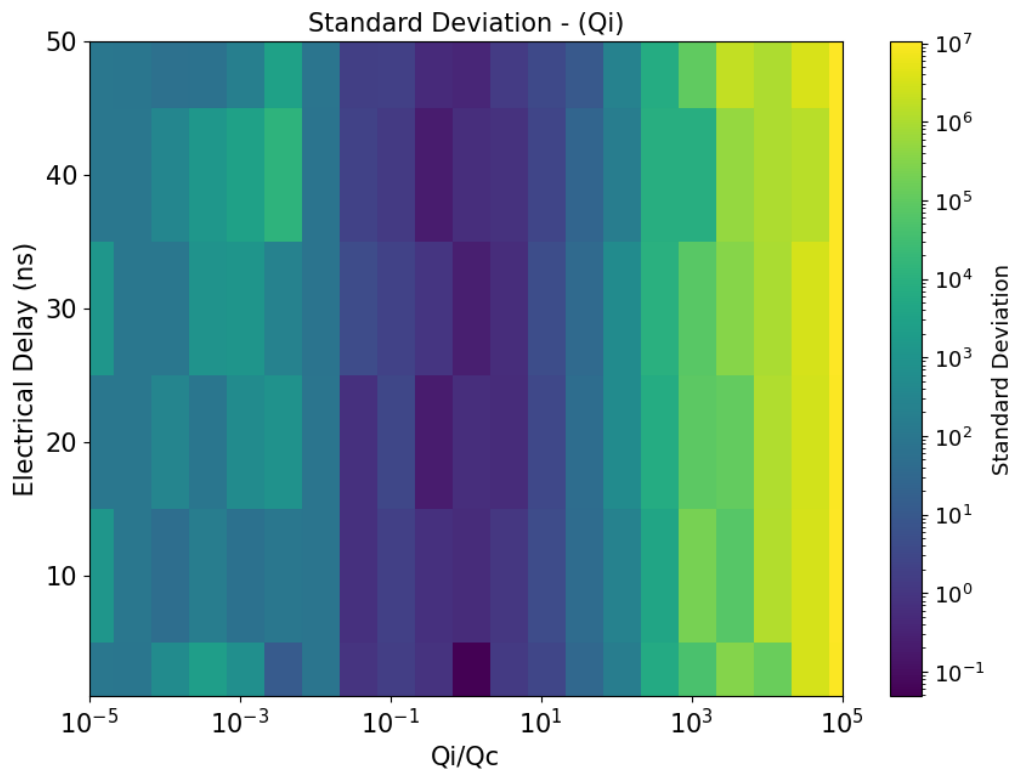


Figure 4.10: Standard deviation of the fitted Q_i as a function of Q_i/Q_c and electrical delay, with noise and 80 averages.

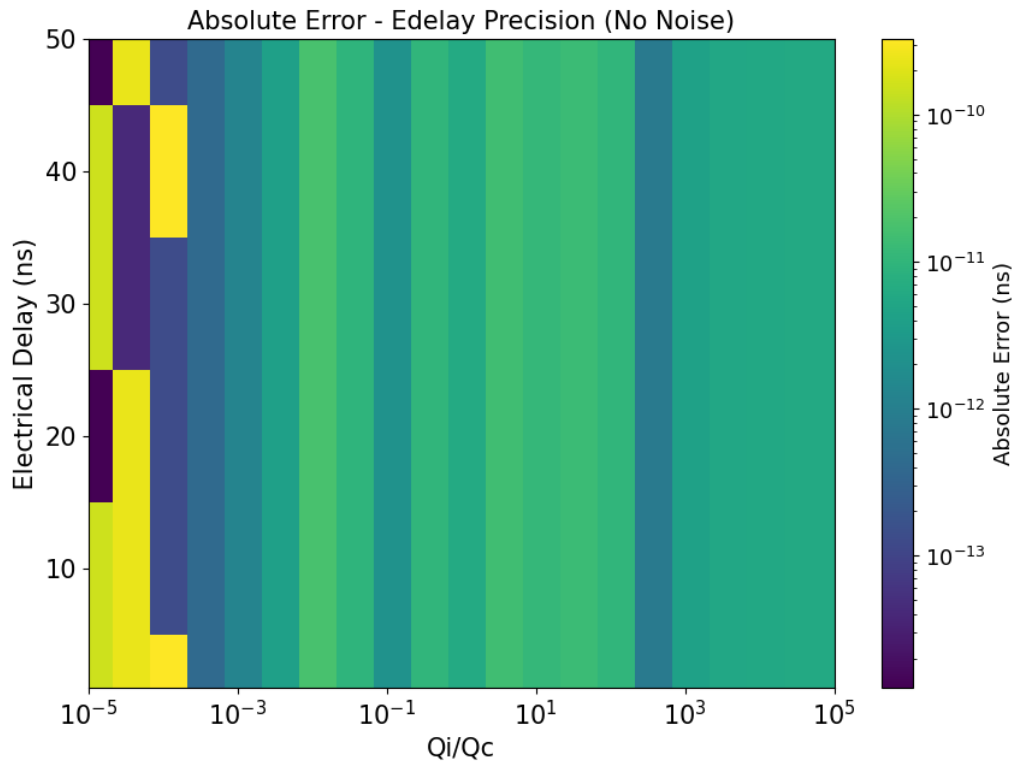


Figure 4.11: Absolute error in the fitted electrical delay as a function of Q_i/Q_c and simulated delay, without noise.

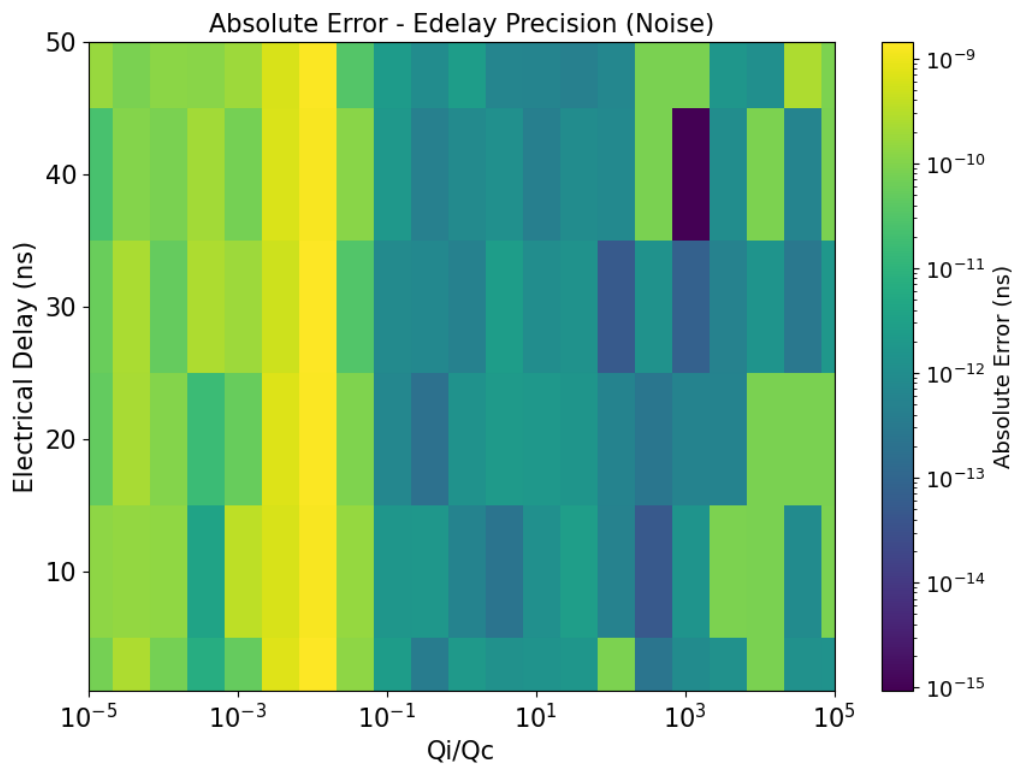


Figure 4.12: Absolute error in the fitted electrical delay as a function of Q_i/Q_c and simulated delay, with noise.

4.5.3.1 Some Observations from the plots

The plots above characterize the algorithm's behavior across a wide range of $\frac{Q_i}{Q_c}$ ratios, encompassing both the undercoupled regime ($Q_i \ll Q_c$) and the overcoupled regime ($Q_i \gg Q_c$). The absolute errors in Q_c (Figure 4.7, 4.8) and Q_i (Figure 4.9, 4.10) follow exactly the same radius-sensitivity behavior derived in Verification I: sensitive in the undercoupled regime, most accurate near critical coupling, and comparatively stable again in the overcoupled regime. For the absolute errors in electrical delay (Figure 4.11, 4.12), in the critical regime $Q_i \approx Q_c$, for all considered values of electrical delay, the absolute errors in the presence and absence of noise are of the order 10^{-11} s and 10^{-12} s, respectively; this behavior shows that the algorithm robustly recovers the value of the electrical delay from the data, both with and without noise, in the critical regime, whereas larger errors are to be expected in the undercoupled and overcoupled regimes.

4.5.4 Conclusion

The fitting algorithm performs reliably here, extracting resonator parameters from complex S_{21} data along with an accurate estimate of the electrical delay. As the analysis shows, the accuracy of the extracted quality factors depends strongly on the coupling regime. In particular, the regime $Q_i \approx Q_c$ yields the most accurate estimates of Q_i , Q_c , and the electrical delay; the extraction of these parameters is governed by the errors in the prediction of the resonance circle radius r ; when $Q_i \approx Q_c$, $r \approx 0.25$. For predictions of Q_c , the extraction becomes increasingly sensitive to small uncertainties in r in the undercoupled regime ($Q_i \ll Q_c$), while in the overcoupled regime ($Q_i \gg Q_c$) the absolute error(s) converge to Q_l . For predictions of Q_i , the situation is reversed: the absolute error(s) converge to Q_l in the undercoupled regime ($Q_i \ll Q_c$), while in the overcoupled regime ($Q_i \gg Q_c$) the parameter extraction becomes increasingly sensitive to small uncertainties in r , leading to larger absolute error. The errors in the extracted resonator parameters are largely independent of the electrical delay value; instead, they are primarily influenced by the radius of the resonance circle, indicating that the performance of the fitting algorithm is robust with respect to variations in electrical delay.

The picture so far characterizes the τ -robustness across the full range of Q_i/Q_c at a single SNR; the next section completes it by fixing the quality factors at a representative value and varying the SNR instead.

4.6 Verification - III: Electrical Delay (τ), Fixed Quality Factors

4.6.1 Motivation

Whereas the previous section swept the coupling ratio Q_i/Q_c at a fixed SNR, the complementary question is how the extraction degrades with noise at fixed quality factors. To this end, a representative case with $Q_i = 1e^4$ and $Q_c = 1e^3$ is analyzed as a function of the signal-to-noise ratio (SNR) [Figure 4.13], probing the performance of the fitting algorithm across a range of SNR values for the same set of electrical delays.

4.6.2 Parameters

These are the simulation and analysis parameters used for this verification.

Table 4.5: Verification III: simulation parameters

	Parameter Type	Noise Simulations	No Noise Simulations	Units
Electrical delay	Input	(10, 20, 30, 40, 50)	(10, 20, 30, 40, 50)	ns
Q_i	Input	$1e^4$	$1e^4$	-
Q_c	Input	$1e^3$	$1e^3$	-
N_{avg}	Input	80	1	-
Noise Type	Input	Gaussian	None	-
Signal-to-noise Ratio	Input	21 points log-spaced over [1, 1000]	N/A	-
Statistical Measure	Output	Standard Deviation	Absolute Error	-

4.6.3 Plots

Having fixed the quality factors at a representative value ($Q_i = 10^4$, $Q_c = 10^3$), this section instead sweeps the signal-to-noise ratio: Figure 4.13 shows the extracted Q_i , Q_c , and Q_l directly versus SNR, Figure 4.14 shows how their standard deviation responds to electrical delay at a fixed SNR, and Figures 4.15 and 4.16 show that same standard deviation swept instead over Q_i/Q_c and over SNR.

Q Factors vs SNR for Different Electrical Delays

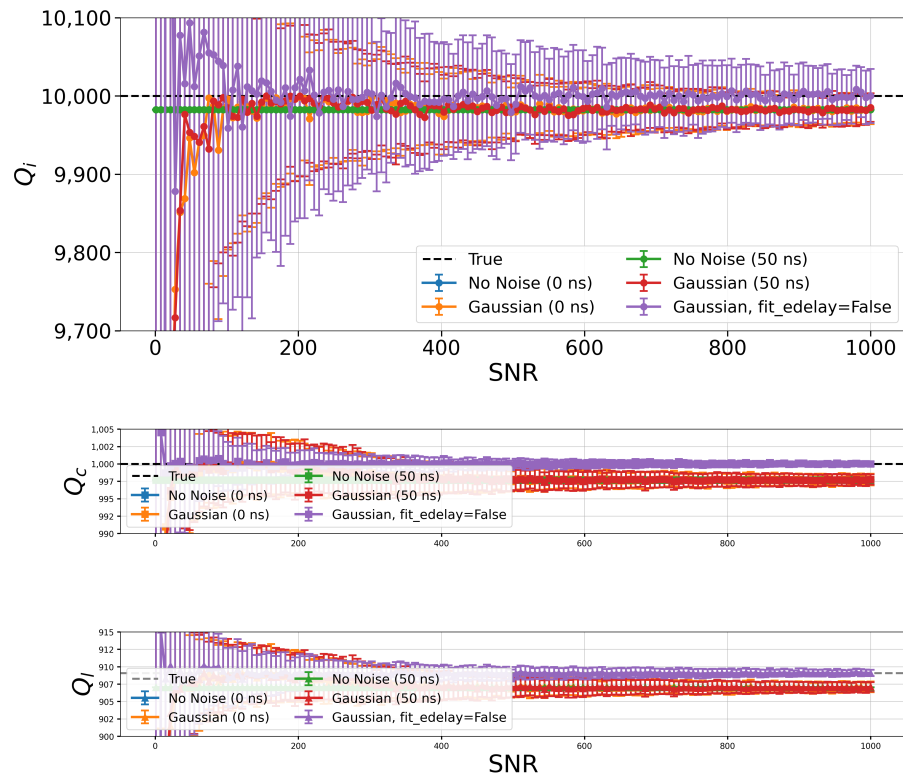


Figure 4.13: Extracted quality factors Q_i , Q_c , and Q_l as a function of SNR, averaged over 25 iterations. Curves compare noiseless and Gaussian-noise data at electrical delays of 0 ns and 50 ns.

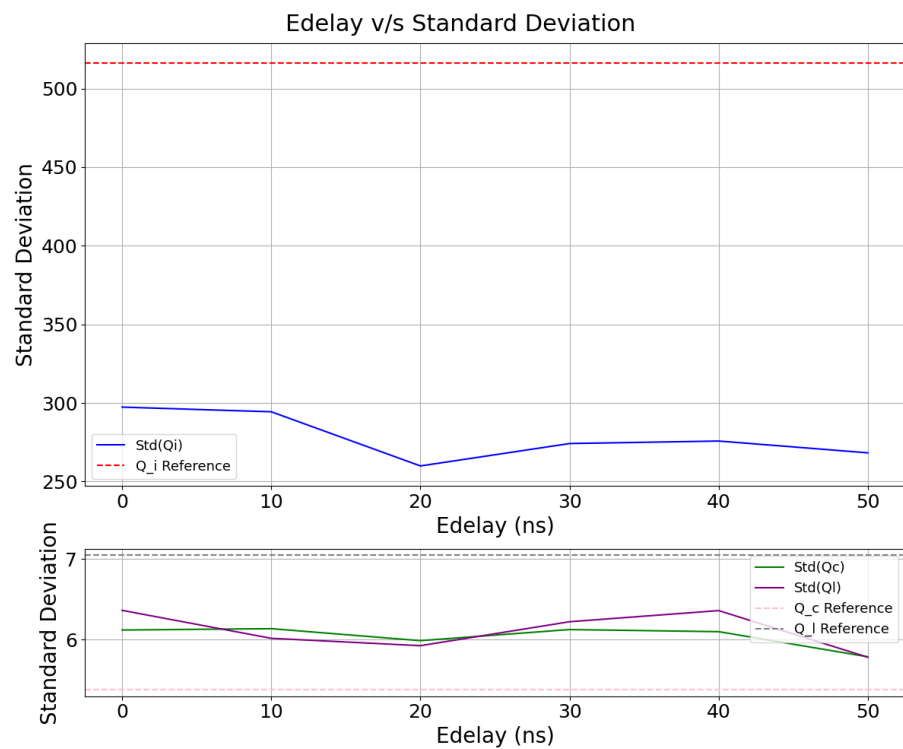


Figure 4.14: Standard deviation of the extracted quality factors Q_i , Q_c , and Q_l as a function of electrical delay at SNR = 65.

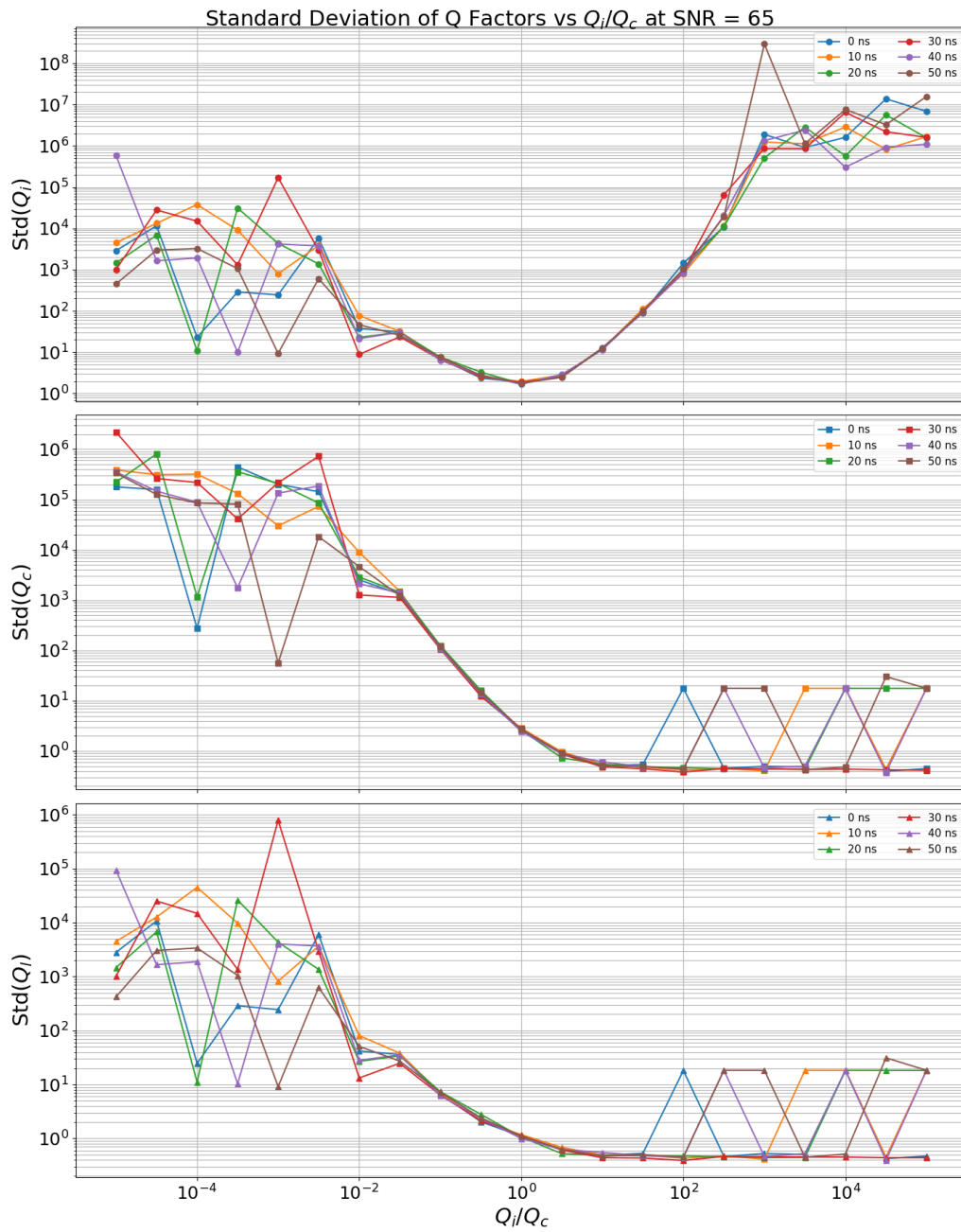


Figure 4.15: Standard deviation of the extracted quality factors Q_i , Q_c , and Q_l as a function of the coupling ratio Q_i/Q_c at fixed SNR = 65.

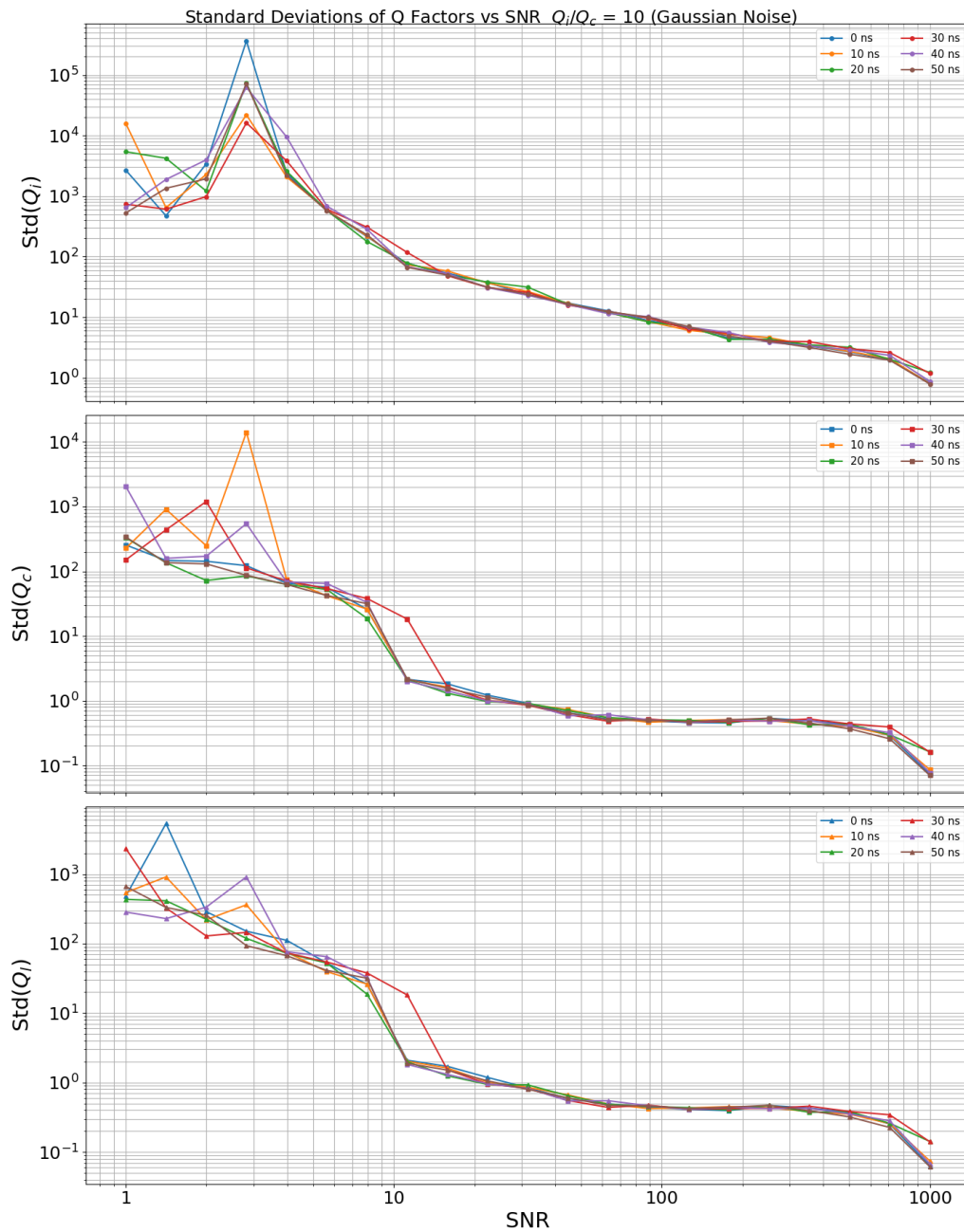


Figure 4.16: Standard deviation of the extracted quality factors Q_i , Q_c , and Q_l as a function of SNR.

4.6.3.1 Some Observations from the plots

As expected, the fitting algorithm performs poorly at low SNR, exhibiting large uncertainties as noise dominates the signal (Figure 4.13). However, as the SNR increases, the fitted parameters converge toward their true values. To further assess this convergence, the standard deviation of the extracted Q factors is plotted as a function of SNR in Figure 4.16; at high SNR values (at fixed $Q_i/Q_c = 10$), the standard deviations approach their minimum values, largely independent of the electrical delay. The error bars seen in Figure 4.13 remain high, which is explained by the span chosen for these

simulations: for spans larger than 100 MHz, less variability is observed at higher SNR. A span of 100 MHz is nevertheless chosen to ensure reliable information retrieval from the resonator, since the simulations place the frequency points linearly around the resonance (standard point distribution (SPD)) [8]. This follows the relation $\Delta F / \Delta f_r = 2$ [8], where ΔF is the span and $\Delta f_r = f_r / Q_l$; with $f_r = 5 \times 10^9$ Hz and $Q_l = 100$, this gives $\Delta F = 100$ MHz.

An additional detail in Figure 4.13 is the curve labeled *fit_edelay* = *False* (purple line). In this case, the fitting routine is executed without accounting for electrical delay, as the simulated S_{21} data contains no delay component; consequently, this curve accurately recovers the true parameters without systematic offset. In contrast, when *fit_edelay* = *True* (green line) is used on data without electrical delay, the algorithm introduces a systematic offset in the extracted Q_i , Q_c , and Q_l values, indicative of overfitting, where the model attempts to fit a non-existent feature in the data.

Figure 4.14 shows the impact of enabling the *fit_edelay* parameter on the standard deviation of the extracted resonator parameters as a function of electrical delay. For Q_i and Q_l , the standard deviation decreases when *fit_edelay* = *True* (blue and green solid lines), indicating improved robustness of the fit in the presence of electrical delay. However, this trend is not observed for Q_c (pink solid line), where the standard deviation does not systematically decrease, suggesting that fitting the electrical delay has a more limited effect on the accuracy of the extraction of Q_c .

In Figure 4.15, at a fixed SNR = 65, the standard deviation is minimized in the regime $Q_i \approx Q_c$, for all extracted parameters Q_i , Q_c and Q_l . For Q_c and Q_l , large standard deviations are observed in the undercoupled regime, likely due to inaccuracies in fitting the circle parameters a, b, c, d [7]; in contrast, in the overcoupled regime, the standard deviation reduces and approaches a nearly constant value. The extraction of Q_i shows high standard deviations in both regimes: in the undercoupled regime, this is again attributed to inaccuracies in fitting the parameters a, b, c, d [7], while in the overcoupled regime ($Q_i \gg Q_c$, i.e. $\kappa_i \ll \kappa_c$), it becomes increasingly difficult to distinguish between losses due to coupling and intrinsic mechanisms, leading to increased uncertainty in the extracted Q_i .

4.6.4 Conclusion

The fitting algorithm again performs reliably, extracting resonator parameters from complex S_{21} data together with an accurate estimate of the electrical delay. The accuracy of the extracted quality factors matches the results expected from [6], confirming that the model behaves as anticipated.

4.7 Verification - IV: Impedance mismatch Phase (ϕ)

4.7.1 Motivation

The second non-ideality of the hanger model, Equation (3.19), is the impedance mismatch phase ϕ , which Section 3.4 introduced as the rotation of the coupling quality factor into the complex plane, $Q_c \rightarrow |Q_c|e^{-i\phi}$. Physically, practical superconducting devices contain multiple material interfaces at which the propagating signal experiences impedance discontinuities due to changes in the medium, and these discontinuities introduce the additional phase shift ϕ in the signal. Such impedance mismatches are not limited to on-chip interfaces (wire bonds etc.) but also occur at various points in the measurement chain, including connectors, amplifiers, and other components. This makes it essential to benchmark the effect of this phase on the S_{21} parameter. To evaluate the performance of the fitting algorithm, an impedance mismatch phase $\phi \in (-\frac{\pi}{2}, \frac{\pi}{2})$ is introduced into synthetic S_{21} data. The robustness of the algorithm is then assessed by extracting the quality factors Q_i and Q_c . As in the previous sections, the algorithm is tested both in the absence and presence of Gaussian noise to mimic realistic measurement conditions [Figure 4.17–4.20]. Finally, the accuracy of the fitted phase ϕ is analyzed [Figure 4.21, 4.22].

4.7.2 Parameters

The parameters used for the simulations and analysis in this verification are listed below.

Table 4.6: Verification IV: simulation parameters

	Parameter Type	Noise Simulations	No Noise Simulations	Units
ϕ	Input	$(-\pi/2, \pi/2)$	$(-\pi/2, \pi/2)$	rad
Q_i	Input	100	100	-
Q_i/Q_c	Input	$(1e^{-5}, 1e^5)$	$(1e^{-5}, 1e^5)$	-
N_{avg}	Input	80	1	-
Noise Type	Input	Gaussian	None	-
Signal-to-noise Ratio	Input	65	N/A	-
Statistical Measure	Output	Standard Deviation	Absolute Error	-

4.7.3 Plots

As in Verification II, the absolute error [Figure 4.17, 4.19] and standard deviation [Figure 4.18, 4.20] of the extracted quality factors are plotted below, with a discussion

of the observed trends following.

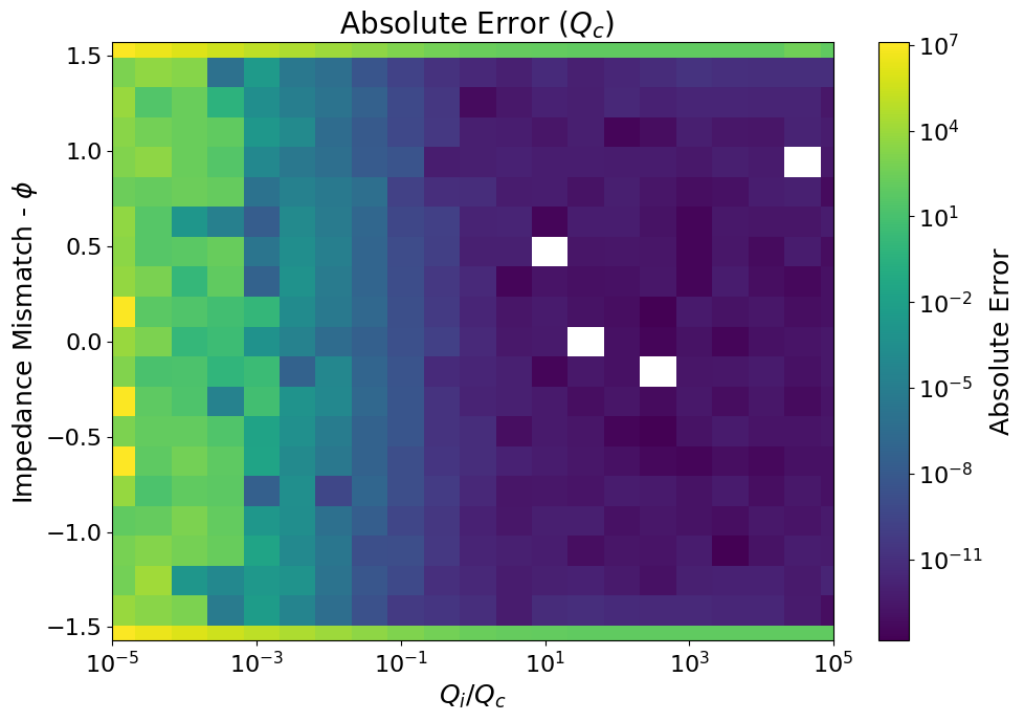


Figure 4.17: Absolute error in the fitted Q_c as a function of Q_i/Q_c and mismatch phase ϕ , without noise.

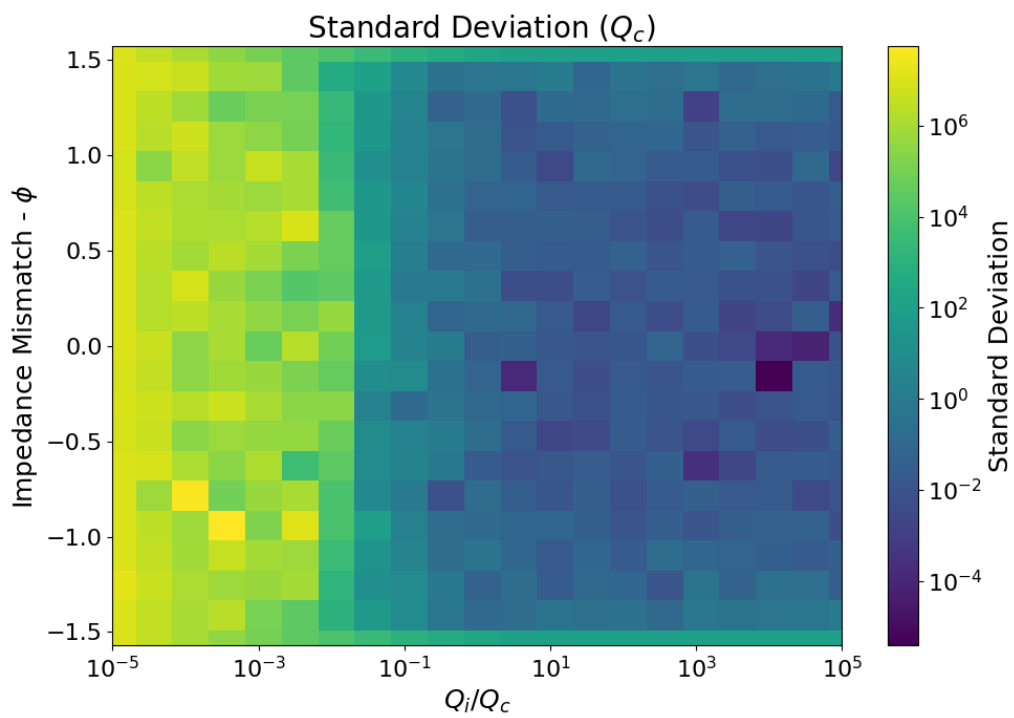


Figure 4.18: Absolute error in the fitted Q_c as a function of Q_i/Q_c and mismatch phase ϕ , with noise.

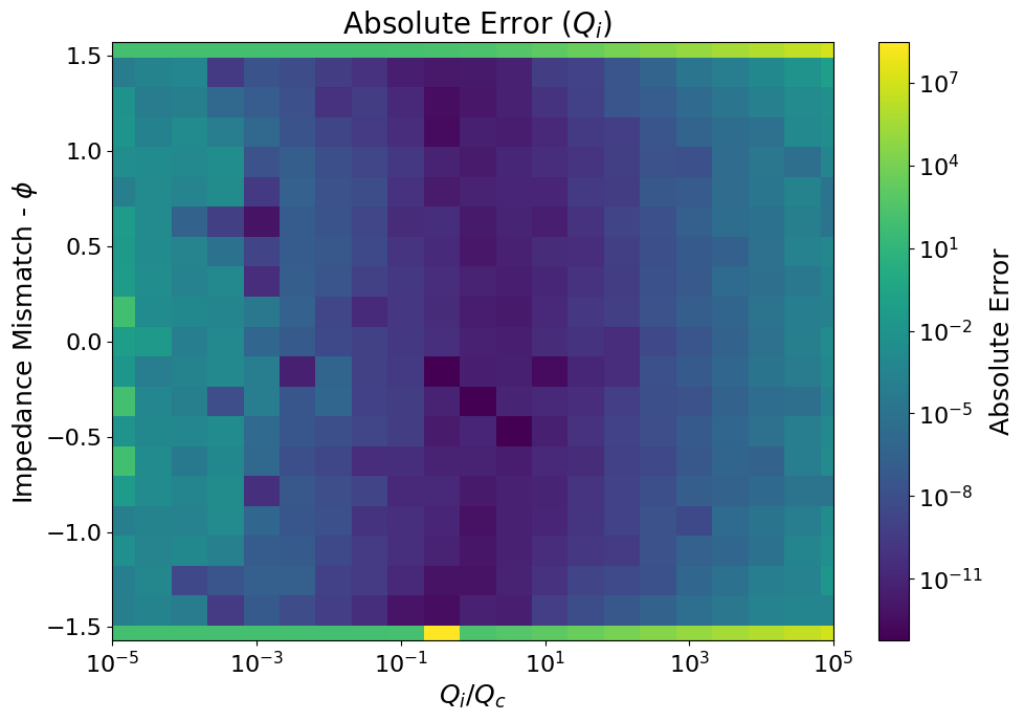


Figure 4.19: Absolute error in the fitted Q_i as a function of Q_i/Q_c and mismatch phase ϕ , without noise.

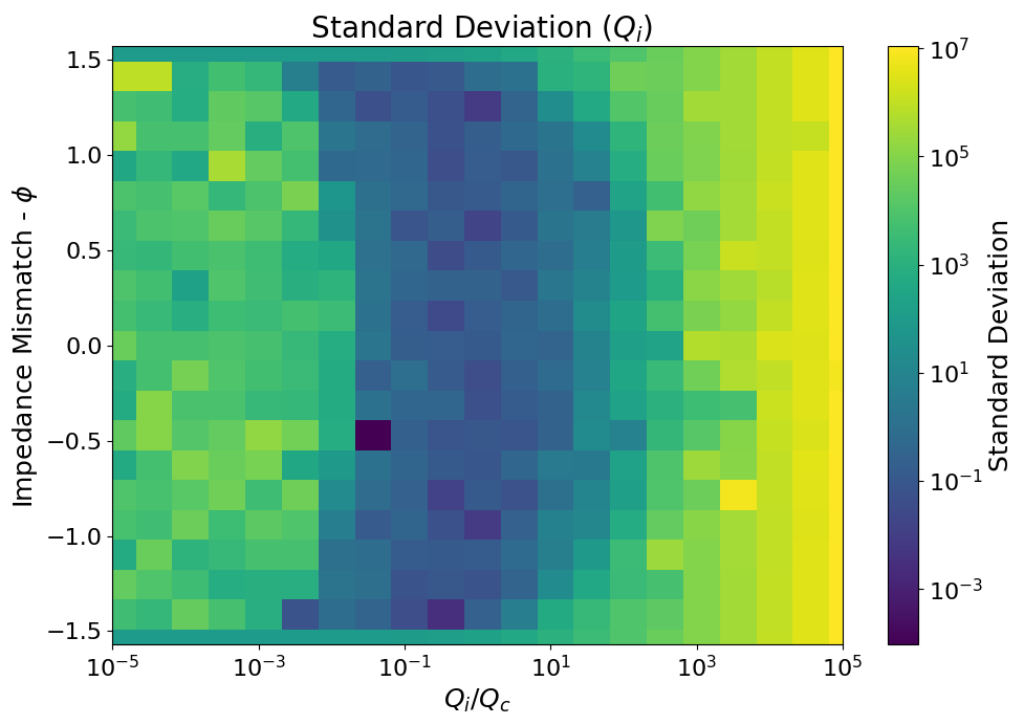


Figure 4.20: Absolute error in the fitted Q_i as a function of Q_i/Q_c and mismatch phase ϕ , with noise.

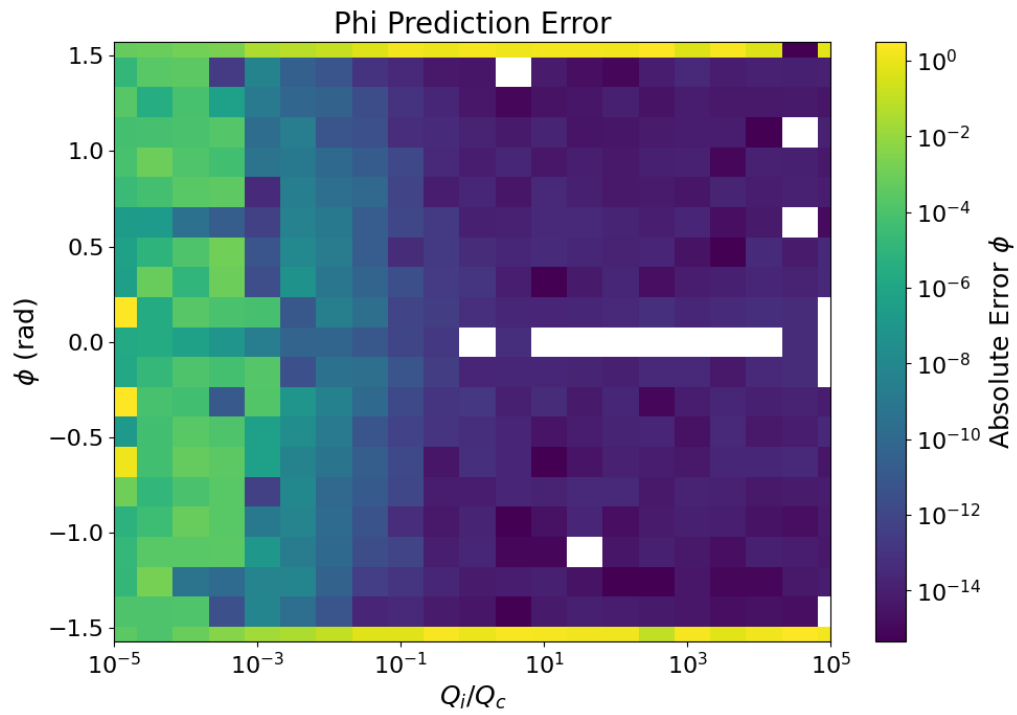


Figure 4.21: Absolute error in the fitted impedance-mismatch phase ϕ as a function of Q_i/Q_c and simulated ϕ , without noise.

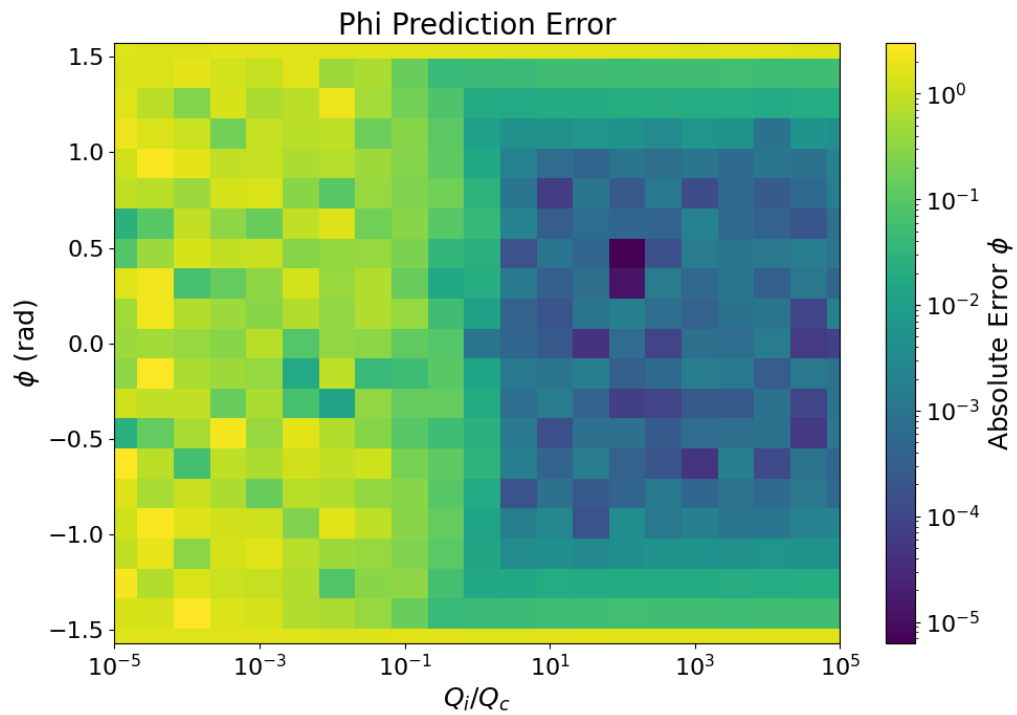


Figure 4.22: Absolute error in the fitted impedance-mismatch phase ϕ as a function of Q_i/Q_c and simulated ϕ , with noise.

4.7.3.1 Some Observations from the plots

The absolute errors in Q_c (Figure 4.17, 4.18) and Q_i (Figure 4.19, 4.20) again follow the same radius-sensitivity behavior derived in Verification I: sensitive to r in the undercoupled regime, most accurate near critical coupling ($r \approx 0.25$), and comparatively stable again in the overcoupled regime.

Near the boundary of the domain $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, very high absolute errors appear across all of these plots (Figure 4.17–4.22) as ϕ approaches $\pm\frac{\pi}{2}$. This is because only the real part of the complex coupling term contributes to the linewidth, $\text{Re}\left(\frac{1}{Q_c}\right) = \frac{\cos(\phi)}{|Q_c|}$ – the purely-reactive-mismatch limit anticipated in Section 3.4. As ϕ approaches $\pm\frac{\pi}{2}$, the real part of the coupling goes to zero, making it difficult to distinguish Q_i and Q_c and leading to higher absolute errors.

For the absolute errors in the impedance mismatch phase itself (Figure 4.21, 4.22), in the overcoupled regime, as $r \rightarrow 0.5$, the sensitivity of the Q_c extraction converges to $\frac{dQ_c}{dr} \approx -2Q_l$, making the extraction of $\text{Re}\left(\frac{1}{Q_c}\right)$ accurate, and hence the extraction of ϕ accurate as well. In the critical regime ($Q_i \approx Q_c$) the fitting algorithm recovers the value of the impedance mismatch phase from the data very accurately, because, with $Q_i \approx Q_c$, the radius of the resonance circle $r \rightarrow 0.25$, which leads to accurate extraction of the a, b, c, d parameters [7] and, in turn, an accurate extraction of ϕ in this regime. In the undercoupled regime, a gradual increase in the absolute error in the prediction of ϕ is observed: as $Q_i/Q_c \rightarrow 0$, the radius of the resonance circle $r \rightarrow 0$, which leads to inaccurate extraction of the a, b, c, d parameters [7], in turn impacting the prediction of ϕ .

4.7.4 Conclusion

As in the previous verifications, the fitting algorithm performs reliably, extracting resonator parameters from complex S_{21} data together with an accurate estimate of the impedance-mismatch phase. The analysis again shows that the accuracy of the extracted quality factors depends strongly on the coupling regime. In particular, the regime $Q_i \approx Q_c$ continues to yield the most stable and accurate parameter estimates, consistent with every previous verification in this chapter. The performance of the fitting algorithm is more strongly influenced by the estimation of the radius r than by the phase ϕ itself: across a wide range of ϕ , the algorithm performs consistently, except near the boundary values of $\pm\frac{\pi}{2}$, where $\text{Re}\left(\frac{1}{Q_c}\right) = \frac{\cos(\phi)}{|Q_c|} \rightarrow 0$, meaning the coupling becomes purely imaginary; as a result, the fitting becomes highly sensitive, leading to significantly larger absolute errors in ϕ specifically.

4.8 Verification - V: Background

4.8.1 Motivation

The hanger model, Equation (3.19), describes the off-resonance background by the constant complex amplitude $ae^{i\alpha}$. In real experiments, however, a frequency-dependent background is often observed, arising from non-idealities such as cables, wire bonds, and other components in the measurement chain. To evaluate the performance of the resonator fitting algorithm under these realistic conditions, this final verification replicates such a background and incorporates it into the synthetic data. The background response S_{21}^{bg} is constructed from experimental data obtained from the chips NTC-0002-02-K, D, GZ, I, H, J (see [here](#)). The background amplitude and phase are independently modeled by fitting second-degree polynomials to short, overlapping frequency windows: the Chunk Size is the width of each such window and the Overlap Size is how much consecutive windows overlap, both needed to normalize the background piecewise before fitting; a detailed step-by-step description of this fitting procedure is provided in Appendix A. An additional important aspect of this analysis is estimating the experimental signal-to-noise ratio (SNR) associated with the data used to generate the background, so that the simulations can be performed under comparable SNR; a detailed discussion is provided in Appendix B, which also lists chip-specific SNR values.

The magnitude of the background is modeled as

$$|S_{21}^{\text{bg}}| = f(\Delta\omega) = m_0 + m_1\Delta\omega + m_2\Delta\omega^2 \quad (4.3)$$

while the phase (ϕ_{bg}) of the background is fitted as

$$\angle S_{21}^{\text{bg}} = \phi_{\text{bg}}^{\text{fit}} = \arctan\left(\frac{m_0^a + m_1^a\Delta\omega + m_2^a\Delta\omega^2}{m_0^b + m_1^b\Delta\omega + m_2^b\Delta\omega^2}\right), \quad \Delta\omega = \omega - \omega_0 \quad (4.4)$$

The final modeled response, including the background, is then given by

$$S_{21}^{\text{model+bg}} = S_{21}^{\text{model}} \times |S_{21}^{\text{bg}}| \exp(i\phi_{\text{bg}}^{\text{fit}}) \quad (4.5)$$

4.8.2 Parameters

The simulation parameters for this verification are summarized below.

Table 4.7: Verification V: simulation parameters

	Parameter Type	Noise Simulations	No Noise Simulations	Units
Chunk Size	Input	10	10	MHz
Overlap Size (overlap between n and n+1 chunk)	Input	8	8	MHz
Span (Δf_0)	Input	20	20	kHz
Q_i	Input	2.5×10^5	2.5×10^5	-
Q_i/Q_c	Input	$(1e^{-5}, 1e^5)$	$(1e^{-5}, 1e^5)$	-
N_{avg}	Input	80	1	-
Noise Type	Input	Gaussian	None	-
Signal-to-noise Ratio	Input	Depends on chip	N/A	-
Statistical Measure	Output	Standard Deviation	Absolute Error	-

4.8.3 Plots

The following plots compare the recovered Q_c and Q_i values under noiseless and noisy conditions, both with and without an impedance-mismatch background. They also compare alternative background models to assess how strongly the chosen correction influences the fitted quality factors.

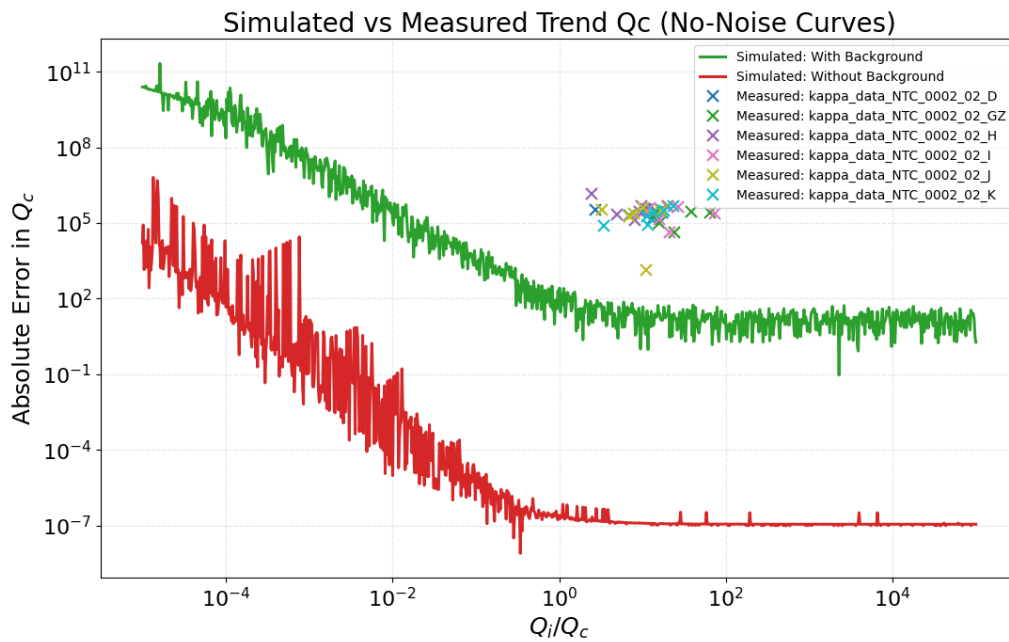


Figure 4.23: Absolute error in the fitted Q_c as a function of Q_i/Q_c , comparing synthetic signals with and without the measured background in the absence of Gaussian noise.

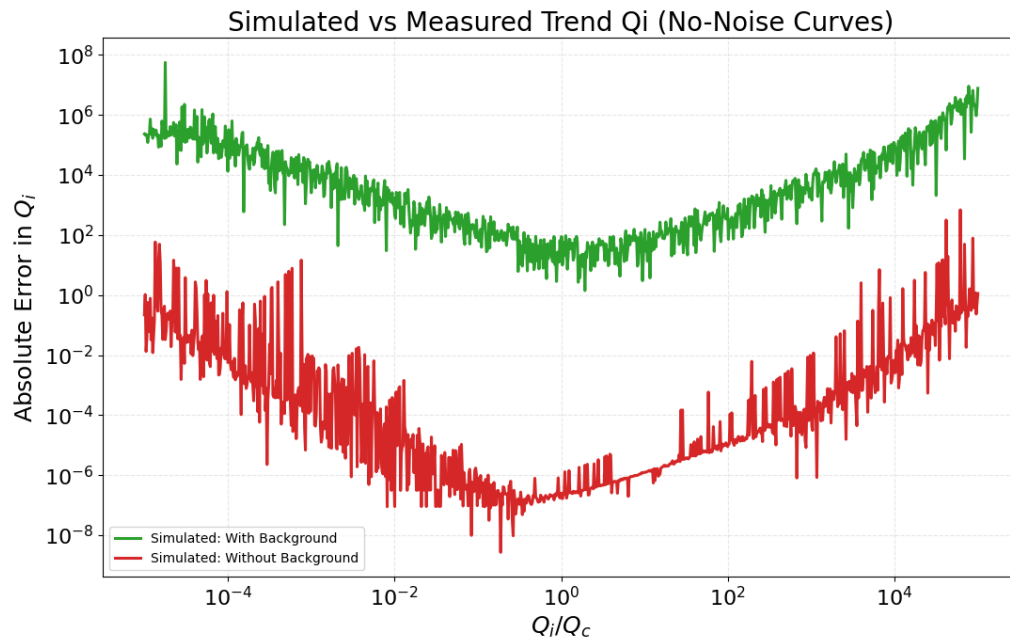


Figure 4.24: Absolute error in the fitted Q_i as a function of Q_i/Q_c , comparing synthetic signals with and without the measured background in the absence of Gaussian noise.

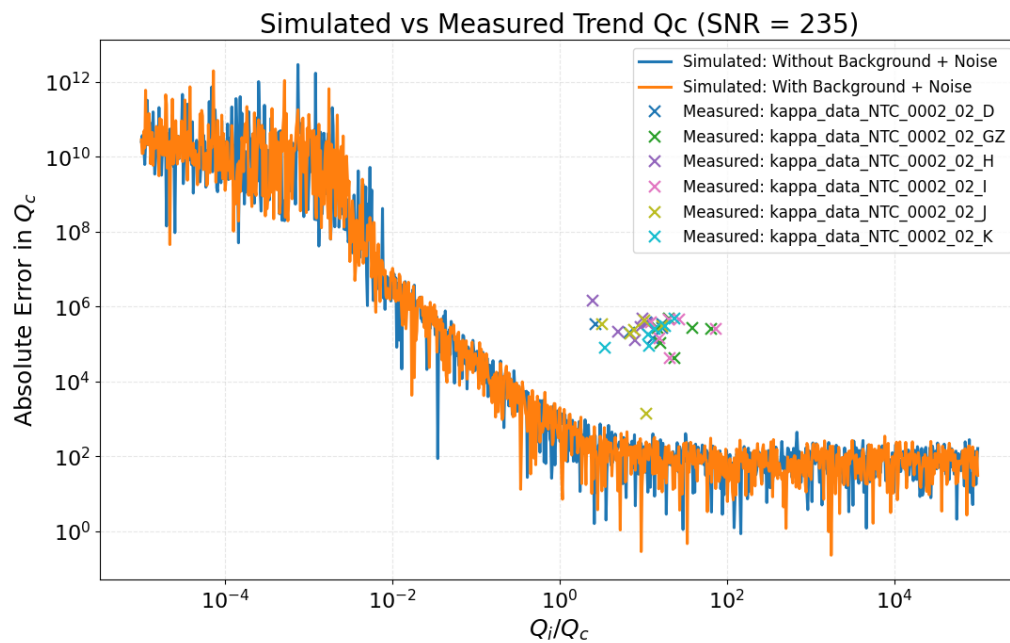


Figure 4.25: Absolute error in the fitted Q_c as a function of Q_i/Q_c , comparing synthetic signals with and without the measured background in the presence of Gaussian noise.

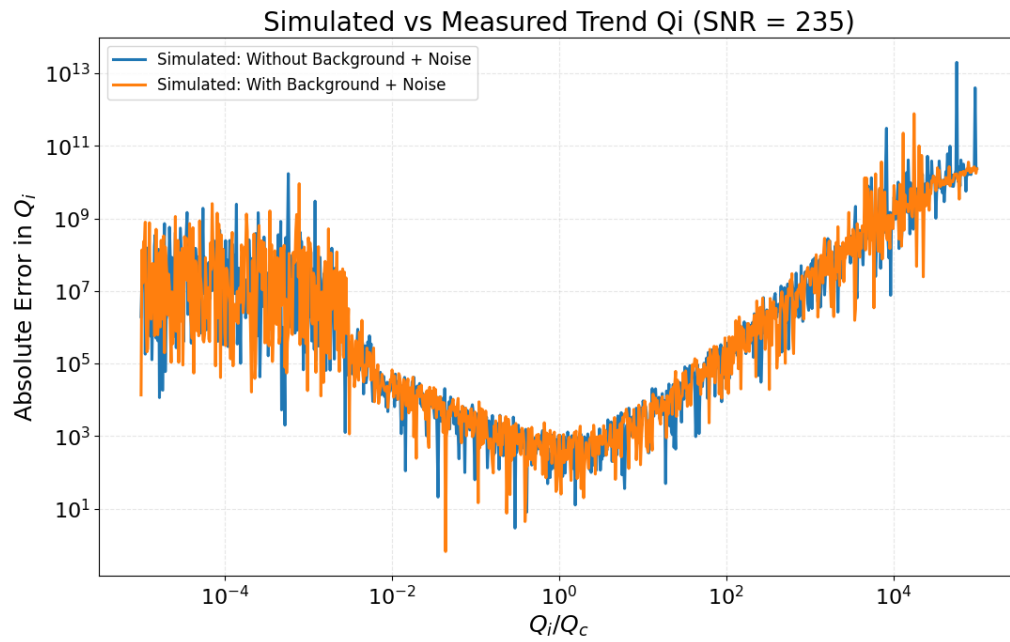


Figure 4.26: Absolute error in the fitted Q_i as a function of Q_i/Q_c , comparing synthetic signals with and without the measured background in the presence of Gaussian noise.

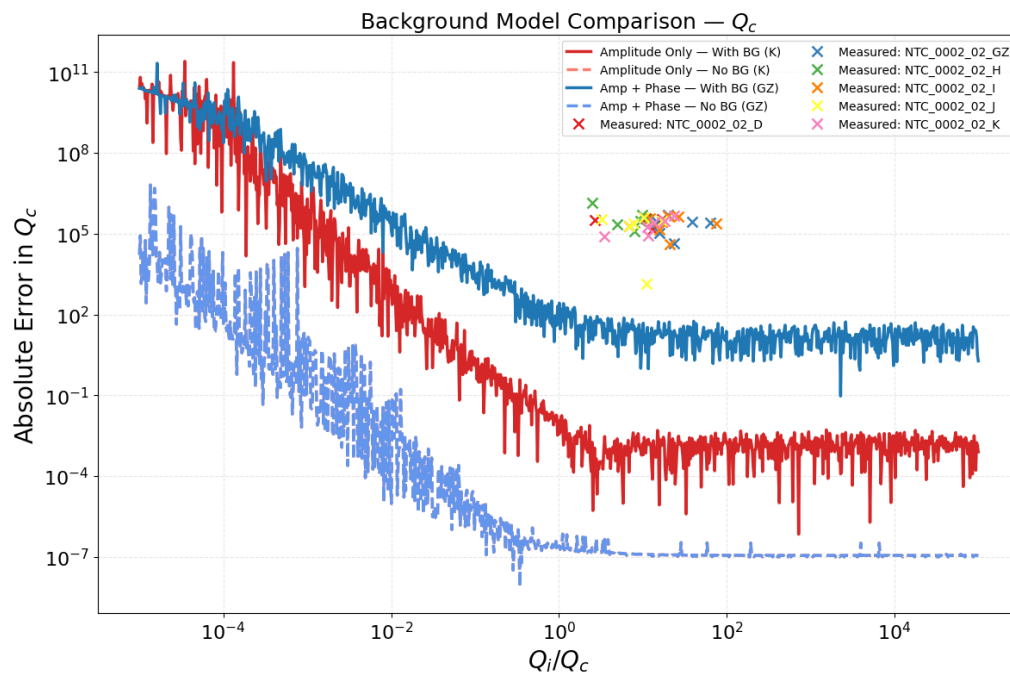


Figure 4.27: Background Development Model Comparison - Q_c

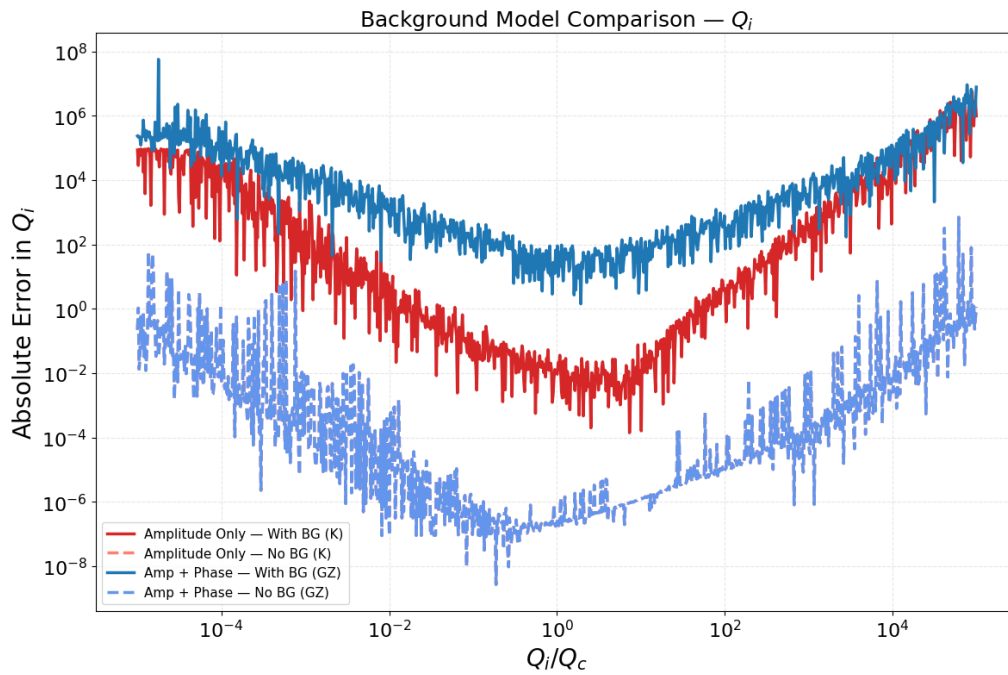


Figure 4.28: Background Development Model Comparison - Q_i

4.8.3.1 Some Observations from the plots

Figure 4.23 shows the absolute error in Q_c in the absence (red) and presence (green) of background. The addition of background increases the error by several orders of magnitude. However, when compared to the discrepancy between SPICE simulations and fitted ABDB measurement data (indicated by crosses), the simulated errors remain insufficient to account for the observed differences. To further investigate, Gaussian noise is introduced into the simulations (Figure 4.25), both in the absence (blue) and presence (orange) of background. Even with the addition of noise, the simulated errors still do not reproduce the level of discrepancy observed between simulations and measurements (crosses marked).

Figure 4.24 shows the absolute error in Q_i in the absence (red) and presence (green) of background. The inclusion of background significantly increases the error across the entire range of Q_i/Q_c , with the largest deviations observed away from the critically coupled regime. In Figure 4.26, Gaussian noise is introduced into the synthetic data, both without background (blue) and with background (orange); the addition of noise further increases the variability and overall error in the extracted Q_i , particularly in the extreme coupling regimes.

An important point to note is the role of fitting the background phase. In Figures 4.27 and 4.28, the background modeled using only amplitude fitting (solid red curves) is compared with the case where both amplitude and phase are fitted (solid blue curves). The comparison reveals a clear difference in the order of magnitude of the absolute errors, highlighting the importance of including the background phase in the modeling process.

4.8.4 Conclusion

The plots above show that background effects can induce large absolute errors, particularly in the undercoupled and overcoupled regimes. The region of primary interest, $Q_i \approx Q_c$, continues to exhibit optimal algorithm performance, with the lowest absolute error and standard deviations consistent with every preceding verification in this chapter. Although the inclusion of background (both amplitude and phase), and the further addition of Gaussian noise, increases the absolute error in the extracted values of Q_i and Q_c , these effects alone are insufficient to explain the discrepancy between simulated and fitted/measured quality factors (Q_c). These findings indicate that more sophisticated or physically complete models are required to fully account for the observed discrepancies between simulations and measurements, precisely the motivation for the first-principles impedance-mismatch modeling developed in Chapter 5.

Chapter 5

SPICE and Physical Modeling

Chapter 4 showed us that the fitting algorithm even after equipping the synthetic data with a realistic frequency-dependent background and noise, the fitting algorithm's error budget remained too small to account for the discrepancy between simulated and measured coupling quality factors. So if the extraction tool is not the major source of errors, the physical model needs further analysis. This chapter therefore moves from the measurement side of the pipeline to the modelling of the physical system itself. Here we model, from first principles, how the environment around a superconducting resonator affects it. This environment includes the feedline the resonator couples to, the wirebonds that connect the chip to its Printed Circuit Board (PCB) and the parasitic resonances of the box the chip sits in.

This chapter first lays out how these environmental elements are modeled and what assumptions are required (Section 5.1). It first describes the wirebond model and shows how impedance mismatch can affect the resonator's loss rates. It then derives the resonator's external and loaded loss rates in the presence of a box mode from first principles, using the quantum Langevin equation and input-output theory realized through the SLH framework (Sections 5.3.2 and 5.3.3). A wide numerical sweep across coupling capacitances and frequencies then tests this prediction (Section 5.3.6). Appendix G closes the loop with an SLH-based fitting routine applied directly to simulated S_{21} data. The experimental discrepancy motivating the analysis was introduced in Figures 1.2 and 1.3 of the thesis introduction.

5.1 Introduction and Circuit Model

This section introduces the range of ways the physical system can be modeled, starting from a simple LC circuit up to the mixed-coupled system that describes the setup most accurately.

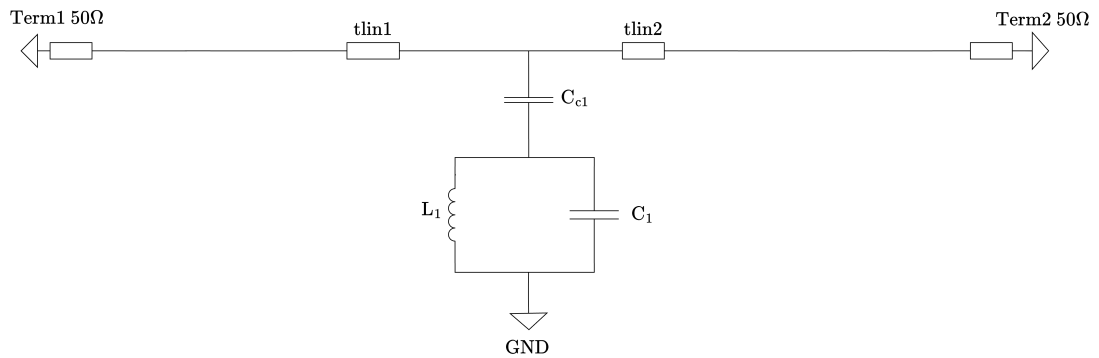
5.1.1 The Circuit Family

The base circuit is the lumped-element hanger established in Chapters 2 and 3: a parallel LC resonator (C_1, L_1) capacitively coupled through a coupling capacitor C_{c1} to a feedline terminated on both sides by $50\ \Omega$ loads. This acts as the base circuit against which all the environmental factors are tested and benchmarked. What changes with every model is what this resonator *sees through* the capacitive coupling and how it

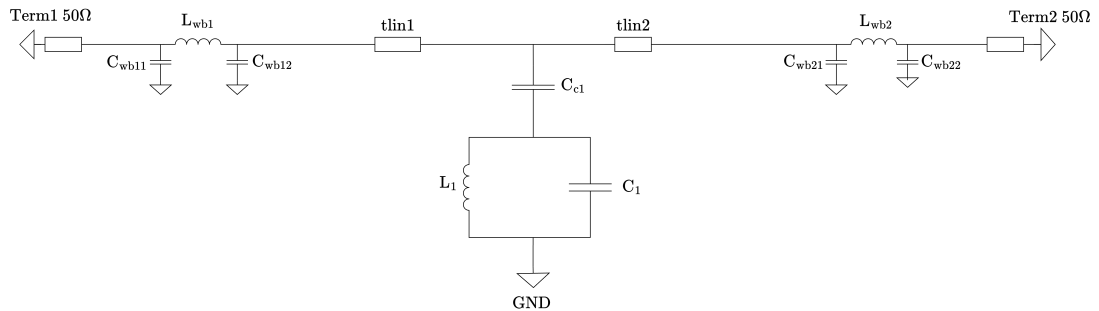
changes the resonator's own external coupling. Below we present the models we tested and worked with.

Table 5.1 shows the KiCAD schematic of each circuit tested, in order of increasing complexity: the bare resonator-plus-feedline circuit, the same circuit with the wirebond's full Π -network in place, and the three box-mode topologies (direct, indirect, mixed) built on top of it.

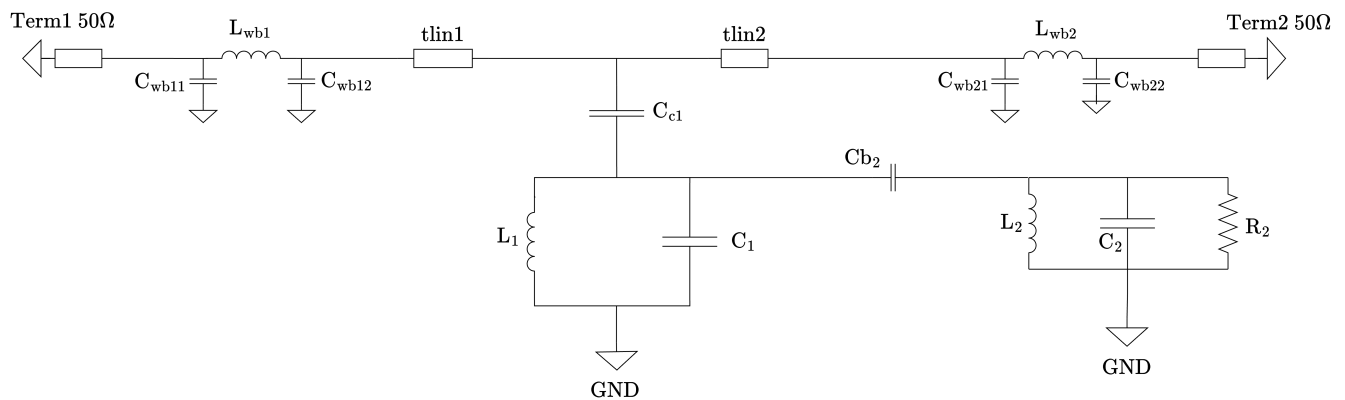
Table 5.1: The circuit used for further analysis of the system, from the bare resonator-feedline circuit through the three box-mode coupling topologies.



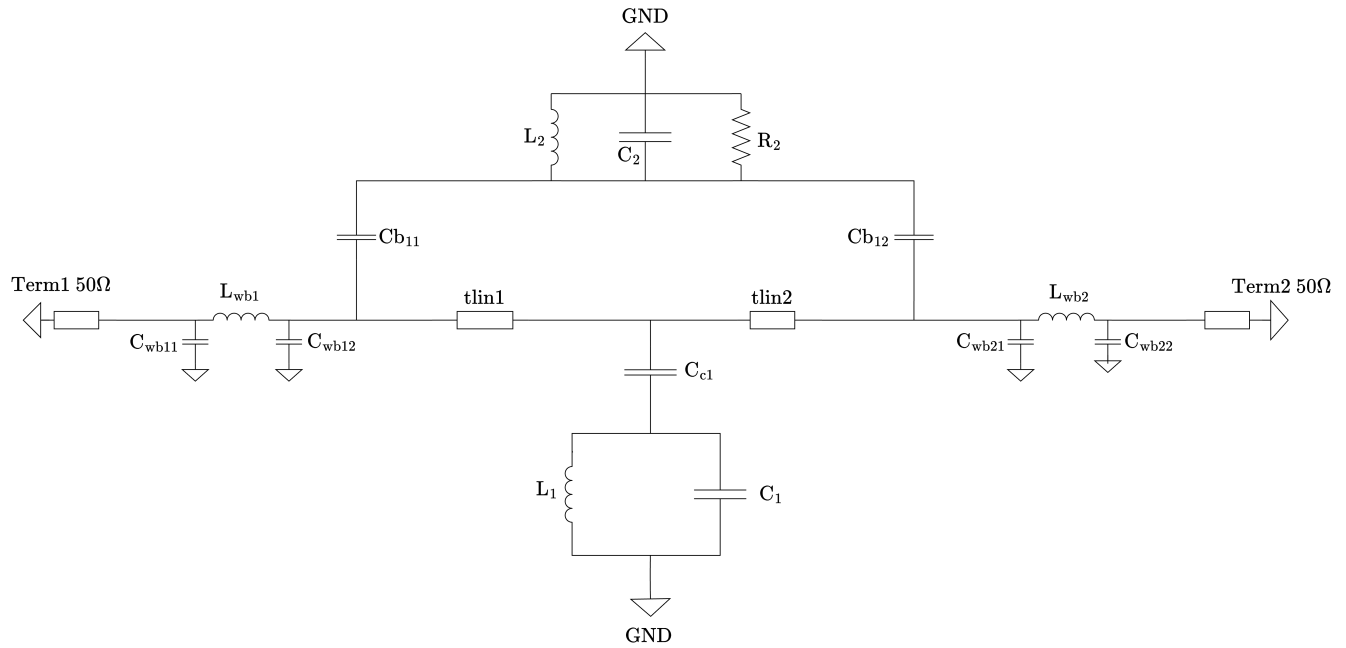
(a) Bare LC + feedline: the resonator coupled through C_{c1} directly into two transmission-line-terminated $50\ \Omega$ loads.



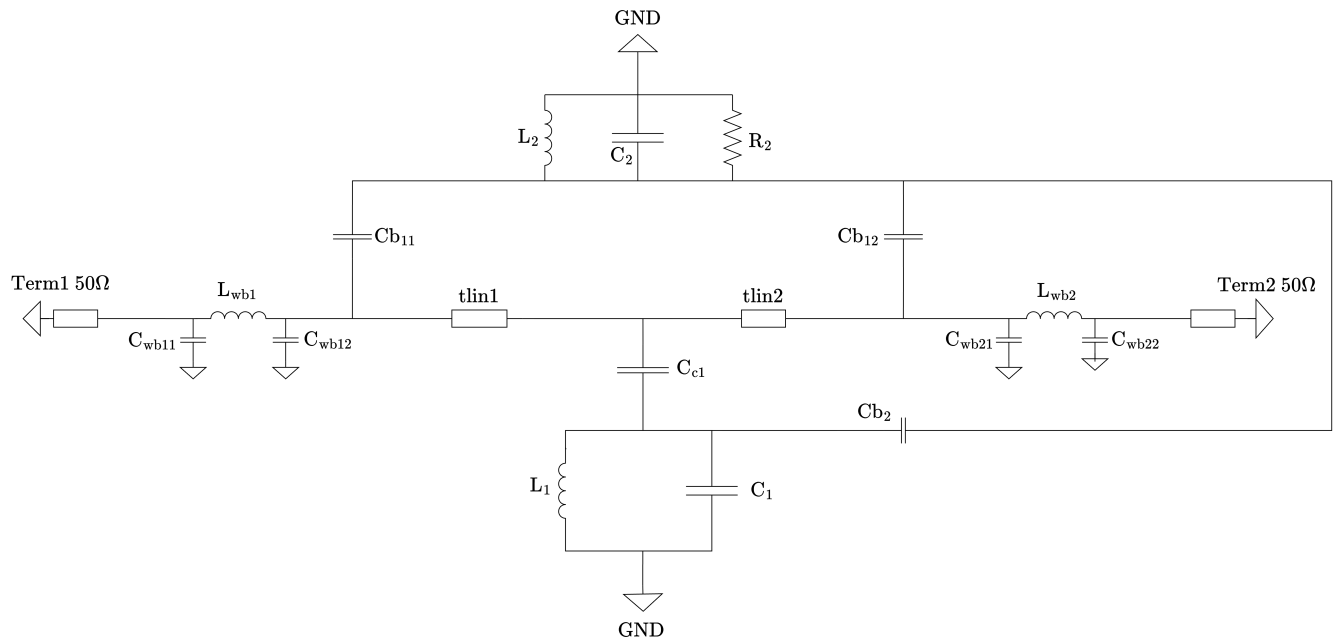
(b) Wirebonded system: each termination arm now carries the full wirebond Π -network of Section 5.2 (L_{wb} flanked by two shunt C_{wb}).



(c) Direct box coupling: the box circuit (C_2 , L_2 , R_2) couples straight into the resonator node through a single capacitor.



(d) Indirect box coupling: the box has no connection to the resonator node at all, only connected via the bare feedline at two points instead.



(e) Mixed box coupling: the box retains both paths at once – a feedline connection and a direct node coupling.

5.2 The Wirebonds: The Π Model

A wirebond is physically a short suspended wire with capacitance to ground at each landing pad, modeled as a symmetric Π -network: a shunt capacitance $Y = j\omega C_{wb}$, a series impedance $Z_s = j\omega L_{wb}$, and a second, identical shunt capacitance Y , as shown below Figure 5.2.

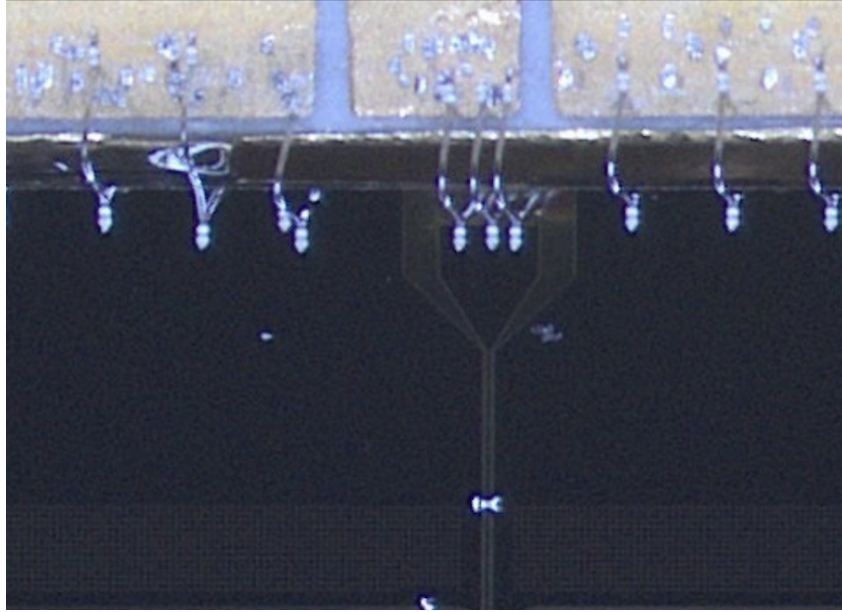


Figure 5.1: A physical wirebond connecting a chip pad to PCB

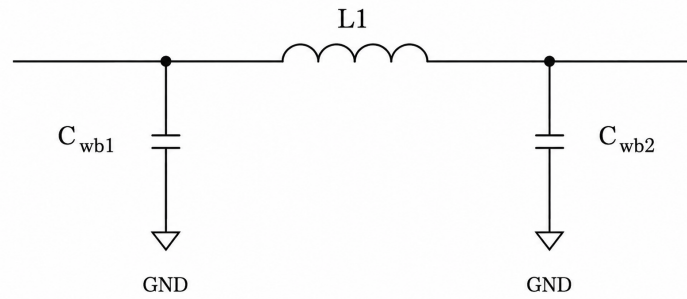


Figure 5.2: Lumped element model for wirebonds

Assigning each element its elementary ABCD block and multiplying in physical order gives its ABCD matrix,

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix} \begin{pmatrix} 1 & Z_s \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix} = \begin{pmatrix} 1 + YZ_s & Z_s \\ 2Y + Y^2Z_s & 1 + YZ_s \end{pmatrix}. \quad (5.1)$$

$A = D$ reflects the network's physical symmetry, and $AD - BC = (1 + YZ_s)^2 - Z_s(2Y + Y^2Z_s) = 1$ confirms reciprocity. Terminating the far port in an arbitrary load gives the exact input impedance

$$Z_{\text{in}} = \frac{A Z_{\text{load}} + B}{C Z_{\text{load}} + D} = \frac{(1 - \omega^2 L_{\text{wb}} C_{\text{wb}}) Z_{\text{load}} + j\omega L_{\text{wb}}}{j\omega C_{\text{wb}} (2 - \omega^2 L_{\text{wb}} C_{\text{wb}}) Z_{\text{load}} + (1 - \omega^2 L_{\text{wb}} C_{\text{wb}})}. \quad (5.2)$$

Whenever $\omega^2 L_{\text{wb}} C_{\text{wb}} \ll 1$ and the shunt reactance $1/\omega C_{\text{wb}}$ dominates the line impedance, Equation (5.2) collapses to the thin-cap limit

$$Z_{\text{in}} \approx Z_{\text{load}} + j\omega L_{\text{wb}}. \quad (5.3)$$

The capacitances drop out: the bond is, to excellent accuracy, a pure series inductor adding a reactance $X = \omega L_{\text{wb}}$ on top of whatever the far side presents. This is the formal justification for the series-inductor wirebond model used throughout this chapter, consistent with the electrical-length criterion of Equation (2.7). Equation (5.3) also supplies the controllable definition of “mismatch strength” used in every sweep below. If the far side is a matched feedline, $Z_{\text{load}} = Z_0$, the matched reference cancels in the mismatch, $|Z_{\text{in}} - Z_0| = |jX| = \omega L_{\text{wb}}$: the mismatch *is* the bond reactance. A target mismatch X at a design frequency f therefore fixes the bond inductance directly,

$$L_{\text{wb}} = \frac{X}{2\pi f}, \quad (5.4)$$

In the next section we perform some simulations to better our understanding of the system.

5.2.1 Simulations

Here we perform simulations to understand, in the presence of impedance mismatch (using L_{wb} as the tuning knob), what happens to κ_c as extracted by three independent routes: the analytical nodal solver developed in this thesis, the SPICE eigenmode solver, and the previously benchmarked S_{21} fitting routine of Chapter 4.

We first investigate the per-arm loss contributions from each termination of the circuit. These simulations are performed on the circuit of Table 5.1(b), under two conditions; with equal transmission-line lengths on both arms (symmetric) and unequal lengths (asymmetric) to show how the per-arm contribution to κ_c depends on feedline length as a function of the wirebond inductance.

5.2.1.1 Parameters

Table 5.2 lists the fixed circuit parameters, used for these sweeps of the wirebond inductance against the external loss rate κ_c .

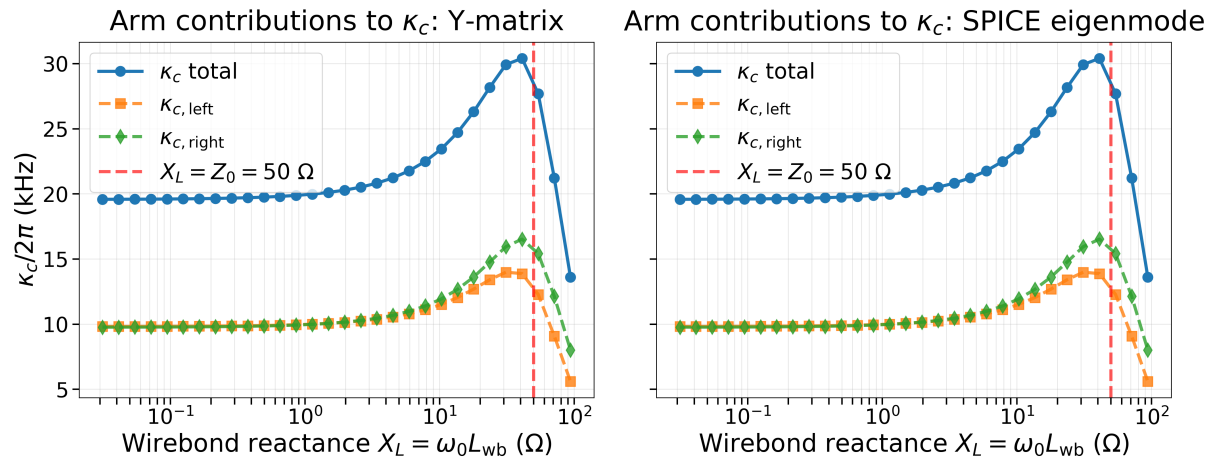
We also perform an error analysis on this model: we compare the nodal solver’s analytical prediction against SPICE, and separately against the S_{21} fitting routine applied to the simulated S_{21} trace, to identify which of the two possibly could dominate the discrepancy.

Table 5.2: Fixed circuit parameters for the wirebond-inductance sweeps of Figures 5.3 - 5.6.

Element	Symbol	Value
Resonator capacitance	C_1	100 fF
Resonator inductance	L_1	10.1 nH
Coupling capacitance	$C_2 \equiv C_c$	0.72 fF
Wirebond shunt capacitances (both arms)	$C_{wb1,1}=C_{wb1,2}=C_{wb2,1}=C_{wb2,2}$	10 fF
Arm-1 wirebond inductance (fixed)	L_2	7.5 pH
Arm-2 wirebond inductance (swept)	$L_{wb} \equiv L_3$	1–300 pH
Feedline characteristic impedance	Z_c (tlin1, tlin2)	50 Ω
Feedline effective permittivity	ϵ_{eff} (tlin1, tlin2)	5.627
tlin1 / tlin2 length (symmetric case)	–	4360 / 4360 μm
tlin1 / tlin2 length (asymmetric case)	–	3060 / 4360 μm

5.2.1.2 Plots

Figures 5.3 - 5.4 show the per-arm κ_c sweeps, and Figures 5.5 - 5.6 the corresponding method-comparison errors.

**Figure 5.3:** Per-arm contribution to κ_c vs. wirebond reactance $X_L = \omega_0 L_{wb}$, asymmetric transmission-line lengths.

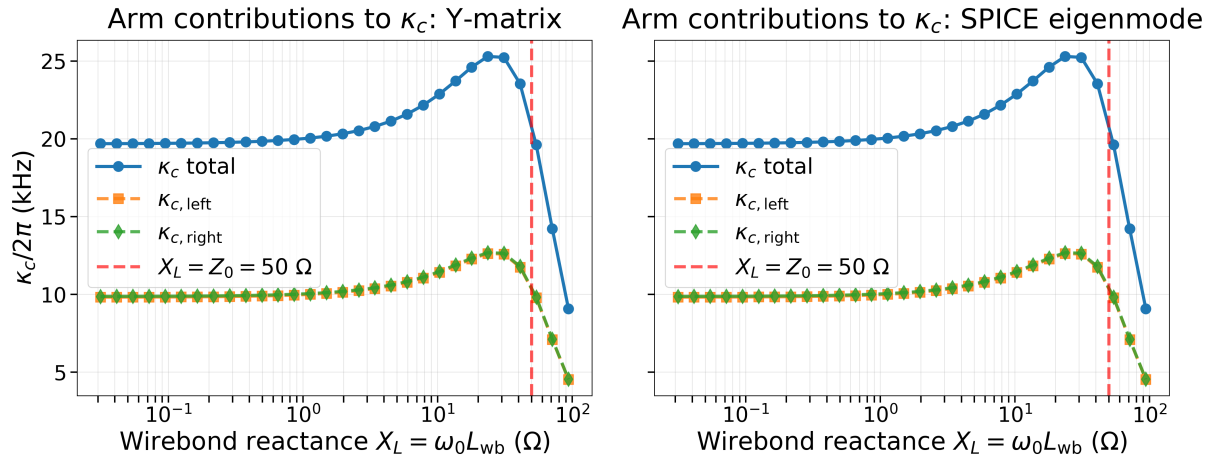


Figure 5.4: Per-arm contribution to κ_c vs. wirebond reactance $X_L = \omega_0 L_{wb}$, symmetric transmission-line lengths.

Method comparison errors

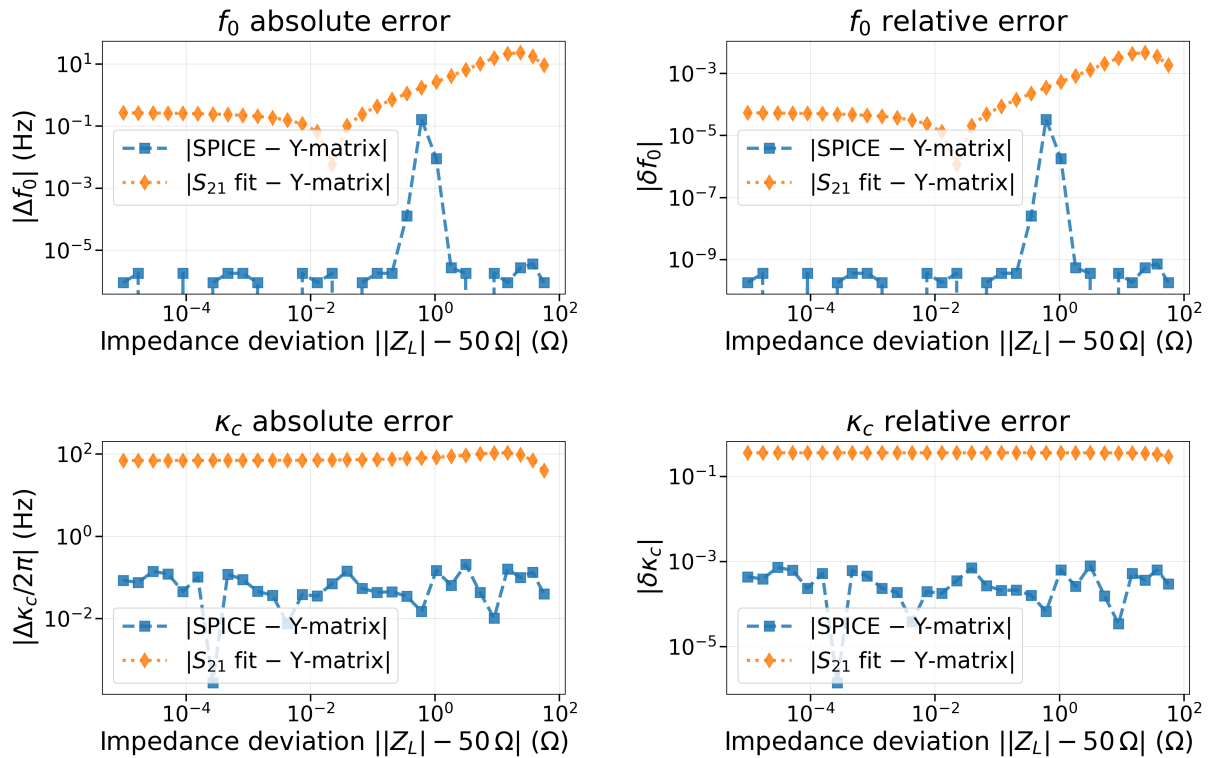


Figure 5.5: Method-comparison errors, asymmetric case: f_0 and κ_c absolute and relative errors.

Method comparison errors

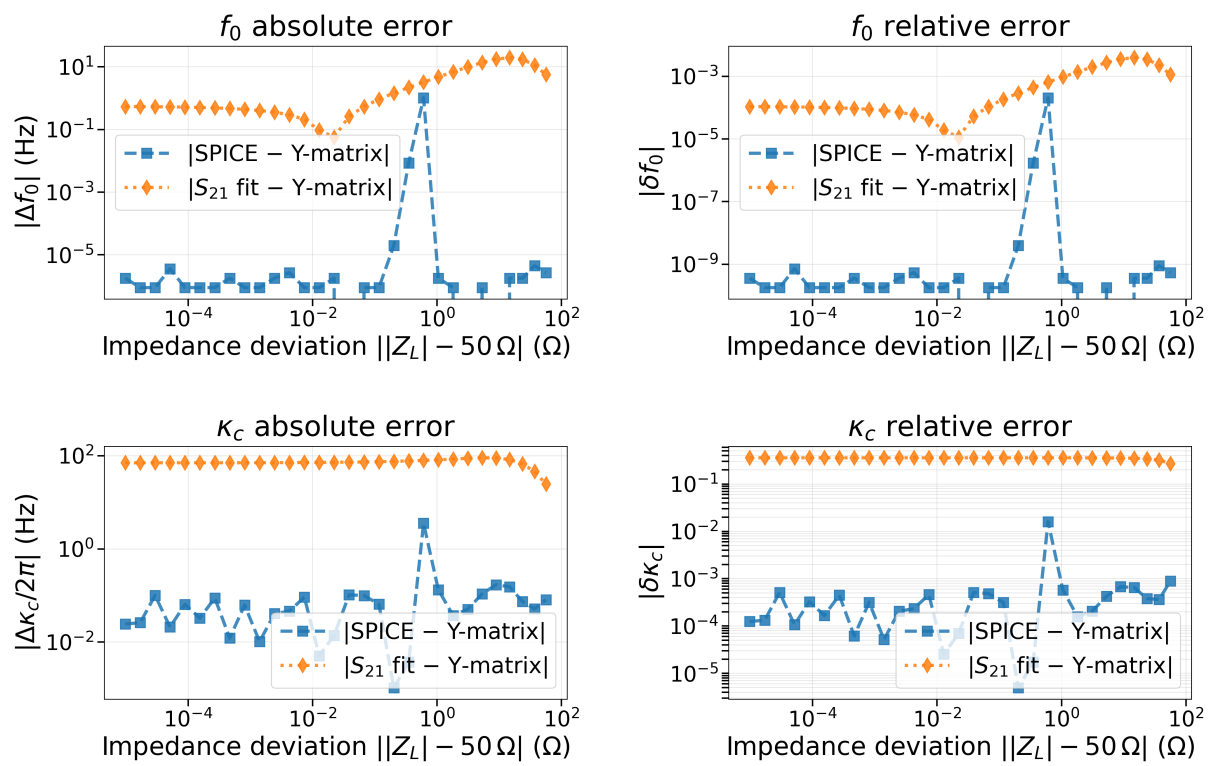


Figure 5.6: Method-comparison errors, symmetric case: f_0 and κ_c absolute and relative errors.

5.2.1.3 Observations

SPICE and the nodal (Y-matrix) solver agree with each other to extremely high precision across the entire sweep in both the symmetric and asymmetric cases (Figures 5.5 and 5.6). The S_{21} fitting routine, by contrast, disagrees with the nodal solver by a much larger margin that stays roughly constant across the whole sweep (same two figures): a relative error in κ_c of order 30–40%, versus the 10^{-3} – 10^{-6} level of SPICE–nodal agreement. The dominant source of error in this comparison is therefore the fitting routine, not disagreement between the two circuit solvers. The asymmetric and symmetric cases (Figures 5.3 and 5.4 respectively) tell the same story quantitatively, differing only in the per-arm split: with equal transmission-line lengths the two arms' contributions to κ_c stay locked together at every X_L , while unequal lengths let them diverge once X_L exceeds a few ohms. In both cases the per-arm κ_c rises from its matched baseline to a peak near $X_L \approx Z_0 = 50\ \Omega$ before falling off sharply beyond it. Furthermore, HFSS electromagnetic simulations of a resonator coupled to a feedline, including wirebonds between the capacitive pad and the chip, yielded a wirebond inductance of approximately 670 pH. This value was therefore adopted in all subsequent simulations.

A wirebond is not, however, the only unmodeled element between a resonator and its measured κ_c . Every chip also sits inside a package that supports electromagnetic resonances of its own; Section 5.3 treats this box mode's effect on κ_c as the second, independently modeled piece of the same unmodeled-environment problem.

5.3 Box Modes

A superconducting chip does not live in free space: its package supports electromagnetic resonances of its own “box modes” which can couple parasitically to the on-chip circuits. A box mode is not a design accident to be tolerated; the package is engineered so that this resonance sits far from the resonator’s own design frequency, and the question this section answers is exactly how much is the box mode coupling to the resonator on the chip. Within the lumped-circuit picture of this chapter, the box mode is one more LC circuit, lossy due to it having its own internal loss channel and its own coupling channels to the shared feedline, so it is modeled as a second parallel RLC circuit sitting alongside the resonator, as shown in Figure 5.7.

Three coupling topologies are studied, *direct* coupling 5.1(c), in which the box couples capacitively to the resonator node; *indirect* coupling 5.1(d), in which it couples to the feedline instead; and *mixed* coupling 5.1(e), which retains both paths. The rest of this section derives, from first principles, how each of these three paths perturbs the resonator’s own κ_c : first analytically, treating the resonator as a mode that is continuously driven by the input tone on the feedline and continuously damped by the photons it leaks back into that same feedline and into its own internal loss channel (Sections 5.3.2–5.3.5), and then numerically, cross-checking the analytic result against the nodal solver. (Section 5.3.6).

5.3.1 The Box Mode: A Parallel RLC

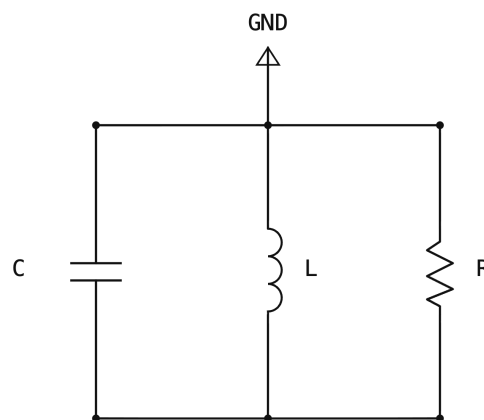


Figure 5.7: Box mode represented as a parallel RLC circuit.

We write $\hat{a}$ for the resonator mode and $\hat{b}$ for the box mode. The box is modelled as a lossy RLC circuit with its own internal losses (via a resistor) and its own coupling capacitor(s) to the shared feedline, so it has an internal rate and one or more external rates. The resonator connects the feedline once, through C_{c1} , giving its external rate

$\kappa_{r,c}$; its internal loss resistor (if any) gives $\kappa_{r,i}$; these are the same κ_c, κ_i used throughout this thesis, subscripted here with an r only because a second mode is now in play. The box mode connects the line at up to two points, through capacitors C_{b11}, C_{b12} (the *indirect* path), giving two external rates $\kappa_{b11}, \kappa_{b12}$; its own internal loss resistor R_2 gives $\kappa_{b,i} = 1/(R_2 C_2)$. A third, separate capacitor C_b (the *direct* path) couples the box mode straight to the resonator node, bypassing the feedline entirely. The *direct*, *indirect*, and *mixed* topologies of this chapter are simply the $C_{b1}=C_{b2}=0, C_b=0$, and general cases of this one circuit.

5.3.2 The Quantum Langevin Equation

Next, we use the idea of the feedline acting as a bath and the LC resonator acting as the system; the interaction between them can then be described using the quantum theory of a damped harmonic oscillator coupled to a continuum of bath modes. Writing $\hat{a}_r$ for a cavity mode of bare frequency ω_r coupled to a bath at rate κ , the *quantum Langevin equation* and its associated input-output relation are [9]

$$\dot{\hat{a}}_r = -i\omega_r \hat{a}_r - \frac{\kappa}{2} \hat{a}_r - \sqrt{\kappa} \hat{a}_{\text{in}}(t), \quad \hat{a}_{\text{out}}(t) = \hat{a}_{\text{in}}(t) + \sqrt{\kappa} \hat{a}_r(t), \quad (5.5)$$

where $\hat{a}_{\text{in}}, \hat{a}_{\text{out}}$ are the freely propagating input and output bath fields, normalized so that $[\hat{a}_{\text{in}}(t), \hat{a}_{\text{in}}^\dagger(t')] = \delta(t - t')$. Equation (5.5) is the single-mode, single-channel special case of the machinery used below; its complete derivation, from the microscopic system-bath Hamiltonian through the Markov approximation to Equation (5.5), is given in Appendix E.

5.3.3 SLH Framework and Reduction of the Resonator-Box Network

The device under consideration now has multiple resonators and feedline components linked by propagating microwave fields, so the interactions between them can no longer be captured by a single system-bath Hamiltonian at a time. Such interconnected systems, known as quantum input-output networks, are naturally described by the SLH framework [9], which is built around exactly the two properties a network description needs: modularity, so that every component keeps its own self-contained description regardless of how large the surrounding network grows, and hierarchy, so that a composite network built from simple parts inherits the same algebraic structure as the parts themselves. The SLH framework—comprising the scattering matrix $\mathbf{S}$, coupling operators $\hat{\mathbf{L}}$, and Hamiltonian $\hat{H}$ —provides a systematic algebraic description of open quantum systems. Each component is represented by the triple

$$G = (\mathbf{S}, \hat{\mathbf{L}}, \hat{H}).$$

By composing these triples according to the network wiring diagram, the framework reproduces the physics obtained from a microscopic system-bath derivation without requiring a full derivation of the entire network. The modularity of this framework makes it very convenient for software integration and computational efficiency.

The remainder of this section converts the system shown in Table 5.1(e) into SLH blocks and derives its external loss rate $\kappa_{r,c}$.

5.3.3.1 The SLH Composition Rules

A component is defined by its triple $G = (S, \hat{\mathbf{L}}, \hat{H})$: a scattering matrix, a vector of coupling operators, and a Hamiltonian. Four rules combine any number of triples into one.

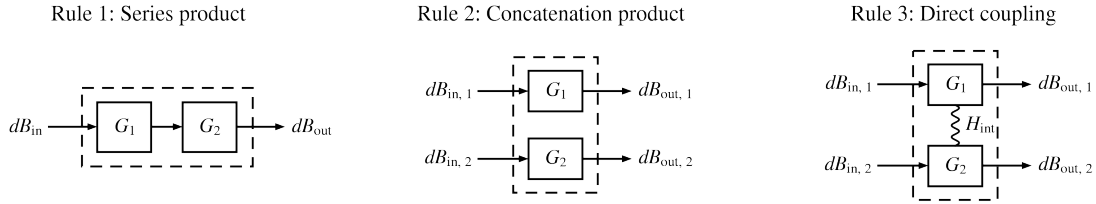


Figure 5.8: The three SLH composition rules used before feedback reduction. The series product connects the output of one component to the input of the next; concatenation places independent components in parallel; and direct coupling adds a Hamiltonian interaction between two otherwise separate components.

Rule 1 (Series product or cascade rule) We begin with the cascading of the output from one localized network, represented by $G_1 = (S_1, \hat{\mathbf{L}}_1, \hat{H}_1)$, into the input of another, represented by $G_2 = (S_2, \hat{\mathbf{L}}_2, \hat{H}_2)$. Both systems must have the same number of input fields, and the i th output field of G_1 is the i th input to G_2 . The result of this connection, the *series product* of G_1 and G_2 , denoted $G_2 \triangleleft G_1$, is given by

$$\begin{aligned}
 G_T &= (S_T, \hat{\mathbf{L}}_T, \hat{H}_T) \\
 &= G_2 \triangleleft G_1 \\
 &= (S_2, \hat{\mathbf{L}}_2, \hat{H}_2) \triangleleft (S_1, \hat{\mathbf{L}}_1, \hat{H}_1) \\
 &= \left(S_2 S_1, \hat{\mathbf{L}}_2 + S_2 \hat{\mathbf{L}}_1, \hat{H}_1 + \hat{H}_2 + \frac{1}{2i} (\hat{\mathbf{L}}_2^\dagger S_2 \hat{\mathbf{L}}_1 - \hat{\mathbf{L}}_1^\dagger S_2^\dagger \hat{\mathbf{L}}_2) \right). \quad (5.6)
 \end{aligned}$$

The effective Hamiltonian of the combined system picks up a dependence on both components' coupling operators, and this term is what makes $G_2 \triangleleft G_1 \neq G_1 \triangleleft G_2$: the fields linking the two components are directional (Figure 5.8).

Rule 2 (Concatenation product). Now consider the parallel grouping of two components with independent input-output fields, $G_1 = (S_1, \hat{\mathbf{L}}_1, \hat{H}_1)$ and $G_2 = (S_2, \hat{\mathbf{L}}_2, \hat{H}_2)$, with no ports wired together. The result of this parallel grouping, the *concatenation*

product of G_1 and G_2 , denoted $G_1 \boxplus G_2$ (Figure 5.8), is given by

$$\begin{aligned}
 G_T &= (S_T, \hat{\mathbf{L}}_T, \hat{H}_T) \\
 &= G_1 \boxplus G_2 \\
 &= (S_1, \hat{\mathbf{L}}_1, \hat{H}_1) \boxplus (S_2, \hat{\mathbf{L}}_2, \hat{H}_2) \\
 &= \left(\begin{pmatrix} S_1 & 0 \\ 0 & S_2 \end{pmatrix}, \begin{pmatrix} \hat{\mathbf{L}}_1 \\ \hat{\mathbf{L}}_2 \end{pmatrix}, \hat{H}_1 + \hat{H}_2 \right). \tag{5.7}
 \end{aligned}$$

Rule 3 (Direct coupling). The next composition rule is *direct coupling*, a generalization of the concatenation product. Consider G_1 and G_2 again in parallel, but now with a direct Hamiltonian interaction between the two systems, $\hat{H}_{\text{int}}$, an operator on $\mathcal{H}_1 \otimes \mathcal{H}_2$, the tensor product of the two systems' Hilbert spaces this is exactly how the box mode's direct coupling g_d to the resonator (bypassing the feedline entirely) enters below (Figure 5.8):

$$\begin{aligned}
 G_T &= (S_T, \hat{\mathbf{L}}_T, \hat{H}_T) \\
 &= G_1 \boxplus G_2 \\
 &= (S_1, \hat{\mathbf{L}}_1, \hat{H}_1 + \hat{H}_{\text{int}}) \boxplus (S_2, \hat{\mathbf{L}}_2, \hat{H}_2) \\
 &= \left(\begin{pmatrix} S_1 & 0 \\ 0 & S_2 \end{pmatrix}, \begin{pmatrix} \hat{\mathbf{L}}_1 \\ \hat{\mathbf{L}}_2 \end{pmatrix}, \hat{H}_1 + \hat{H}_2 + \hat{H}_{\text{int}} \right) \tag{5.8}
 \end{aligned}$$

Rule 4: Feedback Reduction

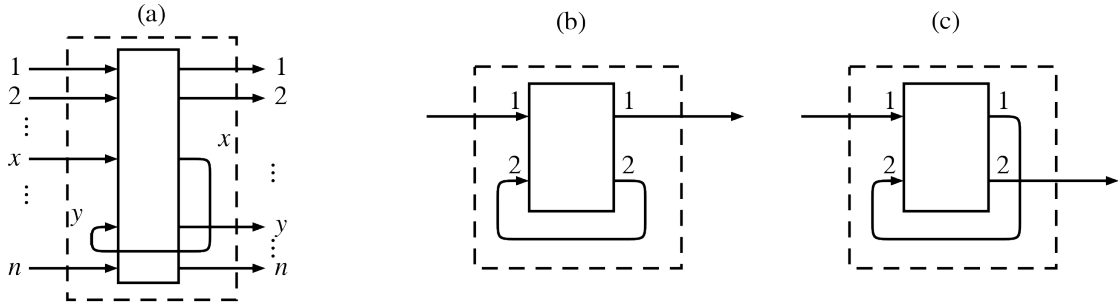


Figure 5.9: SLH feedback reduction. An output port of a component is connected to one of its input ports to form an internal loop; eliminating that internal connection produces a reduced triple containing only the external input and output ports.

Rule 4 (Feedback reduction). Wiring a component's own output port y back into one of its own input ports x (an internal loop, as happens whenever a feedline segment connects two other components in a closed network) requires one further rule, the feedback reduction (Figure 5.9). Partitioning $G = (S, \hat{\mathbf{L}}, \hat{H})$'s port indices into $\{x, y\}$ and their complements $\{\bar{x}, \bar{y}\}$,

$$S_{\text{red}} = S_{\bar{x}\bar{y}} + S_{\bar{x}y}(\mathbf{1} - S_{xy})^{-1}S_{x\bar{y}}, \quad \hat{\mathbf{L}}_{\text{red}} = \hat{\mathbf{L}}_{\bar{x}} + S_{\bar{x}y}(\mathbf{1} - S_{xy})^{-1}\hat{\mathbf{L}}_x, \tag{5.9}$$

$$\hat{H}_{\text{red}} = \hat{H} + \frac{1}{2i} \left(\hat{\mathbf{L}}^{\dagger} S_{:,y} (\mathbf{1} - S_{xy})^{-1} \hat{\mathbf{L}}_x - \text{h.c.} \right). \tag{5.10}$$

We shall explicitly use these rules to describe the system as presented in 5.1(e)

5.3.3.2 Mixed Coupling of SLH Triples

Before specializing to the box-resonator network of interest, it is useful to see explicitly what a directly-coupled pair of SLH systems looks like for two generic modes. Consider two resonator systems, $G_1 = (S_1, \hat{\mathbf{L}}_1, \hat{H}_1 + \hat{H}_{\text{int}})$ and $G_2 = (S_2, \hat{\mathbf{L}}_2, \hat{H}_2)$, each with two output ports, with

$$\begin{aligned} S_k &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \\ \hat{\mathbf{L}}_k &= \begin{pmatrix} \sqrt{\kappa_{k1}} \hat{a}_k \\ \sqrt{\kappa_{k2}} \hat{a}_k \end{pmatrix}, \\ \hat{H}_k/\hbar &= \Delta_k \hat{a}_k^\dagger \hat{a}_k, \\ \hat{H}_{\text{int}}/\hbar &= g_{12} (\hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1), \end{aligned} \quad (5.11)$$

for $k = 1, 2$. Each mode couples out through two independent channels (e.g. two capacitive couplings) at rates κ_{k1}, κ_{k2} , sits at detuning Δ_k from the drive frequency, and the two modes exchange excitations directly at rate g_{12} . This is exactly the structure of Rule 3 (direct coupling) above, and the template instantiated below for G_{box} and G_{res} , with the direct coupling g_d playing the role of g_{12} .

Figure 5.10 shows how the physical circuit maps onto four elementary SLH systems, concatenated in physical layout order box–tl1–resonator–tl2. We consider the box mode $G_{\text{box}} = (S_{\text{box}}, \hat{\mathbf{L}}_{\text{box}}, \hat{H}_{\text{box}})$, the resonator $G_{\text{res}} = (S_{\text{res}}, \hat{\mathbf{L}}_{\text{res}}, \hat{H}_{\text{res}})$, and the two feedline segments $G(\beta\ell_1), G(\beta\ell_2)$, with

$$G_{\text{box}} = \left(\mathbf{1}_4, \begin{pmatrix} \sqrt{\kappa_{b2}/2} \hat{b} \\ \sqrt{\kappa_{b1}/2} \hat{b} \\ \sqrt{\kappa_{b1}/2} \hat{b} \\ \sqrt{\kappa_{b2}/2} \hat{b} \end{pmatrix}, \Delta_b \hat{b}^\dagger \hat{b} + g_d (\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}) \right), \quad (5.12)$$

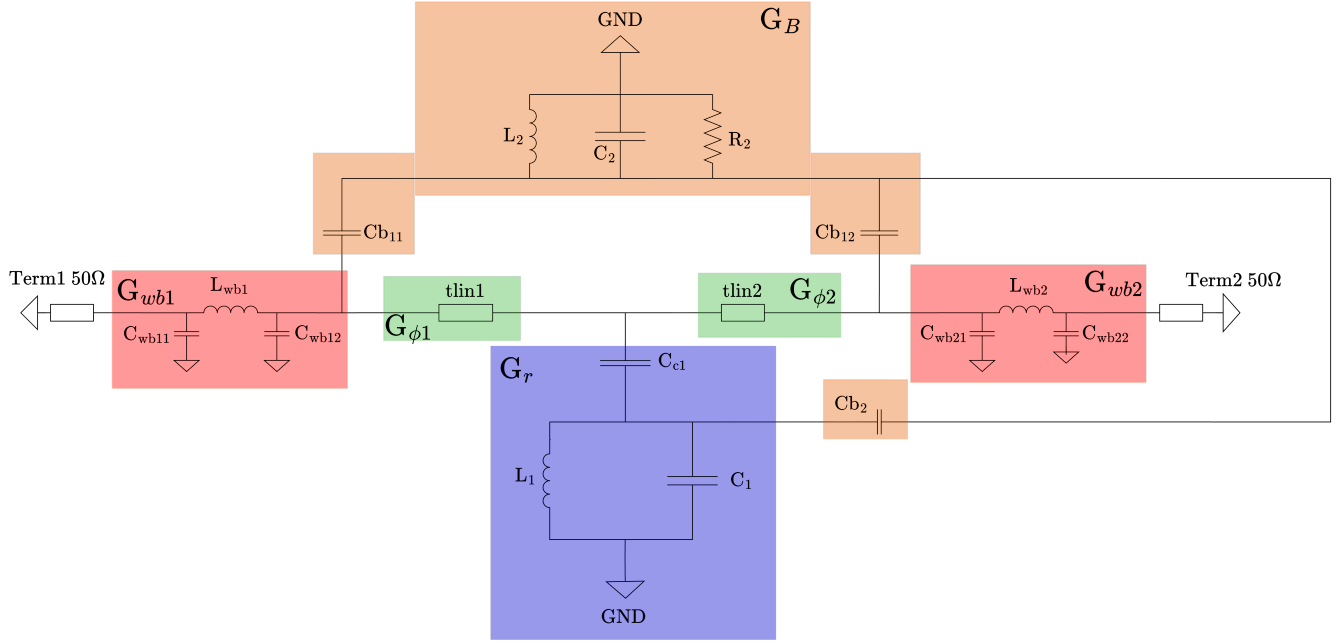
$$G(\beta\ell_j) = \left(\begin{pmatrix} e^{i\beta\ell_j} & 0 \\ 0 & e^{i\beta\ell_j} \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix}, 0 \right), \quad j = 1, 2, \quad (5.13)$$

$$G_{\text{res}} = \left(\mathbf{1}_2, \begin{pmatrix} \sqrt{\kappa_{r,c}/2} \hat{a} \\ \sqrt{\kappa_{r,c}/2} \hat{a} \end{pmatrix}, \Delta_r \hat{a}^\dagger \hat{a} \right). \quad (5.14)$$

G_{box} 's four ports are two copies each of the two physical channels C_{b1} (ports 2, 3) and C_{b2} (ports 1, 4), one copy for each propagation direction along the line. This is the preliminary system we start with; next, since there are internal connections, we need to rewire it to get the reduced form of the triple describing the system.

5.3.3.3 Concatenation and Feedback Reduction

Concatenating Equations (5.12)–(5.14) via Equation (5.7), in the physical layout order of Figure 5.10(b), numbers the ten ports box 1–4, tl1 5–6, resonator 7–8, tl2 9–10, and



The Flow of the signal in the chain can be shown as follows

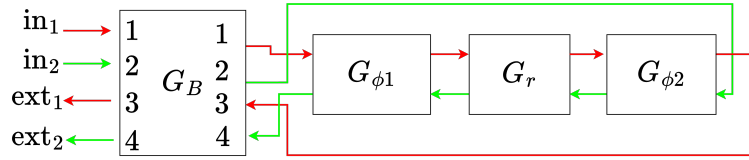


Figure 5.10: Mapping of the mixed-coupling circuit to its elementary SLH triples. The box mode and resonator are represented by G_{box} and G_{res} , while the two intervening feedline sections are phase-shift components $G(\beta\ell_1)$ and $G(\beta\ell_2)$. These four components are concatenated in their physical order before the internal ports are connected by feedback reduction.

gives one ten-port triple $G_T = (S_T, \hat{\mathbf{L}}_T, \hat{H}_T)$,

$$S_T = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & e^{i\beta\ell_1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & e^{i\beta\ell_1} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & e^{i\beta\ell_2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & e^{i\beta\ell_2} \end{pmatrix}, \quad (5.15)$$

$$\hat{\mathbf{L}}_T = \begin{pmatrix} \sqrt{\kappa_{b2}/2} \hat{b} \\ \sqrt{\kappa_{b1}/2} \hat{b} \\ \sqrt{\kappa_{b1}/2} \hat{b} \\ \sqrt{\kappa_{b2}/2} \hat{b} \\ 0 \\ 0 \\ \sqrt{\kappa_{r,c}/2} \hat{a} \\ \sqrt{\kappa_{r,c}/2} \hat{a} \\ 0 \\ 0 \end{pmatrix}, \quad \hat{H}_T = \Delta_r \hat{a}^\dagger \hat{a} + \Delta_b \hat{b}^\dagger \hat{b} + g_d(\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}); \quad (5.16)$$

concatenation alone carries no connectivity, S_T is block diagonal and $\hat{H}_T$ a bare sum, exactly Equation (5.7) applied three times. The physical layout enters only once the ten ports are wired together.

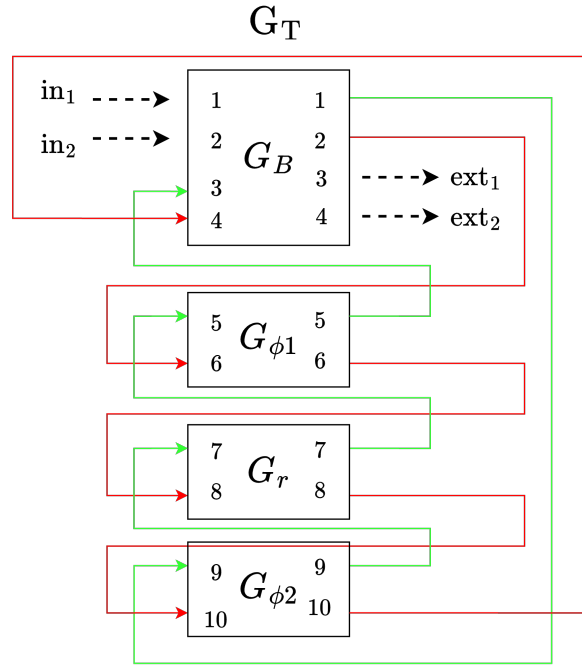


Figure 5.11: The four elementary triples of Equations (5.12)–(5.14), concatenated side by side and wired in physical layout order. Following each mover through the network gives two counter-propagating four-hop chains: the left-movers, $1 \rightarrow 9 \rightarrow 7 \rightarrow 5 \rightarrow 3$, and the right-movers, $2 \rightarrow 6 \rightarrow 8 \rightarrow 10 \rightarrow 4$.

These eight directed hops are not yet in a form the feedback rule, Equations (5.9)–(5.10), can close directly; we need a permutation matrix to rewire the connections, relabeling ports so that every wire becomes a same-index self-loop, out-slot k into in-slot k for

$k = 1, \dots, 8$, with the two physical external ports left exposed on slots 9, 10:

$$P_{\text{in}} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (5.17)$$

$$P_{\text{out}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (5.18)$$

Because a permutation matrix is a pure scatterer, applying Rule 1 twice, $G'_T = (P_{\text{out}}, 0, 0) \triangleleft G_T \triangleleft (P_{\text{in}}, 0, 0)$, giving the permuted triple $G'_T = (S'_T, \hat{\mathbf{L}}'_T, \hat{H}'_T)$ with $\hat{H}'_T = \hat{H}_T$ and

$$S'_T = P_{\text{out}} S_T P_{\text{in}} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ e^{i\beta\ell_2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & e^{i\beta\ell_1} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & e^{i\beta\ell_1} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & e^{i\beta\ell_2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}, \quad (5.19)$$

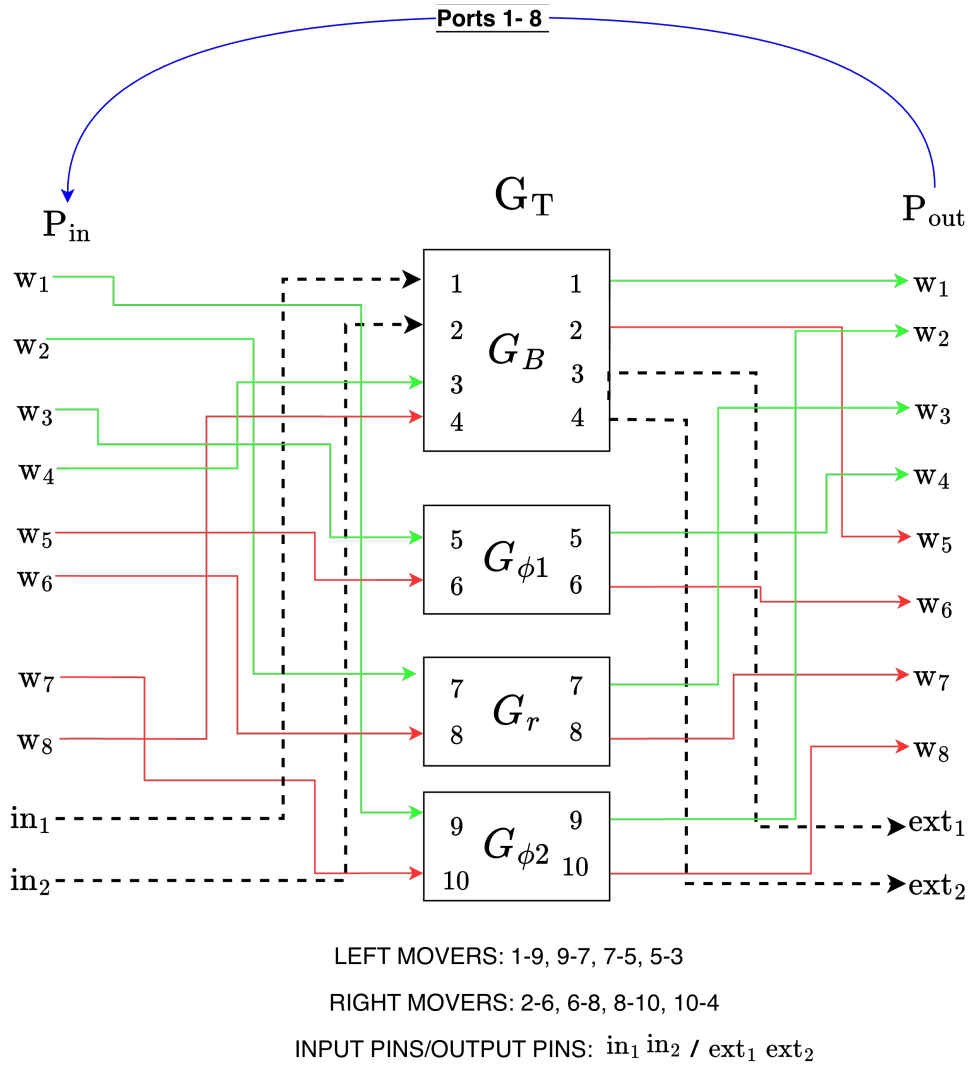


Figure 5.12: The P_{in}/P_{out} relabeling. Left: the eight internal wires of Figure 5.11 in their original order. Right: after relabeling with P_{in} and P_{out} , each wire's output (slot k) feeds directly into its own input (slot k), for $k = 1, \dots, 8$, turning every connection into a simple self-loop. Only the two external ports, on slots 9 and 10, remain exposed.

$$\hat{\mathbf{L}}'_T = P_{\text{out}} \hat{\mathbf{L}}_T = \begin{pmatrix} \sqrt{\kappa_{b2}/2} \hat{b} \\ 0 \\ \sqrt{\kappa_{r,c}/2} \hat{a} \\ 0 \\ \sqrt{\kappa_{b1}/2} \hat{b} \\ 0 \\ \sqrt{\kappa_{r,c}/2} \hat{a} \\ 0 \\ \sqrt{\kappa_{b1}/2} \hat{b} \\ \sqrt{\kappa_{b2}/2} \hat{b} \end{pmatrix}. \quad (5.20)$$

Further restricting to the eight internal slots ($x = y = \{1, \dots, 8\}$) and the two external ones ($\bar{x} = \bar{y} = \{9, 10\}$) in the notation of Equations (5.9)–(5.10) reads off directly as the four blocks and two coupling vectors the feedback rule needs:

$$S_{xy} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ e^{i\beta\ell_2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & e^{i\beta\ell_1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & e^{i\beta\ell_1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & e^{i\beta\ell_2} & 0 \end{pmatrix}, \quad \hat{\mathbf{L}}_x = \begin{pmatrix} \sqrt{\kappa_{b2}/2} \hat{b} \\ 0 \\ \sqrt{\kappa_{r,c}/2} \hat{a} \\ 0 \\ \sqrt{\kappa_{b1}/2} \hat{b} \\ 0 \\ \sqrt{\kappa_{r,c}/2} \hat{a} \\ 0 \end{pmatrix}, \quad (5.21)$$

$$S_{x\bar{y}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad S_{\bar{x}y} = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad S_{\bar{x}\bar{y}} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad \hat{\mathbf{L}}_{\bar{x}} = \begin{pmatrix} \sqrt{\kappa_{b1}/2} \hat{b} \\ \sqrt{\kappa_{b2}/2} \hat{b} \end{pmatrix}. \quad (5.22)$$

To compute the reduced matrices we need $(\mathbf{1} - S_{xy})^{-1}$, which when computed gives

$$(\mathbf{1} - S_{xy})^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ e^{i\beta\ell_2} & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ e^{i\beta\ell_2} & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ e^{i\Theta} & e^{i\beta\ell_1} & e^{i\beta\ell_1} & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & e^{i\beta\ell_1} & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & e^{i\beta\ell_1} & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & e^{i\Theta} & e^{i\beta\ell_2} & e^{i\beta\ell_2} & 1 \end{pmatrix}, \quad \Theta \equiv \beta\ell_1 + \beta\ell_2 \quad (5.23)$$

Assembling Equation (5.9) from these blocks, row 1 of $S_{\bar{x}y}$ selects row 4 of the inverse, whose (4, 1) entry is $e^{i\Theta}$, meeting column 1 of $S_{x\bar{y}}$ (itself 1) to give $(S_{\text{red}})_{11} = e^{i\Theta}$; row 2

picks up $(8, 5) = e^{i\Theta}$ against column 2 of $S_{x\bar{y}}$ identically, giving us

$$S_{\text{red}} = S_{\bar{x}\bar{y}} + S_{\bar{x}y}(\mathbf{1} - S_{xy})^{-1}S_{x\bar{y}} = e^{i\Theta} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (5.24)$$

The same product $(\mathbf{1} - S_{xy})^{-1}\hat{\mathbf{L}}_x$, evaluated on rows 4 and 8 (the only rows $S_{\bar{x}y}$ selects), gives us

$$\hat{\mathbf{L}}_{\text{red}} = \hat{\mathbf{L}}_{\bar{x}} + S_{\bar{x}y}(\mathbf{1} - S_{xy})^{-1}\hat{\mathbf{L}}_x = \begin{pmatrix} e^{i\beta\ell_1}\sqrt{\kappa_{r,c}/2}\hat{a} + (\sqrt{\kappa_{b1}/2} + e^{i\Theta}\sqrt{\kappa_{b2}/2})\hat{b} \\ e^{i\beta\ell_2}\sqrt{\kappa_{r,c}/2}\hat{a} + (\sqrt{\kappa_{b2}/2} + e^{i\Theta}\sqrt{\kappa_{b1}/2})\hat{b} \end{pmatrix} \quad (5.25)$$

Writing $T_1 \equiv \hat{\mathbf{L}}_{\text{red}}^\dagger \hat{\mathbf{w}}$ for the row-column product Equation (5.10) generates ($\hat{\mathbf{w}}$ being the field the propagator of Equation (5.23) delivers to each internal slot), the Hamiltonian correction reads

$$\hat{H}_{\text{red}} = \hat{H}'_T + \frac{1}{2i} \left(\hat{\mathbf{L}}_T^\dagger S_{:,y}(\mathbf{1} - S_{xy})^{-1}\hat{\mathbf{L}}_x - \text{h.c.} \right) = \underbrace{\hat{H}_1 + \hat{H}_2 + \hat{H}_{\text{int}}}_{=\hat{H}'_T=\hat{H}_T} + \frac{1}{2i}(T_1 - T_1^\dagger), \quad (5.26)$$

with $\hat{H}_1 \equiv \Delta_r \hat{a}^\dagger \hat{a}$, $\hat{H}_2 \equiv \Delta_b \hat{b}^\dagger \hat{b}$, $\hat{H}_{\text{int}} \equiv g_d(\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a})$ the three terms already carried, unchanged, in $\hat{H}_T$ – only the last term, $\frac{1}{2i}(T_1 - T_1^\dagger)$, is emergent. Splitting T_1 into its Hermitian and anti-Hermitian parts gives exactly the coherent piece that belongs in $\hat{H}_{\text{red}}$:

$$g_\ell = \frac{\sqrt{\kappa_{r,c}}}{2} \left(\sqrt{\kappa_{b1}} \sin \beta \ell_1 + \sqrt{\kappa_{b2}} \sin \beta \ell_2 \right), \quad \delta_b = \sqrt{\kappa_{b1}\kappa_{b2}} \sin \Theta. \quad (5.27)$$

T_1 's anti-Hermitian remainder is a purely *dissipative* pairing between $\hat{a}$ and $\hat{b}$ – it does not belong in a Hamiltonian at all, and does not vanish; it resurfaces as Γ_{ab} once $\Phi^\dagger \Phi$ is built below (Equation (5.35)). Adding the hand-inserted g_d to the emergent g_ℓ gives $G \equiv g_d + g_\ell$ and the reduced Hamiltonian

$$\hat{H}_{\text{red}} = \Delta_r \hat{a}^\dagger \hat{a} + (\Delta_b + \delta_b) \hat{b}^\dagger \hat{b} + G(\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}). \quad (5.28)$$

Collecting Equations (5.24), (5.25), and (5.28), the box-resonator network reduces to the single two-port triple $G_{\text{red}} = (S_{\text{red}}, \hat{\mathbf{L}}_{\text{red}}, \hat{H}_{\text{red}})$:

$$\boxed{S_{\text{red}} = e^{i\Theta} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \hat{\mathbf{L}}_{\text{red}} = \begin{pmatrix} e^{i\beta\ell_1}\sqrt{\kappa_{r,c}/2}\hat{a} + (\sqrt{\kappa_{b1}/2} + e^{i\Theta}\sqrt{\kappa_{b2}/2})\hat{b} \\ e^{i\beta\ell_2}\sqrt{\kappa_{r,c}/2}\hat{a} + (\sqrt{\kappa_{b2}/2} + e^{i\Theta}\sqrt{\kappa_{b1}/2})\hat{b} \end{pmatrix},} \quad (5.29)$$

$$\boxed{\hat{H}_{\text{red}} = \Delta_r \hat{a}^\dagger \hat{a} + (\Delta_b + \delta_b) \hat{b}^\dagger \hat{b} + G(\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}).} \quad (5.30)$$

Every entry above traces back to a single physical cause: S_{red} is pure transmission with the network's total electrical length Θ ; $\hat{\mathbf{L}}_{\text{red}}$'s two rows are each a sum of three propagation paths to that exit; and $\hat{H}_{\text{red}}$ carries the two detunings plus the coherent coupling $G = g_d + g_\ell$, with δ_b shifting the box's own frequency. This triple is what feeds the multi-mode Langevin equation next.

5.3.3.4 The Multi-Mode Langevin Equation

Generalizing Equation (5.5) to a vector of modes coupled to several channels at once, any SLH triple $(S, \hat{\mathbf{L}}, \hat{H})$ satisfying (1) every entry of S a plain number, (2) $\hat{\mathbf{L}} = \Phi \hat{\mathbf{a}}$ linear in the modes, and (3) $\hat{H} = \hat{\mathbf{a}}^\dagger \Omega \hat{\mathbf{a}}$ quadratic and photon-number conserving, promotes to a Langevin equation and input–output relation by the same four-step recipe [9]:

1. Obtain the reduced triple $(S_{\text{red}}, \hat{\mathbf{L}}_{\text{red}}, \hat{H}_{\text{red}})$.
2. Write $\hat{\mathbf{L}}_{\text{red}} = \Phi \hat{\mathbf{a}}$; read off Φ .
3. Write $\hat{H}_{\text{red}} = \hat{\mathbf{a}}^\dagger \Omega \hat{\mathbf{a}}$; read off Ω .
4. Compute $A = -i\Omega - \frac{1}{2}\Phi^\dagger \Phi$, $B = -\Phi^\dagger S$ (with $C = \Phi$, $D = S$ directly), giving

$$\dot{\hat{\mathbf{a}}} = A\hat{\mathbf{a}} + B\hat{\mathbf{b}}_{\text{in}}, \quad \hat{\mathbf{b}}_{\text{out}} = \Phi\hat{\mathbf{a}} + S\hat{\mathbf{b}}_{\text{in}}. \quad (5.31)$$

Following this recipe on the certified triple of Equations (5.29)–(5.30) (channel order: 1, 2 = feedline; 3 = resonator bath; 4 = box bath; mode order $\hat{a}, \hat{b}$):

Step 2 – matching each row of $\hat{\mathbf{L}}_{\text{red}}$ against $\hat{L}_j = \Phi_{ja}\hat{a} + \Phi_{jb}\hat{b}$, after appending the resonator's and box's internal-loss channels as two further trivial (1×1) SLH components via Equation (5.7) (contributing $\sqrt{\kappa_{r,i}}\hat{a}$ and $\sqrt{\kappa_{b,i}}\hat{b}$):

$$\Phi = \begin{pmatrix} e^{i\beta\ell_1}\sqrt{\kappa_{r,c}/2} & \sqrt{\kappa_{b1}/2} + e^{i\Theta}\sqrt{\kappa_{b2}/2} \\ e^{i\beta\ell_2}\sqrt{\kappa_{r,c}/2} & \sqrt{\kappa_{b2}/2} + e^{i\Theta}\sqrt{\kappa_{b1}/2} \\ \sqrt{\kappa_{r,i}} & 0 \\ 0 & \sqrt{\kappa_{b,i}} \end{pmatrix}. \quad (5.32)$$

Step 3 – matching $\hat{H}_{\text{red}}$ against $\hat{H} = \sum_{j,k} \hat{a}_j^\dagger \Omega_{jk} \hat{a}_k$:

$$\Omega = \begin{pmatrix} \Delta_r & G \\ G & \Delta_b + \delta_b \end{pmatrix}, \quad G \equiv g_d + g_\ell. \quad (5.33)$$

Step 4 – computing A, B (noting $S = \text{diag}(e^{i\Theta}, e^{i\Theta}, 1, 1)$): $\Phi^\dagger \Phi$, computed entry by entry from the rows of Equation (5.32), gives the resonator's own diagonal entry directly, $(\Phi^\dagger \Phi)_{11} = \kappa_{r,c}/2 + \kappa_{r,c}/2 + \kappa_{r,i} = \kappa_{r,c} + \kappa_{r,i}$; the box's diagonal entry, each modulus-squared row contributing $\kappa_{b,\text{eff}}/2$,

$$(\Phi^\dagger \Phi)_{22} = \left| \sqrt{\frac{\kappa_{b1}}{2}} + e^{i\Theta} \sqrt{\frac{\kappa_{b2}}{2}} \right|^2 + \left| \sqrt{\frac{\kappa_{b2}}{2}} + e^{i\Theta} \sqrt{\frac{\kappa_{b1}}{2}} \right|^2 + \kappa_{b,i} \equiv \kappa_{b,\text{eff}} + \kappa_{b,i} \quad (5.34)$$

$$\kappa_{b,\text{eff}} \equiv \kappa_{b1} + \kappa_{b2} + 2\sqrt{\kappa_{b1}\kappa_{b2}} \cos \Theta;$$

and the off-diagonal - the row-by-row sum pairing $e^{-i\beta\ell_j}$ from one feedline row with $e^{+i\beta\ell_j}$ from the other, rows 3–4 contributing nothing –

$$(\Phi^\dagger \Phi)_{12} = (\Phi^\dagger \Phi)_{21}^* \equiv \Gamma_{ab} = \sqrt{\kappa_{r,c}} \left(\sqrt{\kappa_{b1}} \cos \beta\ell_1 + \sqrt{\kappa_{b2}} \cos \beta\ell_2 \right), \quad (5.35)$$

so that

$$\Phi^\dagger \Phi = \begin{pmatrix} \kappa_{r,c} + \kappa_{r,i} & \Gamma_{ab} \\ \Gamma_{ab} & \kappa_{b,\text{eff}} + \kappa_{b,i} \end{pmatrix}, \quad (5.36)$$

and

$$A = \begin{pmatrix} -i\Delta_r - \frac{\kappa_{r,c} + \kappa_{r,i}}{2} & -iG - \frac{\Gamma_{ab}}{2} \\ -iG - \frac{\Gamma_{ab}}{2} & -i(\Delta_b + \delta_b) - \frac{\kappa_{b,\text{eff}} + \kappa_{b,i}}{2} \end{pmatrix}, \quad B = -\Phi^\dagger S, \quad C = \Phi, \quad D = S. \quad (5.37)$$

For the elimination in the next section it is convenient to name the off-diagonal and box-diagonal entries,

$$M_2 \equiv A_{12} = -\left(\frac{\Gamma_{ab}}{2} + iG\right), \quad D_2 \equiv -A_{22} = i(\Delta_b + \delta_b) + \frac{\kappa_{b,\text{eff}} + \kappa_{b,i}}{2}, \quad (5.38)$$

with real and imaginary parts

$$\text{Re } M_2 = -\frac{\Gamma_{ab}}{2}, \quad \text{Im } M_2 = -G, \quad \text{Re } D_2 = \frac{\kappa_{b,\text{eff}} + \kappa_{b,i}}{2}, \quad \text{Im } D_2 = \Delta_b + \delta_b. \quad (5.39)$$

5.3.4 Loss Rates: Exact Elimination of the Box Mode

A continuous drive at a fixed reference frequency makes the steady-state mean fields time-independent, $\dot{\hat{\mathbf{a}}} = 0$, so Equation (5.31) gives $-A\hat{\mathbf{a}} = B\hat{\mathbf{f}}_{\text{in}}$ (where $\hat{\mathbf{f}}_{\text{in}}$ is a generalization of $\hat{b}_{\text{in}}$ used to avoid conflict with box variables). Writing the two scalar equations, with $\hat{\beta}_a \equiv (B\hat{\mathbf{f}}_{\text{in}})_a$, $\hat{\beta}_b \equiv (B\hat{\mathbf{f}}_{\text{in}})_b$:

$$-A_{11}\hat{a} - A_{12}\hat{b} = \hat{\beta}_a \quad (\text{i}), \quad -A_{21}\hat{a} - A_{22}\hat{b} = \hat{\beta}_b \quad (\text{ii}). \quad (5.40)$$

Solving (ii) for $\hat{b}$ and substituting into (i) eliminates the box mode exactly – a Gaussian elimination:

$$\hat{b} = -\frac{\hat{\beta}_b + A_{21}\hat{a}}{A_{22}}, \quad \left[-A_{11} + \frac{A_{12}A_{21}}{A_{22}}\right]\hat{a} = \hat{\beta}_a - \frac{A_{12}}{A_{22}}\hat{\beta}_b. \quad (5.41)$$

Defining the self-energy $\Sigma \equiv A_{12}A_{21}/A_{22}$ gives $\hat{a} = [\hat{\beta}_a - (A_{12}/A_{22})\hat{\beta}_b]/(-A_{11} + \Sigma)$, and with $A_{11} = -i\Delta_r - (\kappa_{r,c} + \kappa_{r,i})/2$ from Equation (5.37), the dressed denominator is

$$\mathcal{D}_r \equiv -A_{11} + \Sigma = i\Delta_r + \frac{\kappa_{r,c} + \kappa_{r,i}}{2} + \Sigma. \quad (5.42)$$

A decay rate must be real and non-negative (a resonance cannot leak an imaginary probability) so splitting Σ into real and imaginary parts and assigning each to where it physically belongs,

$$i\Delta_r + \frac{\kappa_{r,c} + \kappa_{r,i}}{2} + \Sigma = i \underbrace{[\Delta_r + \text{Im } \Sigma]}_{\Delta_r^{(\text{eff})}} + \underbrace{\frac{\kappa_{r,c} + \kappa_{r,i} + 2 \text{Re } \Sigma}{2}}_{\kappa_{\text{loaded}}/2}, \quad (5.43)$$

$$\boxed{\kappa_{\text{loaded}} = \kappa_{r,c} + \kappa_{r,i} + 2 \operatorname{Re} \Sigma, \quad \omega_r^{(\text{eff})} = \omega_r + \operatorname{Im} \Sigma.} \quad (5.44)$$

The real part of Σ is genuine extra dissipation, power that really does leak out through the box, while the imaginary part is a coherent frequency pull with no associated energy loss, exactly the resistance/reactance split already used for a driven RLC circuit throughout this thesis.

Explicitly, $\Sigma = A_{12}A_{21}/A_{22} = M_2^2/(-D_2) = -M_2^2/D_2$. Squaring $M_2 = -(\Gamma_{ab}/2 + iG)$,

$$M_2^2 = \left(\frac{\Gamma_{ab}^2}{4} - G^2 \right) + i\Gamma_{ab}G, \quad (5.45)$$

and rationalizing by $D_2^*/|D_2|^2$,

$$\Sigma = -\frac{[(\Gamma_{ab}^2/4 - G^2) + i\Gamma_{ab}G]D_2^*}{|D_2|^2}. \quad (5.46)$$

Expanding term by term and collecting real and imaginary parts (the identity $\operatorname{Re}[(x + iy)/(p + iq)] = (xp + yq)/(p^2 + q^2)$, applied here with $x + iy = -M_2^2$ and $p + iq = D_2$):

$$\operatorname{Re} \Sigma = \frac{(G^2 - \Gamma_{ab}^2/4) \frac{\kappa_{b,\text{eff}} + \kappa_{b,i}}{2} - G\Gamma_{ab}(\Delta_b + \delta_b)}{(\Delta_b + \delta_b)^2 + \frac{(\kappa_{b,\text{eff}} + \kappa_{b,i})^2}{4}}, \quad (5.47)$$

and substituting into Equation (5.44) gives the fully explicit loaded rate

$$\boxed{\kappa_{\text{loaded}} = \kappa_{r,c} + \kappa_{r,i} + \frac{(G^2 - \Gamma_{ab}^2/4)(\kappa_{b,\text{eff}} + \kappa_{b,i}) - 2G\Gamma_{ab}(\Delta_b + \delta_b)}{(\Delta_b + \delta_b)^2 + \frac{(\kappa_{b,\text{eff}} + \kappa_{b,i})^2}{4}}.} \quad (5.48)$$

The correction peaks when $\Delta_b + \delta_b \rightarrow 0$ (the box on resonance with the drive, maximal hybridization) and vanishes as $|\Delta_b + \delta_b| \rightarrow \infty$ (the box mode decouples, recovering the bare resonator).

5.3.5 S_{21} and the External/Internal Split

Next we can compute/derive the S_{21} parameters as well,

Channel-to-port pairing. The wiring assigns the reduced channels as travel *directions*, not physical ports:

$$\begin{aligned} \text{channel 1} &= (\text{in: right-end injection} \rightarrow \text{out: left exit, ports 1-3}), \\ \text{channel 2} &= (\text{in: left-end injection} \rightarrow \text{out: right exit, ports 2-4}). \end{aligned} \quad (5.49)$$

The VNA transmission (drive left connector, read right connector) is therefore drive channel 2, read output 2 has the same reduced index, because the diagonal S_{red} of Equation (5.24) strings opposite physical ends into one channel.

Input–output relation. The exact relation is $\hat{\mathbf{f}}_{\text{out}} = \hat{\mathbf{L}}_{\text{red}} + S_{\text{red}}\hat{\mathbf{f}}_{\text{in}}$, which after appending the two internal-loss channels reads $\hat{\mathbf{f}}_{\text{out}} = \Phi(\hat{\mathbf{a}}_{\hat{b}}) + S\hat{\mathbf{f}}_{\text{in}}$ with Φ, S from Equation (5.32) and $S = \text{diag}(e^{i\Theta}, e^{i\Theta}, 1, 1)$. Writing out all four rows,

$$\begin{aligned}\hat{f}_{\text{out},1} &= e^{i\beta\ell_1} \sqrt{\frac{\kappa_{r,c}}{2}} \hat{a} + \left(\sqrt{\frac{\kappa_{b1}}{2}} + e^{i\Theta} \sqrt{\frac{\kappa_{b2}}{2}} \right) \hat{b} + e^{i\Theta} \hat{f}_{\text{in},1}, \\ \hat{f}_{\text{out},2} &= e^{i\beta\ell_2} \sqrt{\frac{\kappa_{r,c}}{2}} \hat{a} + \left(\sqrt{\frac{\kappa_{b2}}{2}} + e^{i\Theta} \sqrt{\frac{\kappa_{b1}}{2}} \right) \hat{b} + e^{i\Theta} \hat{f}_{\text{in},2}, \\ \hat{f}_{\text{out},3} &= \sqrt{\kappa_{r,i}} \hat{a} + \hat{f}_{\text{in},3}, \quad \hat{f}_{\text{out},4} = \sqrt{\kappa_{b,i}} \hat{b} + \hat{f}_{\text{in},4}.\end{aligned}\tag{5.50}$$

Coherent drive on channel 2 only, $\langle \hat{\mathbf{f}}_{\text{in}} \rangle = (0, \alpha, 0, 0)^\top$ (vacuum on the rest); the network is linear, so every mean is proportional to α , and the ratio defining S_{21} removes it identically. From row 2,

$$S_{21} \equiv \frac{\langle \hat{f}_{\text{out},2} \rangle}{\alpha} = e^{i\Theta} + e^{i\beta\ell_2} \sqrt{\frac{\kappa_{r,c}}{2}} \frac{\langle \hat{a} \rangle}{\alpha} + \left(\sqrt{\frac{\kappa_{b2}}{2}} + e^{i\Theta} \sqrt{\frac{\kappa_{b1}}{2}} \right) \frac{\langle \hat{b} \rangle}{\alpha},\tag{5.51}$$

the direct term $(S\langle \hat{\mathbf{f}}_{\text{in}} \rangle)_2 = e^{i\Theta} \alpha$ being the drive passing straight through the line.

Drive vertices. $\langle \hat{\beta} \rangle \equiv B\langle \hat{\mathbf{f}}_{\text{in}} \rangle = \alpha \times (\text{column 2 of } B)$. Since $B = -\Phi^\dagger S$ and $S_{22} = e^{i\Theta}$, column 2 of B is $-(\text{row 2 of } \Phi)^* \times e^{i\Theta}$

$$\frac{\langle \hat{\beta}_a \rangle}{\alpha} = -\Phi_{2a}^* e^{i\Theta} = -e^{-i\beta\ell_2} \sqrt{\frac{\kappa_{r,c}}{2}} e^{i\Theta} = -e^{i\beta\ell_1} \sqrt{\frac{\kappa_{r,c}}{2}}, \quad \frac{\langle \hat{\beta}_b \rangle}{\alpha} = -\Phi_{2b}^* e^{i\Theta} = -\left(\sqrt{\frac{\kappa_{b2}}{2}} e^{i\Theta} + \sqrt{\frac{\kappa_{b1}}{2}} \right).\tag{5.52}$$

Eliminating $\hat{b}$ from the steady state proceeds exactly as in Equations (5.40)–(5.43), with $\hat{\beta}_a, \hat{\beta}_b$ in place of the generic drive and the same self-energy $\Sigma = A_{12}A_{21}/A_{22}$, giving $\langle \hat{a} \rangle = [\langle \hat{\beta}_a \rangle - (A_{12}/A_{22})\langle \hat{\beta}_b \rangle]/\mathcal{D}_r$ and the same dressed denominator $\mathcal{D}_r = i(\Delta_r + \text{Im } \Sigma) + \kappa_{\text{loaded}}/2$. Driving the network on the feedline's one physical channel and eliminating $\hat{b}$ this way gives $S_{21}(\omega)$ in the constant-plus-single-pole form every hanger measurement is fit to,

$$S_{21}(\omega) = \mathcal{B} - \frac{N}{\mathcal{D}_r},\tag{5.53}$$

where the terms of Equation (5.51) free of $\langle \hat{a} \rangle$ (hence free of Δ_r) form the background, and the $\langle \hat{a} \rangle$ term carries the pole:

$$S_{21} = \underbrace{e^{i\Theta} - \Phi_{2b} \frac{\langle \hat{\beta}_b \rangle}{\alpha A_{22}}}_{\mathcal{B}} + \underbrace{\left(\Phi_{2a} - \Phi_{2b} \frac{A_{21}}{A_{22}} \right) \frac{\langle \hat{a} \rangle}{\alpha}}_{V_e}, \quad N \equiv -V_e \left(\frac{\langle \hat{\beta}_a \rangle}{\alpha} - \frac{A_{12}}{A_{22}} \frac{\langle \hat{\beta}_b \rangle}{\alpha} \right).\tag{5.54}$$

The background. Expanding $\Phi_{2b}\langle \hat{\beta}_b \rangle/\alpha$ into its four products and using $A_{22} = -D_2$,

$$\Phi_{2b} \frac{\langle \hat{\beta}_b \rangle}{\alpha} = -\left(\sqrt{\frac{\kappa_{b2}}{2}} + e^{i\Theta} \sqrt{\frac{\kappa_{b1}}{2}} \right) \left(\sqrt{\frac{\kappa_{b2}}{2}} e^{i\Theta} + \sqrt{\frac{\kappa_{b1}}{2}} \right) = -e^{i\Theta} \left[\frac{\kappa_{b1} + \kappa_{b2}}{2} + \sqrt{\kappa_{b1}\kappa_{b2}} \cos \Theta \right] = -e^{i\Theta} \frac{\kappa_{b,\text{eff}}}{2},\tag{5.55}$$

the middle two of the four products combining via $e^{-i\Theta} + e^{i\Theta} = 2 \cos \Theta$. This is the box's own hanger notch on the bare line, with the resonator absent – as it must be for the Δ_r -free part

$$\mathcal{B} \equiv e^{i\Theta} \left(1 - \frac{\kappa_{b,\text{eff}}/2}{D_2} \right) = e^{i\Theta} \frac{i(\Delta_b + \delta_b) + \kappa_{b,i}/2}{i(\Delta_b + \delta_b) + (\kappa_{b,\text{eff}} + \kappa_{b,i})/2}. \quad (5.56)$$

The residue N . With $A_{21}/A_{22} = M_2/(-D_2)$,

$$V_e = e^{i\beta\ell_2} \sqrt{\frac{\kappa_{r,c}}{2}} + \Phi_{2b} \frac{M_2}{D_2}, \quad \frac{\langle \hat{\beta}_a \rangle}{\alpha} - \frac{A_{12} \langle \hat{\beta}_b \rangle}{A_{22} \alpha} = -e^{i\beta\ell_1} \sqrt{\frac{\kappa_{r,c}}{2}} - \frac{M_2}{D_2} e^{i\Theta} \Phi_{2b}^*, \quad (5.57)$$

and the overall minus of N cancelling the bracket's minus,

$$N = \left(e^{i\beta\ell_2} \sqrt{\frac{\kappa_{r,c}}{2}} + \Phi_{2b} \frac{M_2}{D_2} \right) \left(e^{i\beta\ell_1} \sqrt{\frac{\kappa_{r,c}}{2}} + e^{i\Theta} \Phi_{2b}^* \frac{M_2}{D_2} \right) = T_1 + T_2 + T_3 + T_4, \quad (5.58)$$

which resolves into exactly three physical pathways from the input port to the output port – direct transmission past both modes, $T_1 = e^{i\beta\ell_2} e^{i\beta\ell_1} \kappa_{r,c}/2 = e^{i\Theta} \kappa_{r,c}/2$; a single detour through the box on the way in or out, $T_2 + T_3 = e^{i\Theta} \Gamma_{ab} M_2/D_2$ (the κ_{b1}, κ_{b2} pieces of $T_2 + T_3$ each combine two exponentials into $2 \cos \beta\ell_j$, using $\beta\ell_1 - \Theta = -\beta\ell_2$ and $\beta\ell_2 - \Theta = -\beta\ell_1$); and a full round trip through the box, $T_4 = e^{i\Theta} |\Phi_{2b}|^2 (M_2/D_2)^2 = e^{i\Theta} (\kappa_{b,\text{eff}}/2) (M_2/D_2)^2$ zero, one, and two powers of M_2/D_2 for zero, one, and two box detours, the common $e^{i\Theta}$ being the one net full-line crossing every left-to-right route makes,

$$N = e^{i\Theta} \left[\frac{\kappa_{r,c}}{2} + \Gamma_{ab} \frac{M_2}{D_2} + \frac{\kappa_{b,\text{eff}}}{2} \left(\frac{M_2}{D_2} \right)^2 \right]. \quad (5.59)$$

The complex external rate z_c . Normalizing S_{21} by its own background,

$$\frac{S_{21}}{\mathcal{B}} = 1 - \frac{N/\mathcal{B}}{\mathcal{D}_r} = 1 - \frac{z_c/2}{\mathcal{D}_r}, \quad z_c \equiv \frac{2N}{\mathcal{B}}, \quad (5.60)$$

is the hanger notch normal form a diameter-corrected circle fit resolves. Since $\mathcal{B} = e^{i\Theta} (D_2 - \kappa_{b,\text{eff}}/2)/D_2$ from Equation (5.56), the common $e^{i\Theta}$ cancels and z_c assembles into one fraction,

$$z_c = \frac{\kappa_{r,c} D_2^2 + 2\Gamma_{ab} M_2 D_2 + \kappa_{b,\text{eff}} M_2^2}{D_2 (D_2 - \kappa_{b,\text{eff}}/2)}, \quad (5.61)$$

that plays exactly the role κ_c plays for a resonator with no box mode at all: box decoupled ($M_2 \rightarrow 0$, $\kappa_{b,\text{eff}} \rightarrow 0$, or $|D_2| \rightarrow \infty$) gives $z_c \rightarrow \kappa_{r,c}$, and z_c is manifestly Δ_r -independent, as a rate must be. Taking its real part gives

$$\boxed{\kappa_c^{(\text{eff})} = \text{Re}[z_c], \quad \kappa_i^{(\text{eff})} = \kappa_{\text{loaded}} - \kappa_c^{(\text{eff})}.} \quad (5.62)$$

Equation (5.62) also forms the basis of the fitting module developed in Appendix G. There, the SLH-derived transmission response is fitted directly to S_{21} traces generated independently by SPICE and the nodal solver, and the recovered $\kappa_c^{(\text{eff})}$ is benchmarked against the corresponding reference value.

5.3.6 Simulations

We next perform numerical simulations of the mixed-coupling circuit, cross-checking the analytic predictions of the preceding sections against the nodal solver (Appendix D). Robust identification of the resonator and the box mode is done through the strong coupling regimes using the energy-participation-ratio method of Appendix F. The sweep ranges used below are anchored to a real packaged device rather than chosen arbitrarily: full-wave HFSS simulations of the package give a wirebond inductance $L_{wb} = 670$ pH and a box-to-resonator direct coupling $C_b = 333.7$ fF. The capacitance sweeps of this section are centered on these HFSS-extracted values, and the mismatched-wirebond modeling of Section 5.2 uses the same $L_{wb} = 670$ pH.

Table 5.3 summarizes the nominal lumped-element parameters of the resonator–box circuit. During the sweeps, the tank parameters are retuned so that the resonator is nominally at 7 GHz and the box mode remains pinned at 11 GHz.

Table 5.3: Nominal parameters of the mixed-coupling resonator–box circuit.

Subsystem / coupling	Symbol	Nominal value	Simulation frequency
Resonator inductance	L_1	10.1 nH	7 GHz
Resonator capacitance	C_1	100 fF	
Box-mode inductance	L_2	5.0 nH	11 GHz
Box-mode capacitance	C_2	29 fF	
Resonator–feedline coupling	C_{c1}	0.72 fF	–
Box–feedline coupling	$C_{b11} = C_{b12}$	0.70 fF	–
Direct resonator–box coupling	C_{b2}	0.70 fF	–

5.3.6.1 Capacitive Sweeps

The capacitive sweeps vary the box-to-resonator and box-to-feedline coupling capacitances about these HFSS-extracted reference values while holding the remaining circuit parameters fixed.

Table 5.4: Parameters used for the capacitive sweeps.

Swept quantity	Symbol	Sweep range	Nominal value
Box–feedline capacitance	$C_{b11} = C_{b12}$	1–300 fF	0.70 fF
Direct resonator–box capacitance	C_{b2}	0.1–1.0 fF	0.70 fF

Figures 5.13 and 5.14 show that the discrepancy grows systematically as the box–feedline coupling capacitance increases. Near the upper end of the sweep, $C_{b11} = C_{b12} \approx 300$ fF, the difference between the ideal-circuit value and the value obtained when the box-mode coupling is included reaches approximately 20–30 kHz. This scale is obtained for both the matched and mismatched feedlines; wirebond mismatch

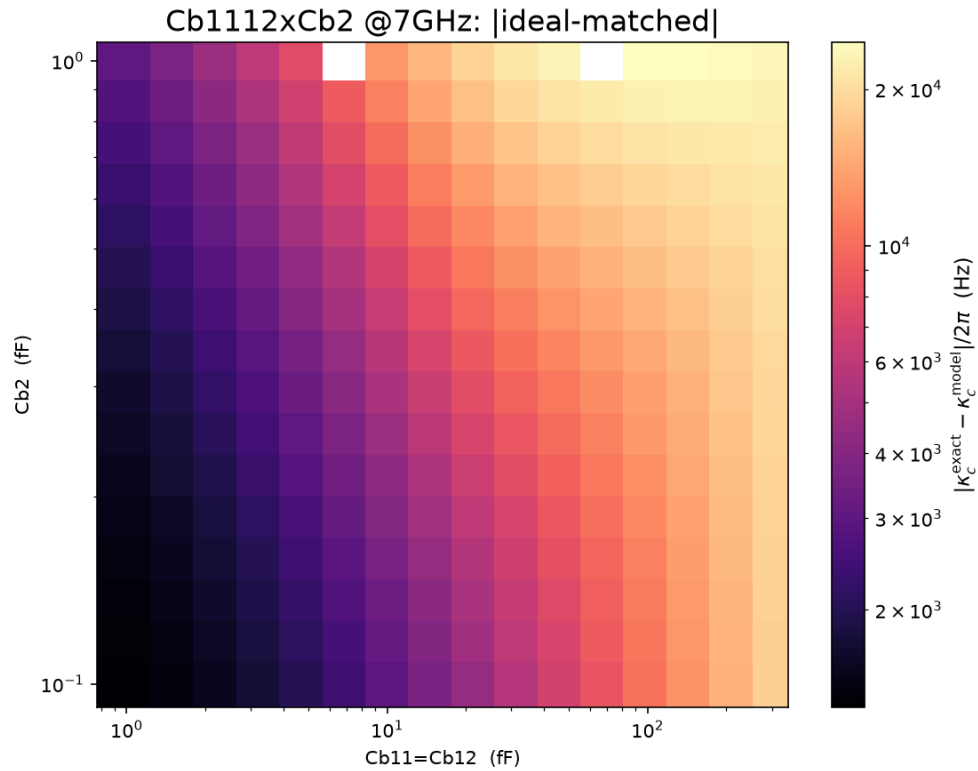


Figure 5.13: Absolute difference between the exact and mixed-coupling model predictions of $\kappa_c/2\pi$ over the $C_{b11} = C_{b12} - C_{b2}$ sweep at $f_r = 7$ GHz for the matched-feedline case.

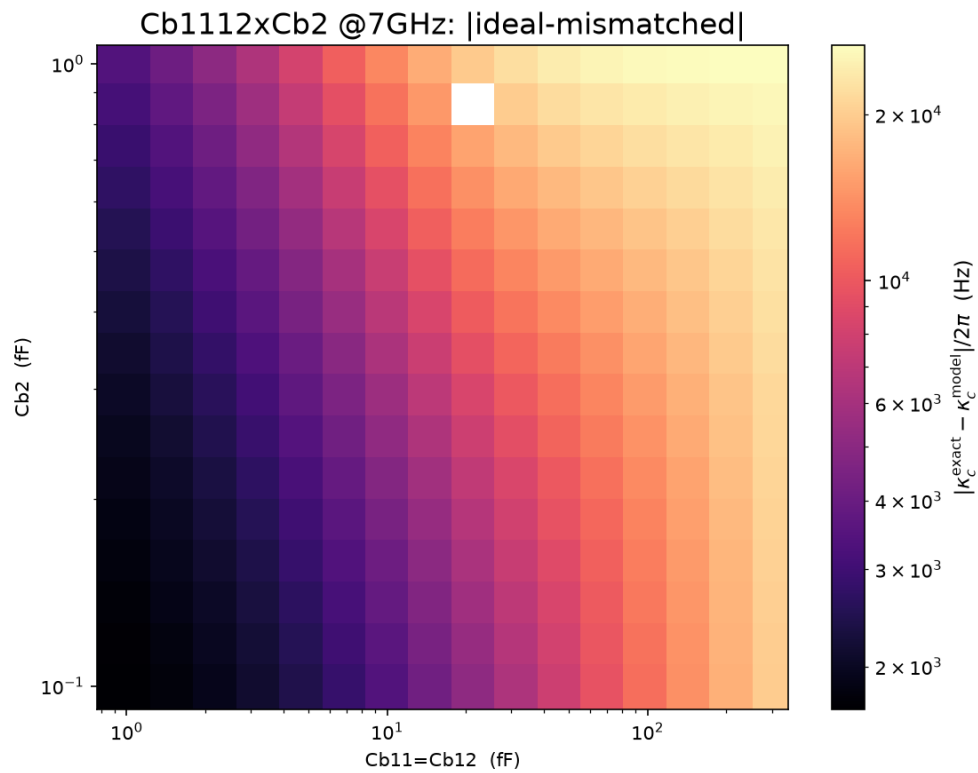


Figure 5.14: Absolute difference between the exact and mixed-coupling model predictions of $\kappa_c/2\pi$ over the $C_{b11} = C_{b12} - C_{b2}$ sweep at $f_r = 7$ GHz for the mismatched-feedline case.

changes the detailed distribution but is not required for the box mode to produce the effect. The calculated magnitude is the same few-tens-of-kilohertz scale observed experimentally in Figures 1.2 and 1.3 in the thesis introduction.

5.3.6.2 Inductive Sweeps

The inductive sweeps vary the wirebond inductance to quantify how the resulting feedline mismatch modifies the extracted resonator loss rates.

Table 5.5: Parameters used for the wirebond-inductance sweeps.

Swept quantity	Symbol	Sweep range	Nominal value
Wirebond inductance	L_{wb}	400–1200 pH	650 pH

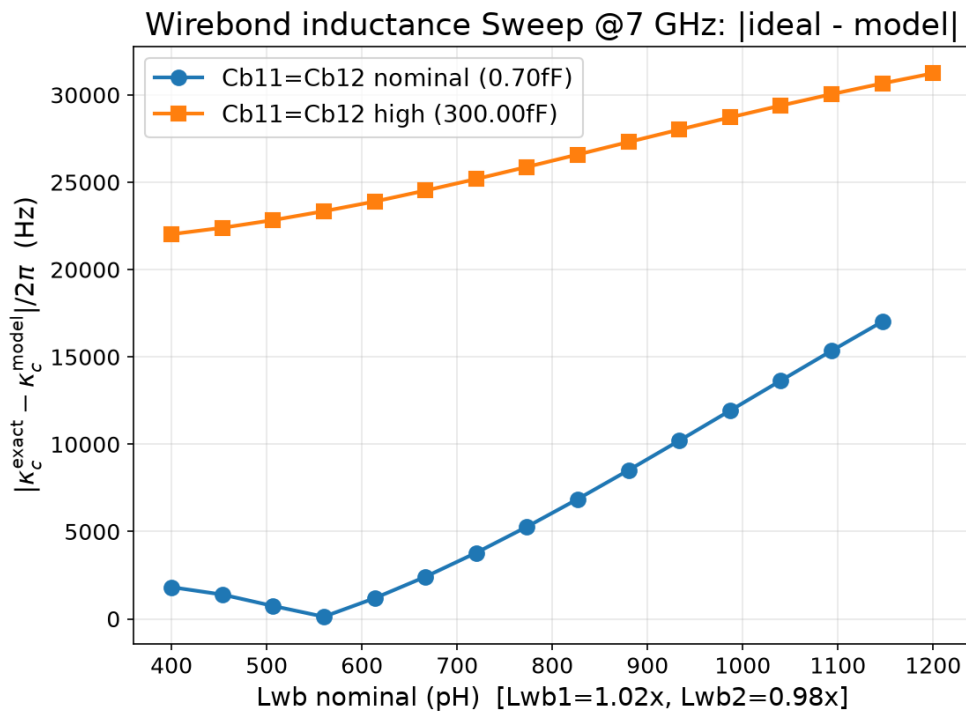


Figure 5.15: Absolute difference between the exact and mixed-coupling model predictions of $\kappa_c/2\pi$ as a function of wirebond inductance at $f_r = 7$ GHz, shown for the nominal and high box–feedline coupling capacitances.

The HFSS extraction corresponds to a two-wirebond connection with an effective inductance of approximately 650 pH. A fabricated connection may instead contain one, two, or three effective wirebonds, motivating the broader 400–1200 pH range used in Figure 5.15. For the nominal box–feedline coupling, the discrepancy remains comparatively small over part of this range. When the coupling capacitance is increased to $C_{b11} = C_{b12} = 300$ fF, however, the predicted error is approximately 15–25 kHz across much of the physically relevant range and continues to increase with inductance. This again overlaps the experimental ballpark of a few tens of kilohertz shown in Figures 1.2 and 1.3 in the thesis introduction.

5.3.6.3 Frequency Sweeps

The frequency sweeps vary the resonator frequency relative to the box-mode frequency, thereby probing the transition from weak off-resonant coupling to the strongly hybridized regime near an avoided crossing.

Table 5.6: Parameters used for the frequency sweeps.

Quantity	Symbol	Sweep range or fixed value	Nominal value
Resonator frequency (swept)	f_r	4–8 GHz	7 GHz
Box-mode frequency (fixed)	f_b	11 GHz	11 GHz

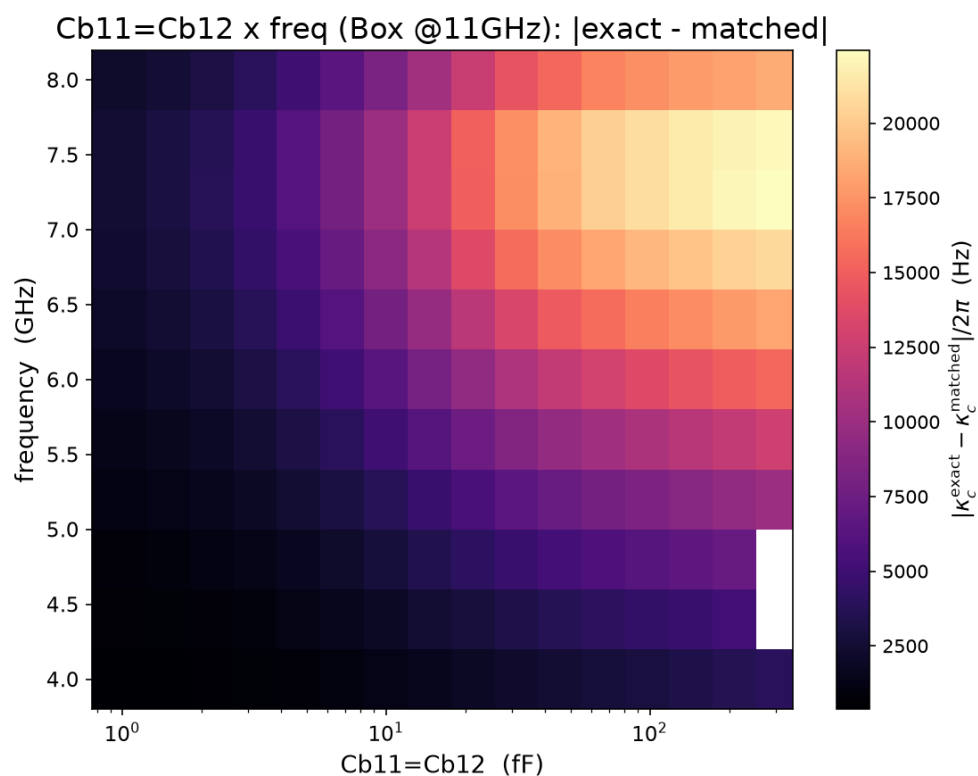


Figure 5.16: Absolute difference in $\kappa_c/2\pi$ across the resonator-frequency and $C_{b11} = C_{b12}$ sweep for the matched-feedline case, with the box mode pinned at 11 GHz.

Taken together, Figures 5.16– 5.19 show that the discrepancy is also strongly frequency dependent. With the box mode fixed at 11 GHz, increasing the resonator frequency strengthens its interaction with the box mode. Around the experimentally relevant region near 7 GHz, the high-capacitance $C_{b11} = C_{b12}$ sweeps reach errors of order 20 kHz in both the matched and mismatched cases. The direct-coupling C_{b2} sweeps show the same trend but a smaller magnitude over the plotted range, indicating that strong box–feedline coupling is the dominant route to the tens-of-kilohertz discrepancy. The frequency sweeps therefore reproduce the scale established by Figures 1.2 and 1.3 in the thesis introduction near the measured resonator frequencies.

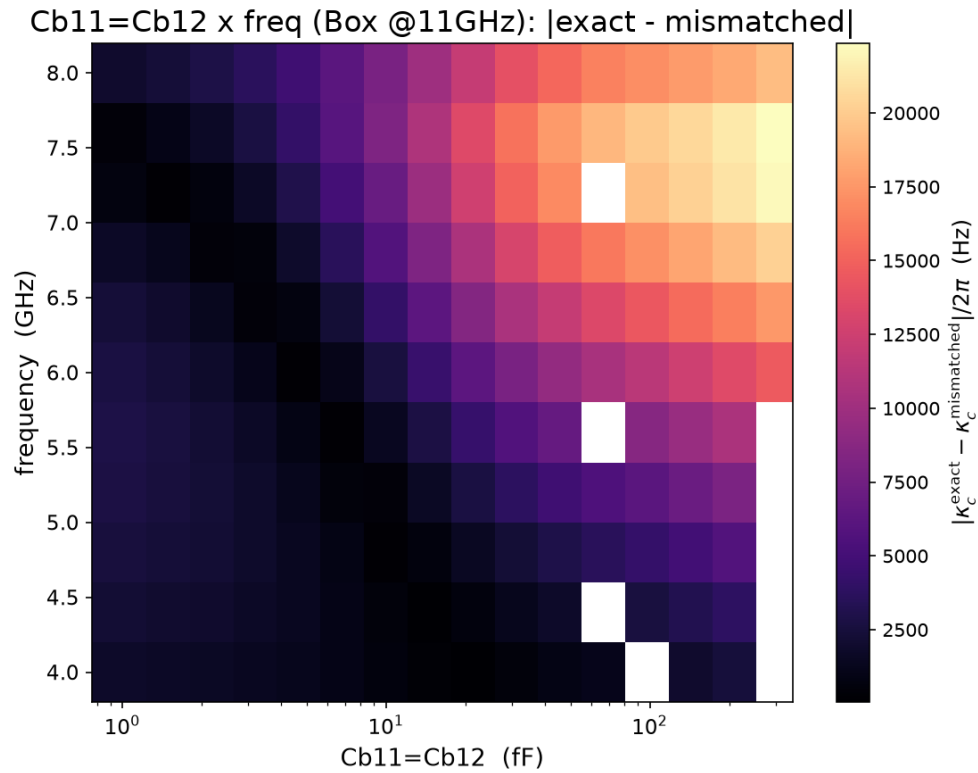


Figure 5.17: Absolute difference in $\kappa_c/2\pi$ across the resonator-frequency and $C_{b11} = C_{b12}$ sweep for the mismatched-feedline case, with the box mode pinned at 11 GHz.

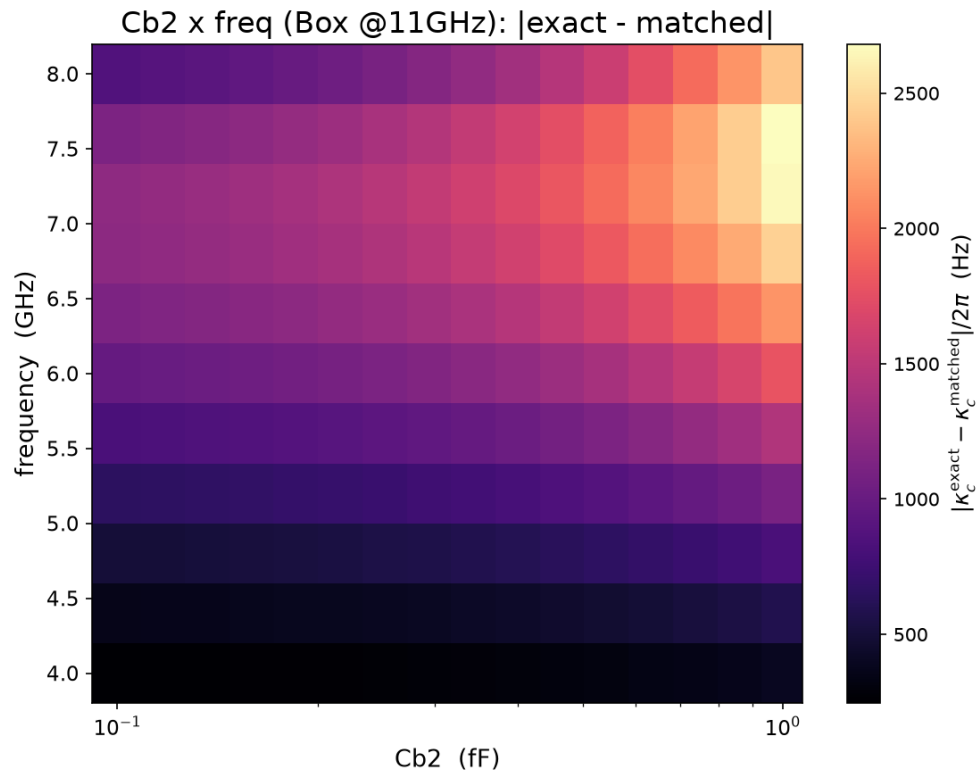


Figure 5.18: Absolute difference in $\kappa_c/2\pi$ across the resonator-frequency and direct resonator-box coupling C_{b2} sweep for the matched-feedline case, with the box mode pinned at 11 GHz.

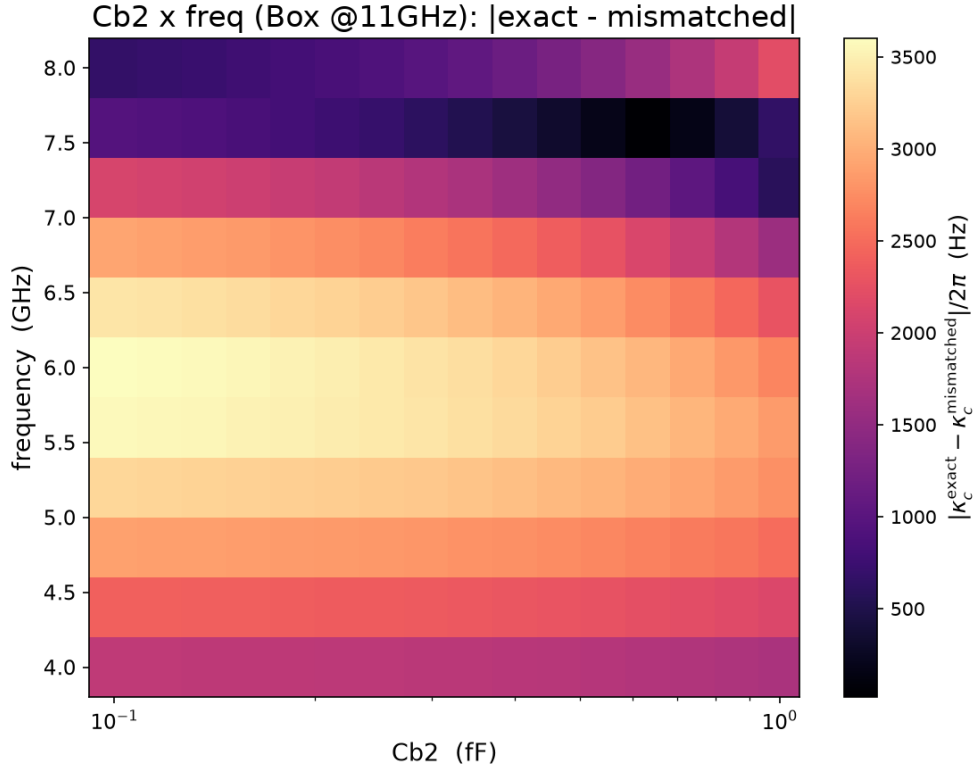


Figure 5.19: Absolute difference in $\kappa_c/2\pi$ across the resonator-frequency and direct resonator-box coupling C_{b2} sweep for the mismatched-feedline case, with the box mode pinned at 11 GHz.

5.4 Conclusion

The capacitive, inductive, and frequency sweeps consistently show that a box mode strongly coupled to the feedline can shift the apparent coupling rate by approximately 10–30 kHz. This agrees in order of magnitude with the discrepancy between the simulated and fitted measurement values introduced in Figures 1.2 and 1.3 of the thesis introduction. The model therefore accounts for the previously missing scale of the discrepancy: coupling through the box mode opens an additional environmental loss channel that is absent from the ideal resonator-feedline circuit.

The model also permits either sign of the change $\Delta\kappa_c = \kappa_c^{(\text{eff})} - \kappa_{r,c}$. The four input-output pathways contained in Equation (5.61)—the direct route, the two routes with one box-mode detour, and the route with two box-mode detours—add coherently. Their propagation phases can therefore produce constructive or destructive interference. Destructive interference can reduce the effective coupling below its bare value and thus yield a negative $\Delta\kappa_c$, without implying a negative physical loss rate.

What the nominal parameter sweeps do not determine is the particular device-to-device pattern of these fluctuations. They produce systematic trends with capacitance, frequency, and wirebond inductance, whereas each fabricated device samples a different combination of phases and coupling parameters. The remaining scatter can

therefore plausibly arise from variations in fabrication, wirebond geometry, box-mode frequency, or other device-specific environmental parameters beyond the nominal circuit used here.

Chapter 6

Summary and Outlook

6.1 Summary

This thesis investigated the discrepancy between coupling rates predicted by simulations and those extracted from measurements of superconducting microwave resonators. The fitting procedure was first benchmarked against noise, electrical delay, impedance mismatch, and measurement background. Although these effects influence the extracted quality factors, they do not reproduce the full discrepancy.

The circuit model was therefore extended to include wirebond-induced feedline mismatch and a parasitic package mode. Quantum Langevin and SLH input-output methods were used to derive the resulting effective loss rates and transmission response, which were validated independently with SPICE and a nodal-admittance solver. The capacitive, inductive, and frequency sweeps show that a box mode strongly coupled to the feedline can shift $\kappa_c/2\pi$ by a few tens of kilohertz. This is consistent with the experimental order of magnitude introduced in Figures 1.2 and 1.3.

The parasitic box mode therefore constitutes an additional environmental loss channel absent from the ideal resonator-feedline circuit. Because the direct and box-mediated input-output pathways add coherently, their relative phases allow both constructive and destructive interference. The model consequently explains both the observed scale and the possible positive or negative sign of $\Delta\kappa_c$.

6.2 Outlook

Future work should combine device-specific package simulations with the measured wirebond geometry rather than using a single nominal circuit for all resonators. Variations in wirebond number, placement, and inductance should be propagated through the model together with uncertainty in the box-mode frequency and coupling capacitances. This would establish whether realistic fabrication and assembly variations account for the remaining scatter.

The fitting module documented in Appendix G also provides a route toward analyzing measured S_{21} data with the box mode included directly in the fit. Applying it across multiple chips, cooldowns, and package configurations would help separate the additional loss channel from fitting uncertainty. Ultimately, incorporating these sensitivity

calculations into the EDA workflow could make package-mode and wirebond effects design parameters rather than corrections introduced after measurement.

Appendix A

Background Fitting Using a Second-Degree Polynomial

To model the background realistically, we use data from measured resonator sweeps. The complete procedure is documented in the accompanying [notebook](#); its steps are summarized below.

- Step 1: Load the `.json.h5` data in a notebook.
- Step 2: Retrieve the resonance frequency and measured linewidth of each resonator.
- Step 3: Mask each resonator over a window of $N \approx 100$ times its measured linewidth κ .
- Step 4: Normalize the signal amplitude and frequency. This is performed in chunks of 10 MHz, with overlap between adjacent intervals to ensure smoothness. For every chunk:
 - 4.1: Compute the derivative of the background phase and extract its median, which corresponds to the electrical delay; renormalizing the phase by this delay yields the true phase of the background.
 - 4.2: Normalize the amplitude by computing the mean of the chunk and dividing the chunk by it.
 - 4.3: Normalize the frequency by computing the central frequency f_0 of the chunk and subtracting it from the chunk.
 - 4.4: Fit the background amplitude with a second-degree polynomial and extract the fitted coefficients m_0, m_1, m_2 .
 - 4.5: Fit the renormalized background phase with a second-degree polynomial and extract the fitted coefficients $m_0^a, m_1^a, m_2^a, m_0^b, m_1^b, m_2^b$.
- Step 5: Use these parameters to generate a background and multiply it with the synthetic signal.
- Step 6: Perform the resonator-fitting subroutine.

The remainder of this appendix describes how the polynomial background enters the fit model.

As shown in [7], the S_{21} parameter can be written as

$$S_{21}(\omega) = \frac{a + b\omega}{c + d\omega}, \quad a, b, c, d \in \mathbb{C} \quad (\text{A.1})$$

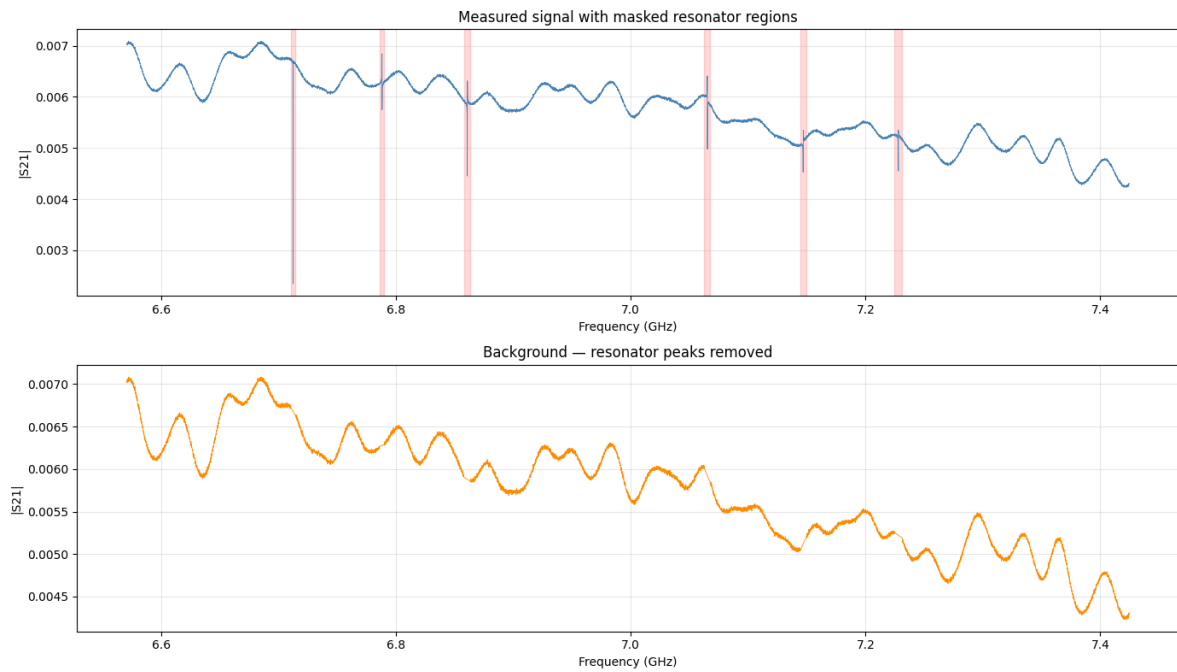


Figure A.1: Step 3 – background extraction and masking, before the polynomial fit.

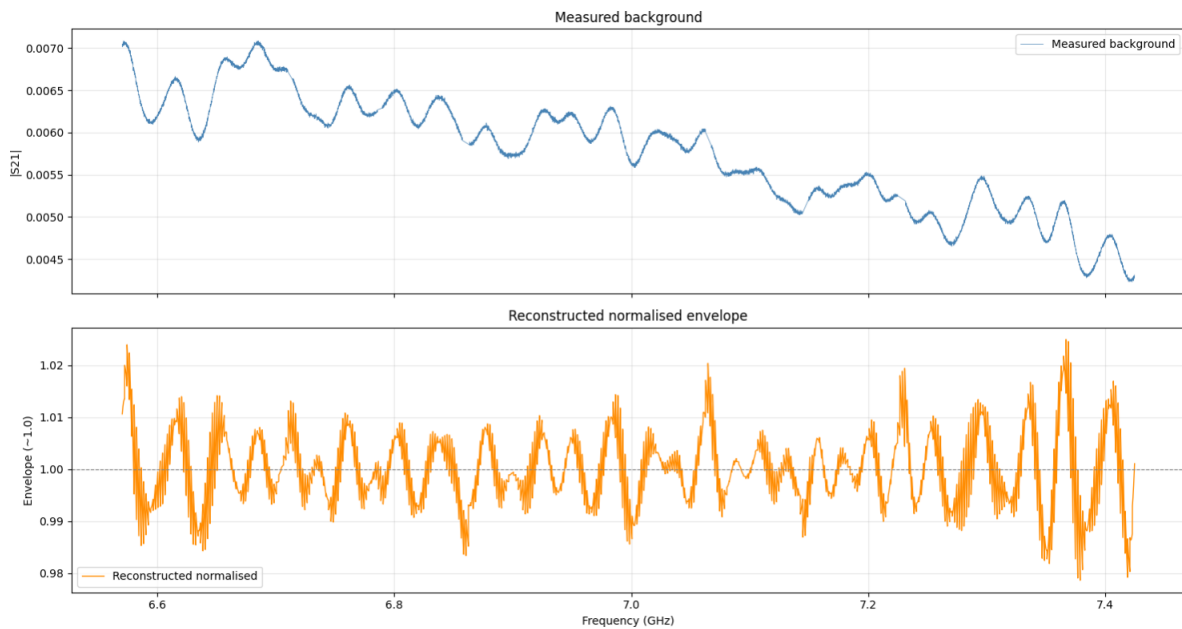


Figure A.2: Steps 4 and 5 – background normalization and reconstruction, after the polynomial fit.

The background is accounted for by a second-degree polynomial, $f(x) = m_0 + m_1x + m_2x^2$, where m_0 , m_1 , and m_2 are the coefficients fitted to the background extracted from the experimental sweeps. The model then becomes

$$S'_{21}(\omega) = (m_0 + m_1\Delta\omega + m_2\Delta\omega^2) \left(\frac{a + b\omega}{c + d\omega} \right) \Rightarrow \frac{a' + b'_1\omega + b'_2\omega^2 + b'_3\omega^3}{c + d\omega}, \quad \Delta\omega = \omega - \omega_0 \quad (\text{A.2})$$

Since the fitting routine extracts a, b, c, d , the relations between $a', b'_1, b'_2, b'_3, c, d$ and a, b, c, d are required. Expanding Equation (A.2) yields

$$a' = a(m_0 + m_1\omega_0 + m_2\omega_0^2) \quad (\text{A.3})$$

$$b'_1 = a(m_1 - 2m_2\omega_0) + b(m_0 - m_1\omega_0 + m_2\omega_0^2) \quad (\text{A.4})$$

$$b'_2 = am_2 + b(m_1 - 2m_2\omega_0) \quad (\text{A.5})$$

$$b'_3 = m_2b \quad (\text{A.6})$$

where ω_0 is the resonance frequency of the resonator (of order gigahertz). Numerical evaluation shows that the constant, first-order, and second-order contributions are of comparable magnitude, whereas the third-order contribution is smaller.

$$|a'| \approx |b'_1\omega| \approx |b'_2\omega^2| > |b'_3\omega^3| \quad (\text{A.7})$$

Consequently, a second-degree polynomial captures the dominant background variation without introducing an unnecessary higher-order fit parameter.

A.1 Why a second-degree polynomial?

The degree of the background polynomial is fixed by the algebra of the S-parameter of a single element. Consider a load $Z(\omega)$ presented to a line of impedance Z_0 , with reflection coefficient $S(\omega) = (Z_0 - Z(\omega))/(Z_0 + Z(\omega))$. For a series inductor, $Z_L(\omega) = i\omega L$:

$$S(\omega) = \frac{Z_0 - i\omega L}{Z_0 + i\omega L} = \frac{(Z_0^2 - \omega^2 L^2) - 2i\omega LZ_0}{Z_0^2 + \omega^2 L^2}, \quad \tan \psi_L(\omega) = \frac{-2\omega LZ_0}{Z_0^2 - \omega^2 L^2}. \quad (\text{A.8})$$

For a shunt capacitor, $Z_C(\omega) = 1/(i\omega C)$:

$$S(\omega) = \frac{Z_0 - \frac{1}{i\omega C}}{Z_0 + \frac{1}{i\omega C}} = \frac{(\omega^2 C^2 Z_0^2 - 1) + 2i\omega CZ_0}{\omega^2 C^2 Z_0^2 + 1}, \quad \tan \psi_C(\omega) = \frac{2\omega CZ_0}{\omega^2 C^2 Z_0^2 - 1}. \quad (\text{A.9})$$

In both cases $\tan \psi$ is a ratio of polynomials no higher than second degree: clearing the complex denominator squares a linear reactance, so the shared denominator is at

most quadratic in ω while the numerator's cross term is one power lower. Cascading further elements – a wirebond, a feedline segment, a connector – preserves this power counting, giving the general form

$$\tan \psi(\omega) = \frac{p_0 + p_1\omega + p_2\omega^2}{q_0 + q_1\omega + q_2\omega^2}, \quad (\text{A.10})$$

with $p_0, p_1, p_2, q_0, q_1, q_2$ distinct from the circle-fit's own a, b, c, d above.

The background amplitude follows the same algebra once the load carries any real dissipation. Writing $Z(\omega) = R + i\omega L$ for a lossy inductor (the lossy-capacitor case is analogous), the magnitude is the ratio of the numerator's and denominator's own magnitudes,

$$|S(\omega)|^2 = \frac{(Z_0 - R)^2 + \omega^2 L^2}{(Z_0 + R)^2 + \omega^2 L^2}, \quad (\text{A.11})$$

Thus a quadratic background is therefore not an arbitrary truncation: it is the exact degree supported by the underlying S-parameter, for both phase and amplitude.

Appendix B

Determination of the Experimental SNR

Following [6], the signal-to-noise ratio (SNR) of a resonance-circle fit is defined as

$$\text{SNR} = \frac{r_0}{\sigma_r} = \frac{\text{Signal}}{\text{Noise}} \quad (\text{B.1})$$

where r_0 is the radius of the resonance circle and σ_r quantifies the noise as the standard deviation of the measured data about the fitted circle. The radius is obtained from the off-resonant and on-resonant transmission, $S_{21}^{f=\infty}$ and $S_{21}^{f=f_r}$, as

$$r_0 = \left| \frac{S_{21}^{f=\infty} - S_{21}^{f=f_r}}{2} \right| \quad (\text{B.2})$$

and the noise is estimated as the standard deviation of the residuals between the measured and fitted transmission,

$$\sigma_r = \sqrt{\frac{1}{N-1} \sum_k |S_{21}^{\text{meas}}(\omega)_k - S_{21}^{\text{fit}}(\omega)_k|} \quad (\text{B.3})$$

This procedure yields the SNR of each measured chip; the implementation is available [here](#). The resulting values are listed in Table B.1. Chip NTC-0002-02-D exhibits a markedly lower SNR than the other chips. Inspection of its room-temperature (RT) wiring reveals that a 20 dB attenuator appears to be missing from one of the VNA ports. This finding is unexpected, since additional attenuation, rather than its absence, would normally be invoked to explain a degraded SNR. It is also possible that the KiCAD documentation is not fully up to date.

Table B.1: Experimental SNR per chip, averaged over the six resonators on each chip.

Chip	Experimental SNR
NTC - 0002 - 02 - GZ	235
NTC - 0002 - 02 - K	280
NTC - 0002 - 02 - D	70
NTC - 0002 - 02 - I	670
NTC - 0002 - 02 - H	128
NTC - 0002 - 02 - J	251

Appendix C

Derivation of the Hanger Transmission Formula via a Directional Coupler

This appendix derives its resonant structure from first principles, realizing the hanger geometry as a weakly-coupled four-port directional coupler with the resonator terminating one of its ports, following the same construction used for continuously-monitored hanger-coupled resonators in [10].

C.1 The Directional Coupler

Consider a symmetric, reciprocal four-port directional coupler with ports 1 (input), 2 (through/output), 3, and 4, cross-coupled by a small parameter $\epsilon \ll 1$ [3, Ch. 7]. To leading order in ϵ , its scattering matrix is

$$V = \begin{pmatrix} 0 & 1 - \epsilon^2/2 & i\epsilon & 0 \\ 1 - \epsilon^2/2 & 0 & 0 & i\epsilon \\ i\epsilon & 0 & 0 & 1 - \epsilon^2/2 \\ 0 & i\epsilon & 1 - \epsilon^2/2 & 0 \end{pmatrix}. \quad (\text{C.1})$$

V is symmetric, as required for a reciprocal network (Section 3.1): ports 1 and 2 form the through line, weakly cross-coupled to ports 3 and 4, as sketched in Figure C.1(a). Panel (b) of the same figure shows the reduced hanger geometry this derivation arrives at: port 3 left open and port 4 terminated by the resonator.

C.2 Reducing to a Three-Port by Opening Port 3

Terminating one port of an N -port network in a load of reflection coefficient Γ reduces it to an $(N - 1)$ -port network,

$$T_{kl} = S_{kl} + \frac{\Gamma S_{kN} S_{Nl}}{1 - \Gamma S_{NN}}, \quad (\text{C.2})$$

Leaving port 3 open ($\Gamma = 1$) and applying Equation (C.2) to Equation (C.1) with $N = 3$ gives the reduced three-port matrix over the remaining ports (1, 2, 4),

$$T = \begin{pmatrix} -\epsilon^2 & 1 - \epsilon^2/2 & i\epsilon(1 - \epsilon^2/2) \\ 1 - \epsilon^2/2 & 0 & i\epsilon \\ i\epsilon(1 - \epsilon^2/2) & i\epsilon & (1 - \epsilon^2/2)^2 \end{pmatrix}. \quad (\text{C.3})$$

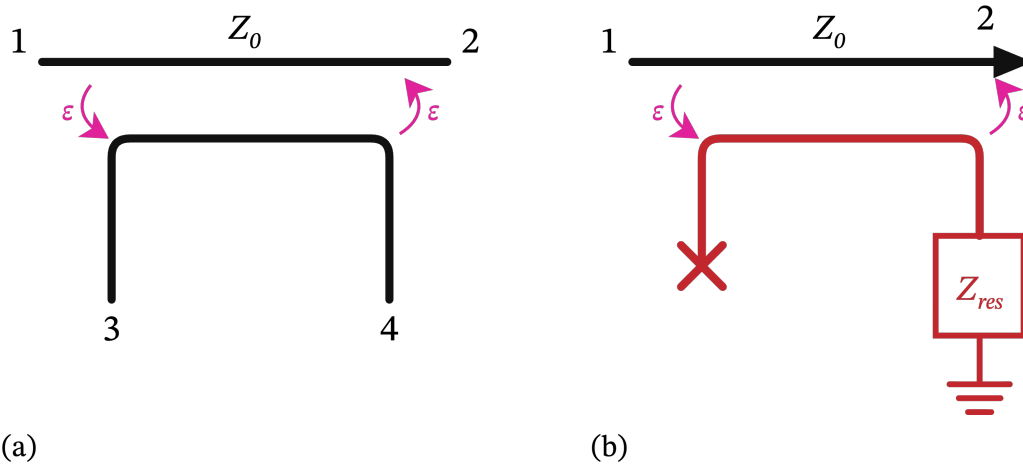


Figure C.1: (a) The four-port directional coupler (b) The hanger geometry realized from it: port 3 is left open, and port 4 is terminated by the resonator.

Port 4 – the third row/column of T – remains available to connect to the resonator.

C.3 Terminating Port 4 with the Resonator

Let $\Gamma_{\text{res}}(\omega)$ be the reflection coefficient seen looking into the resonator from port 4. Applying Equation (C.2) once more, this time terminating port 4 in Γ_{res} , gives the transmission between the two physical ports 1 and 2,

$$S_{21}(\omega) = T_{21} + \frac{\Gamma_{\text{res}} T_{24} T_{41}}{1 - \Gamma_{\text{res}} T_{44}}. \quad (\text{C.4})$$

Taking $\Gamma_{\text{res}}(\omega)$ in the standard near-resonance form of a one-port terminated in a resonator (Section 3.3.2), substituting the entries of Equation (C.3), and locating the resulting pole and zero of Equation (C.4) exactly in ω before expanding in ϵ (rather than expanding Equation (C.4) directly, which breaks down close to resonance) gives, to leading order in ϵ^2 , the compact Möbius form

$$S_{21}(\omega) \approx \frac{\omega - \tilde{\Omega}_0}{\omega - \tilde{\Omega}_0 - i\kappa_e/2}, \quad (\text{C.5})$$

where $\tilde{\Omega}_0 = \Omega_0 + i\kappa_i/2$ is the resonator's own complex pole (Section 3.3.2), and κ_e collects the ϵ -dependence of the coupling: the directional coupler converts the resonator's internal linewidth κ_i into an additional, externally imposed loss channel κ_e , exactly the coupling rate of Equation (3.15). The corresponding coupling quality factor is therefore

$$Q_c = \frac{\omega_r}{\kappa_e}, \quad (\text{C.6})$$

so stronger directional coupling increases κ_e and reduces Q_c .

C.4 From Loss Rates to the Model Function

Writing $\Delta\omega \equiv \omega - \Omega_0$ and $\kappa_l \equiv \kappa_i + \kappa_e$ (the loaded linewidth), the denominator and numerator of Equation (C.5) differ by exactly $i\kappa_e/2$, so

$$S_{21}(\Delta\omega) = 1 + \frac{i\kappa_e/2}{\Delta\omega - i\kappa_l/2}. \quad (\text{C.7})$$

Multiplying numerator and denominator of the second term by i gives

$$S_{21}(\Delta\omega) = 1 - \frac{\kappa_e/\kappa_l}{1 + 2i\Delta\omega/\kappa_l}. \quad (\text{C.8})$$

Finally, converting loss rates to quality factors via $Q_l \equiv \Omega_0/\kappa_l$ and $Q_c \equiv \Omega_0/\kappa_e$ (Section 3.3), so that $\kappa_e/\kappa_l = Q_l/Q_c$ and $2\Delta\omega/\kappa_l = 2Q_l \Delta f/f_r$, recovers

$$S_{21}(f) = 1 - \frac{Q_l}{Q_c} \frac{1}{1 + 2iQ_l \frac{f - f_r}{f_r}}, \quad (\text{C.9})$$

exactly the bracketed Lorentzian term of Equation (3.19). The overall background amplitude/phase $ae^{i\alpha}$ and the cable-delay factor $e^{-i2\pi f\tau}$ are properties of the measurement chain external to the coupler-plus-resonator network derived here, and are discussed separately in Section 3.4.

Appendix D

The Analytical Nodal-Admittance Solver

Nodal is the thesis's own analytical solver, developed using the theory introduced in Section 3.1 and used throughout Chapter 5. It generalizes the hand-derived admittance matrices used for the two-node hanger and three-node box-mode circuits to arbitrary topology: rather than rederiving $[Y(\omega)]$ by hand for every new coupling path, the solver reads a circuit's connectivity directly from its KiCAD netlist and assembles the matrix generically, so a new topology such as the seven-node mixed-coupling circuit of Section 5.3 needs only a new netlist file, not new code.

One net in the netlist is one electrical node; the reference net is dropped, and Kirchhoff's current law is written at every remaining node. Each two-terminal element of admittance Y_e between nodes m, n contributes an independent, symmetric "stamp" to the matrix,

$$Y_{mm} \rightarrow Y_{mm} + Y_e, \quad Y_{nn} \rightarrow Y_{nn} + Y_e, \quad Y_{mn}, Y_{nm} \rightarrow Y_{mn}, Y_{nm} - Y_e, \quad (\text{D.1})$$

with

$$Y_C = j\omega C, \quad Y_L = \frac{1}{j\omega L}, \quad Y_R = \frac{1}{R}; \quad (\text{D.2})$$

a $50\,\Omega$ termination stamps only its own diagonal entry, and an ideal transmission line stamps all four entries of its two nodes from the shorted-line special case of Equation (2.12). Stacking every stamp gives

$$[I] = [Y(\omega)][V], \quad (\text{D.3})$$

exactly Pozar's N -port admittance relation, built from the inside out. By construction, $[Y]$ is complex-symmetric, and purely imaginary once every resistor is removed.

A source-free oscillation of the network is a solution of Equation (D.3) with no injected current, $[I] = 0$, which admits a nonzero node-voltage vector only if the matrix is singular,

$$[Y(\tilde{\omega})][V] = 0 \iff \det[Y(\tilde{\omega})] = 0. \quad (\text{D.4})$$

Each complex root $\tilde{\omega}$ of Equation (D.4) encodes one mode's resonance frequency and total energy-decay rate,

$$\omega_{\text{res}} = \text{Re } \tilde{\omega}, \quad \kappa_{\text{total}} = 2 \text{Im } \tilde{\omega}, \quad (\text{D.5})$$

generalizing the single-branch complex pole of Section 3.3.2 to an arbitrary multi-node circuit. The solver finds these roots by scanning the lossless matrix $[Y] = j[K]$ for real sign changes of $\det K$ to seed each candidate, then refining every seed off the real axis

by Newton’s method on the full lossy matrix; a candidate is kept only if it converges to a finite, positive, non-duplicate frequency with loaded quality factor $Q_l = \omega_{\text{res}}/\kappa_{\text{total}} \geq 1$, a guard that rejects spurious roots such as an open feedline arm’s own standing-wave resonances.

Internal and external loss are separated using the same pole condition: because each resistor enters $[Y(\omega)]$ only through its own stamp Equation (D.1), any chosen subset can be analytically opened, $R \rightarrow \infty$, while every other element is kept. Solving Equation (D.4) with every loss element present gives κ_{total} via Equation (D.5); removing only the internal-loss resistors and re-solving gives the port damping alone,

$$\kappa_c = 2 \operatorname{Im} \tilde{\omega} \Big|_{R_{\text{int}} \rightarrow \infty}, \quad \kappa_i = \kappa_{\text{total}} - \kappa_c. \quad (\text{D.6})$$

The solver was validated circuit by circuit against both the hand-built modules it replaces and the independent scattering-matrix solver `spice` of Section 5.3.6, which shares no code or formalism with it.

Circuit	Reference	$ \Delta f_0/f_0 $	$ \Delta \kappa_c/\kappa_c $
Two-node hanger	hand-built module	0	2×10^{-14}
Box, direct coupling	<code>spice</code>	$\leq 1.6 \times 10^{-7}$	$\leq 2.6 \times 10^{-4}$
Box, indirect coupling	<code>spice</code>	$\leq 1.7 \times 10^{-7}$	≈ 0 (uncoupled)
Box, mixed coupling	<code>spice</code>	$\leq 1.9 \times 10^{-7}$	$\leq 2.9 \times 10^{-3}$

Table D.1: Agreement between the solver and its reference implementations across the circuit family

Appendix E

Derivation of the Quantum Langevin Equation

This appendix derives Equation (5.5), the single-mode quantum Langevin equation and input-output relation used in Section 5.3.2, from a microscopic system-bath Hamiltonian. Every commutator is computed explicitly and every approximation is named at the point it is used, following the standard input-output construction [9].

E.1 Setup

A cavity mode $\hat{a}_r$ of bare frequency ω_r is linearly coupled, in the rotating-wave approximation, to a continuum of bath oscillators $\hat{a}_k$ (frequencies ω_k):

$$\hat{H} = \hat{H}_{\text{sys}} + \hat{H}_{\text{bath}} + \hat{H}_{\text{sb}}, \quad \hat{H}_{\text{sys}} = \hbar\omega_r \hat{a}_r^\dagger \hat{a}_r, \quad \hat{H}_{\text{bath}} = \sum_k \hbar\omega_k \hat{a}_k^\dagger \hat{a}_k, \quad (\text{E.1})$$

$$\hat{H}_{\text{sb}} = -i\hbar \sum_k (f_k \hat{a}_r \hat{a}_k^\dagger - f_k^* \hat{a}_r^\dagger \hat{a}_k), \quad (\text{E.2})$$

with $[\hat{a}_r, \hat{a}_r^\dagger] = 1$, $[\hat{a}_k, \hat{a}_{k'}^\dagger] = \delta_{kk'}$, and every system operator commuting with every bath operator. For any operator $\hat{A}$ with no explicit time dependence, the Heisenberg equation of motion is $d\hat{A}/dt = (i/\hbar)[\hat{H}, \hat{A}]$; this is the only tool used below, together with the two elementary identities $[\hat{a}_r^\dagger \hat{a}_r, \hat{a}_r] = -\hat{a}_r$ and $[\hat{a}_r^\dagger \hat{a}_r, \hat{a}_r^\dagger] = \hat{a}_r^\dagger$ (and the same two lines with $\hat{a}_r \rightarrow \hat{a}_k$).

E.2 Heisenberg Equations for Bath and System

Commuting $\hat{H}$ with a bath mode $\hat{a}_k$ term by term – $[\hat{H}_{\text{sys}}, \hat{a}_k] = 0$ since $\hat{H}_{\text{sys}}$ is built only from $\hat{a}_r, \hat{a}_r^\dagger$; $[\hat{H}_{\text{bath}}, \hat{a}_k] = -\hbar\omega_k \hat{a}_k$, only the $k' = k$ term surviving; and $[\hat{H}_{\text{sb}}, \hat{a}_k] = i\hbar f_k \hat{a}_r$, since annihilation operators of different bath modes commute – gives

$$\dot{\hat{a}}_k = -i\omega_k \hat{a}_k - f_k \hat{a}_r. \quad (\text{E.3})$$

Commuting $\hat{H}$ with the system mode $\hat{a}_r$ analogously – $[\hat{H}_{\text{bath}}, \hat{a}_r] = 0$; $[\hat{H}_{\text{sys}}, \hat{a}_r] = -\hbar\omega_r \hat{a}_r$; and $[\hat{H}_{\text{sb}}, \hat{a}_r] = -i\hbar \sum_k f_k^* \hat{a}_k$ – gives

$$\dot{\hat{a}}_r = -i\omega_r \hat{a}_r - \sum_k f_k^* \hat{a}_k. \quad (\text{E.4})$$

E.3 Solving the Bath Equation Exactly

Equation (E.3) is linear in $\hat{a}_k$ with $\hat{a}_r(t)$ as a forcing term; using the integrating factor $e^{i\omega_k t}$ from an initial time t_0 gives the exact operator identity

$$\hat{a}_k(t) = \hat{a}_k(t_0) e^{-i\omega_k(t-t_0)} - f_k \int_{t_0}^t e^{-i\omega_k(t-t')} \hat{a}_r(t') dt', \quad (\text{E.5})$$

true for whatever $\hat{a}_r(t)$ turns out to be – nothing here is an approximation.

E.4 Substitution and the Memory Integral

Substituting Equation (E.5) into Equation (E.4) and using $f_k^* \cdot (-f_k) = -|f_k|^2$ (the coupling's phase cancels against its own conjugate) splits the result into two pieces of very different character:

$$\dot{\hat{a}}_r = -i\omega_r \hat{a}_r - \underbrace{\sum_k f_k^* e^{-i\omega_k(t-t_0)} \hat{a}_k(t_0)}_{(i)} - \underbrace{\sum_k |f_k|^2 \int_{t_0}^t e^{-i\omega_k(t-t')} \hat{a}_r(t') dt'}_{(ii)}. \quad (\text{E.6})$$

Piece (ii) is a genuine memory integral over a range of delay times; different bath modes k oscillate at different ω_k , and away from $t' = t$ their phases scatter and cancel, concentrating the integral near $t' = t$. Piece (i) has no integral at all – t, t_0 are fixed and $\hat{a}_k(t_0)$ is a fixed, unknown operator – and a sum over a continuum of independently phased oscillators, evaluated at one instant, does not average to anything: it is the operator-valued noise that any open quantum system must be driven by, and it cannot be dropped without breaking the canonical commutator $[\hat{a}_r, \hat{a}_r^\dagger] = 1$ under the resulting non-unitary evolution.

E.5 The Markov Approximation and Fermi's Golden Rule

Taking the coupling frequency-independent over the relevant bandwidth, $f_k \rightarrow f$, and converting the mode sum to a frequency integral, $\sum_k \rightarrow \int d\omega g(\omega)$, turns piece (ii)'s kernel into a delta function (Fermi's Golden Rule):

$$\sum_k |f_k|^2 e^{-i\omega_k(t-t')} \rightarrow f^2 g(\omega_r) \int d\omega e^{-i\omega(t-t')} = 2\pi f^2 g(\omega_r) \delta(t-t') \equiv \kappa \delta(t-t'). \quad (\text{E.7})$$

The delta function sits exactly at the upper boundary of the memory integral's domain, $t' \in [t_0, t]$: modeling $\delta(t-t')$ as a narrow, symmetric bump of width 2η centered at $t' = t$, only the bump's left half, $t' \in (t-\eta, t)$, lies inside the integration range, so the integral captures exactly half the bump's weight. Hence piece (ii) reduces, exactly at this boundary, to $-\frac{\kappa}{2} \hat{a}_r(t)$ – consistent with κ being an *energy* (photon-number) decay rate, since $\dot{\hat{a}}_r = -(\kappa/2) \hat{a}_r$ alone gives $|\hat{a}_r|^2 \sim e^{-\kappa t}$, amplitude decaying at half the energy rate.

E.6 The Quantum Langevin Equation and Input-Output Relation

In the continuum limit, piece (i) is, up to normalization, the freely propagating bath field evaluated *before* it reaches the system. Demanding piece (i) $\rightarrow -\sqrt{\kappa} \hat{a}_{\text{in}}(t)$ fixes that normalization ($\kappa = 2\pi f^2 g(\omega_r)$ gives $f/\sqrt{\kappa} = 1/\sqrt{2\pi}$, tracing $\kappa/2$ and $\sqrt{\kappa}$ back to the same microscopic constant f , rescaled two different ways – the algebraic fingerprint of the fluctuation-dissipation theorem) and defines

$$\hat{a}_{\text{in}}(t) \equiv \frac{-1}{\sqrt{2\pi}} \int d\omega e^{-i\omega(t-t_0)} \hat{a}(\omega, t_0), \quad [\hat{a}_{\text{in}}(t), \hat{a}_{\text{in}}^\dagger(t')] = \delta(t - t'), \quad (\text{E.8})$$

so that Equation (E.6) becomes the boxed quantum Langevin equation

$$\dot{\hat{a}}_r = -i\omega_r \hat{a}_r - \frac{\kappa}{2} \hat{a}_r - \sqrt{\kappa} \hat{a}_{\text{in}}(t). \quad (\text{E.9})$$

Without the noise term, $\hat{a}_r(t) = \hat{a}_r(0)e^{-(i\omega_r + \kappa/2)t}$ gives $[\hat{a}_r(t), \hat{a}_r^\dagger(t)] \rightarrow 0$, violating the uncertainty principle; restoring $\hat{a}_{\text{in}}$ contributes $\kappa \int_0^t e^{-\kappa(t-t')} dt' = 1 - e^{-\kappa t}$ to the same commutator, and $e^{-\kappa t} + (1 - e^{-\kappa t}) = 1$ for all t – damping and noise are tied together, exactly as the fluctuation dissipation theorem requires.

The output field is obtained by repeating Steps 3–5 above, but integrating *backward* from a future time $t_1 > t$ instead of forward from t_0 : this describes the field the bath must carry away so as to be consistent with its state at t_1 , after everything has happened, rather than before. The same boundary argument now applies at the integral's *lower* limit, giving $\dot{\hat{a}}_r = -i\omega_r \hat{a}_r + \frac{\kappa}{2} \hat{a}_r + \sqrt{\kappa} \hat{a}_{\text{out}}(t)$ with $\hat{a}_{\text{out}}$ defined analogously to Equation (E.8). Equating this to Equation (E.9) – two expressions for the same $\dot{\hat{a}}_r$ – and solving for $\hat{a}_{\text{out}}$ gives the input-output relation

$$\hat{a}_{\text{out}}(t) = \hat{a}_{\text{in}}(t) + \sqrt{\kappa} \hat{a}_r(t), \quad (\text{E.10})$$

completing the pair boxed as Equation (5.5) in the main text. Every step above is either an exact operator identity (Steps 1 and 2), an exact linear-ODE solution (Step 3), a single named physical approximation (Step 5: frequency-independent coupling over the relevant band), a definition (Step 6, with its normalization derived rather than assumed), or exact algebra (Step 7); nothing beyond these enters. Generalizing this construction to a vector of modes coupled to several channels at once – needed for the two-mode resonator-box system of Section 5.3.4 – is carried out in Section 5.3.3.

Appendix F

Mode Identification: The Energy Participation Ratio

An eigensolver returns unlabelled resonances. Picking the one closest in frequency to an analytic guess works until two modes hybridize strongly enough to undergo an avoided crossing: near the crossing, frequency no longer distinguishes them, since each becomes a genuine mixture of both bare circuits. What is needed is a quantity that tracks *energy*, not frequency viz. the energy participation ratio (EPR).

The EPR of a mode with respect to a group of elements is the fraction of its stored energy in that group,

$$p_g = \frac{\sum_{e \in g} E_e}{\sum_{e'} E_{e'}}, \quad \sum_g p_g \leq 1, \quad (\text{F.1})$$

the same quantity used in black-box circuit quantization to assign physical identities to eigenmodes [11, 12]. At resonance $[Y(\omega_{\text{res}})]$ is near-singular (Equation (D.4)), and its null vector (the mode's node-voltage pattern) is recovered as the right-singular vector of smallest singular value. The branch voltage ΔV across each element then gives the stored energies

$$E_C = \frac{1}{2} C |\Delta V|^2, \quad E_L = \frac{|\Delta V|^2}{2 \omega^2 L}, \quad (\text{F.2})$$

special cases of the reactance-theorem energy of a lossless one-port, $E = \frac{1}{4} |v|^2 d(\omega \text{Im } y)/d\omega$ [3, §6.2]. Among candidates passing a loaded-Q filter, the mode is identified as the one with the highest EPR for the relevant group (e.g. L_1, C_1 for the resonator) so no frequency guess involved.

Every sweep performed in chapter 5 (Cb11=Cb12, Cb2, (look at circuit in table 5.1 (e)) and wirebond inductance, each run for both the matched and mismatched circuit) identifies the resonator and box modes this way: among eigenmodes clearing a loaded-Q floor ($Q_l > 1$), the resonator is the one with highest p_{res} and the box is the one with highest p_{box} , additionally required to clear its own EPR floor ($\text{EPR} > 0.05$) so that a lossy, low-participation artifact mode cannot be mistaken for it. Table F.1 shows this holding up across the full swept range, retuning the resonator to 7 GHz and the box to 11 GHz at every point.

The resonator's participation ratio is essentially flat at $p_{\text{res}} \approx 0.99$ over three decades of coupling strength showcasing it stays weakly hybridized throughout, so identification never becomes ambiguous. The box's participation ratio falls steadily, from 0.96 at 1 fF to 0.30–0.35 at 300 fF, reflecting genuine physical hybridization as the box couples more strongly to the feedline. The mode remains correctly identified at every single point

Table F.1: Resonator and box mode participation ratios ($p_{\text{res}}, p_{\text{box}}$) as $C_{b11} = C_{b12}$ is swept from 1 to 300 fF. The resonator stays identified with $p_{\text{res}} \approx 0.99$ across the entire range; the box is correctly identified at every point despite p_{box} falling as coupling increases and the box hybridizes more strongly.

$C_{b11}=C_{b12}$ (fF)	Matched		Mismatched	
	p_{res}	p_{box}	p_{res}	p_{box}
1	0.9899	0.9626	0.9899	0.9626
10	0.9900	0.7724	0.9900	0.7724
50	0.9901	0.5160	0.9901	0.5147
100	0.9900	0.4579	0.9900	0.4550
200	0.9901	0.3907	0.9901	0.4010
300	0.9902	0.2971	0.9902	0.3520

(both the resonator target of 7 GHz and the box target of 11 GHz are hit throughout), because EPR ranks candidates by where their energy actually is rather than by how close their frequency happens to sit to a fixed guess.

Appendix G

SLH-Based Fitting Module

This appendix benchmarks the fitting routine developed from the effective coupling formula derived with the SLH framework in Chapter 5. Equation (5.62) expresses the external coupling rate as $\kappa_c^{(\text{eff})} = \text{Re}(z_c)$, including the additional coupling and loss channel introduced by the box mode.

G.1 Benchmark Procedure

For each sweep point, we generate the complex transmission $S_{21}(\omega)$ of the complete coupled circuit. The SLH routine fits Equation (5.53) to this trace and independently extracts $\kappa_{c,\text{SLH}}^{(\text{eff})}$. The benchmark uses the absolute error

$$\epsilon_{\kappa_c} = \frac{1}{2\pi} \left| \kappa_{c,\text{SLH}}^{(\text{eff})} - \kappa_{c,\text{reference}}^{(\text{eff})} \right|, \quad (\text{G.1})$$

where the reference is the nodal solver (see Appendix D). This is an end-to-end test of how accurately the reduced SLH response recovers the effective coupling rate from a full-circuit transmission trace.

G.2 Capacitance-Sweep Benchmark

Figures G.1 and G.2 show the retained 1–300 fF nodal-solver benchmarks at 7 GHz for matched and mismatched feedlines.

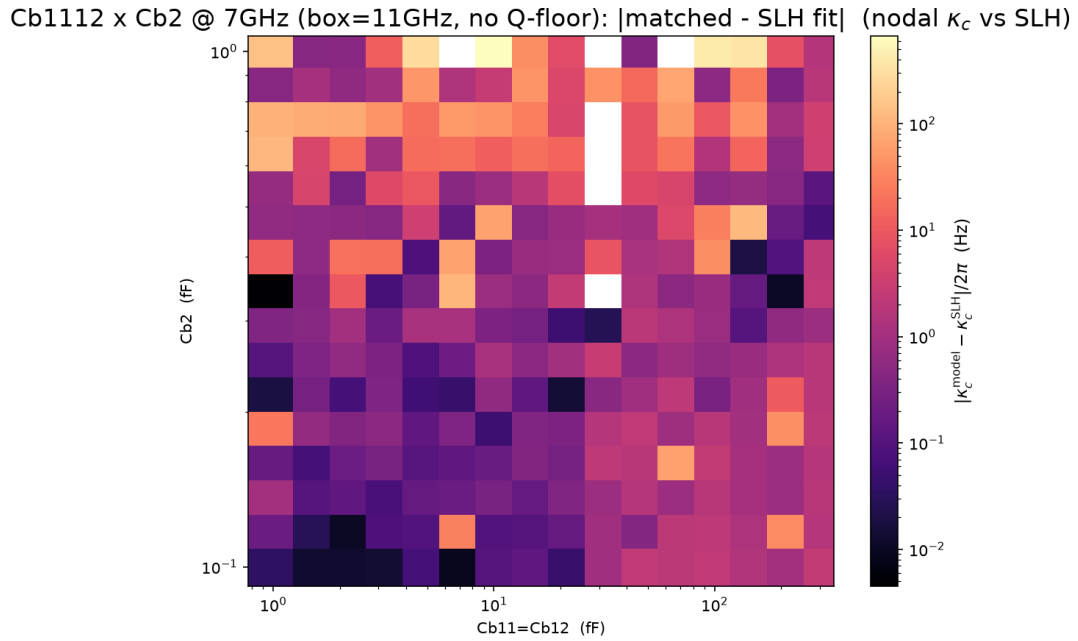


Figure G.1: Matched capacitance sweep: SLH fit compared with the nodal solver.

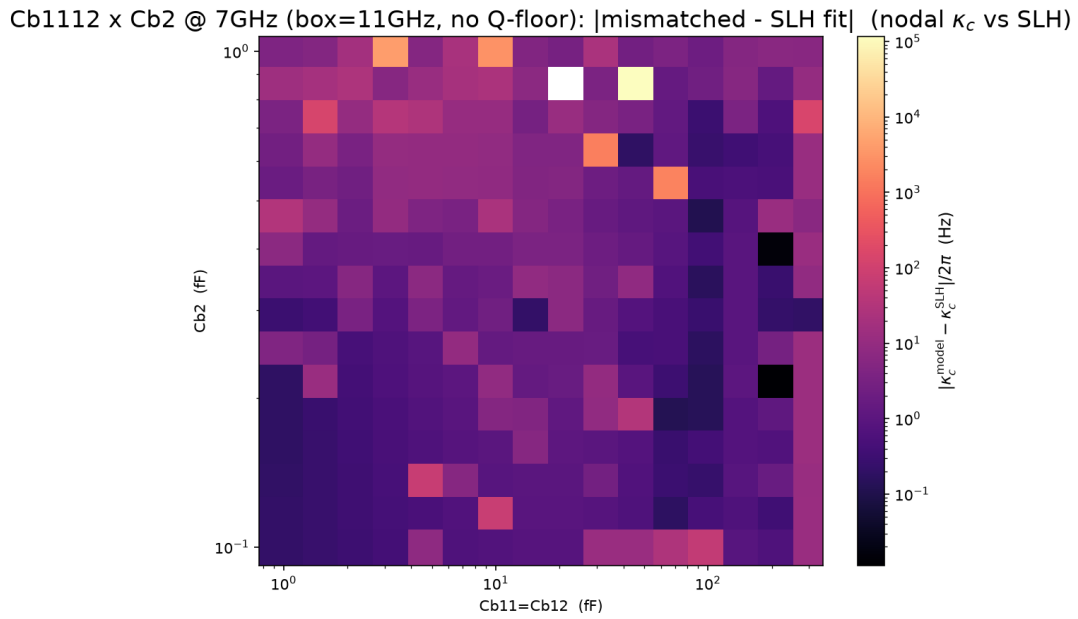


Figure G.2: Mismatched capacitance sweep: SLH fit compared with the nodal solver.

G.3 Frequency-Sweep Benchmark

The frequency benchmark separately varies $C_{b11} = C_{b12}$ and C_{b2} , with the box mode fixed at 11 GHz. The retained plots below show the nodal-solver benchmark for the $C_{b11} = C_{b12}$ sweep.

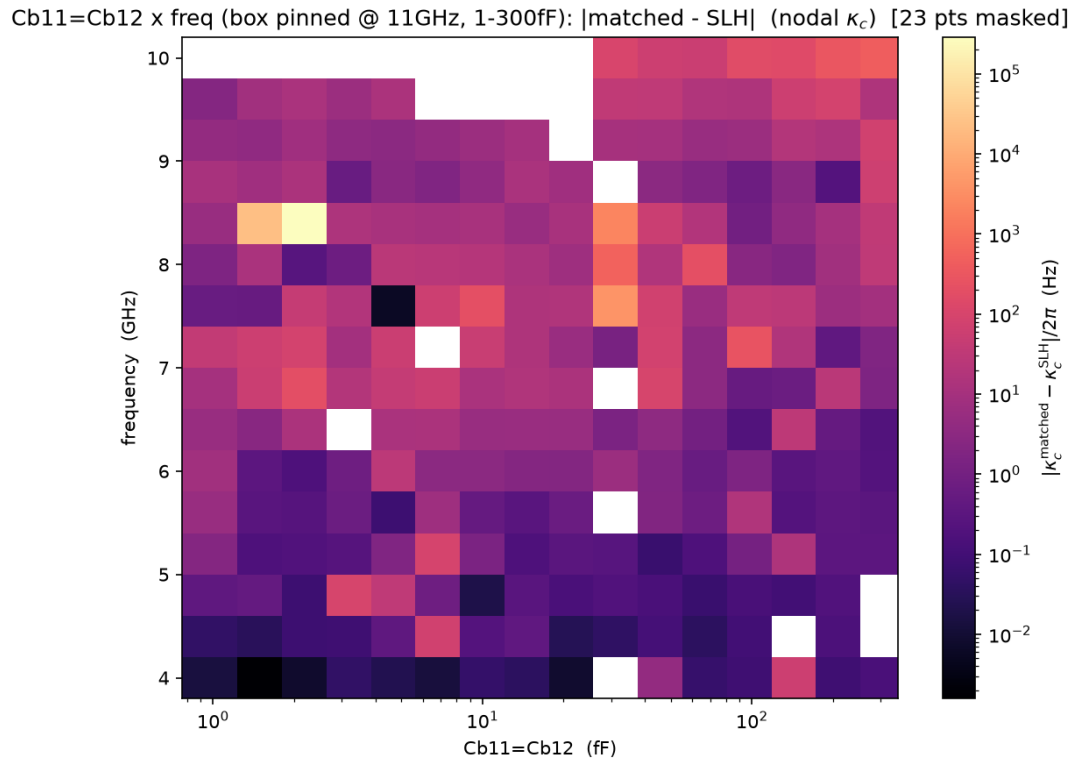


Figure G.3: Matched $C_{b11} = C_{b12}$ sweep: SLH fit versus the nodal solver.

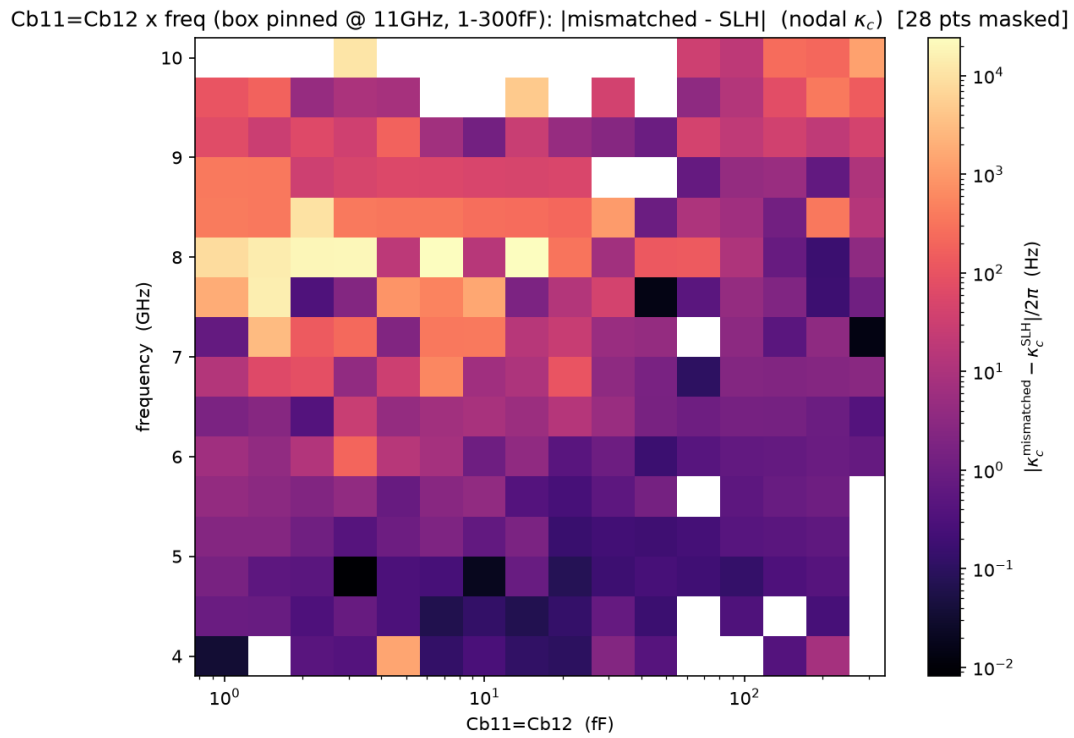


Figure G.4: Mismatched $C_{b11} = C_{b12}$ sweep: SLH fit versus the nodal solver.

The retained comparisons verify the fitting module against the independent nodal solver in both matched and mismatched feedline conditions.

References

- [1] Alice & Bob. *Alice & Bob – Building the First Useful Quantum Computer*. 2026. URL: <https://alice-bob.com/> (visited on 07/21/2026).
- [2] Raphaël Lescanne et al. “Exponential suppression of bit-flips in a qubit encoded in an oscillator”. In: *Nature Physics* 16 (2020), pp. 509–513. eprint: 1907.11729.
- [3] David M. Pozar. *Microwave Engineering*. 4th. Wiley, 2012.
- [4] Jiansong Gao. “The Physics of Superconducting Microwave Resonators”. PhD thesis. California Institute of Technology, 2008.
- [5] M. S. Khalil et al. “An analysis method for asymmetric resonator transmission applied to superconducting devices”. In: *Journal of Applied Physics* 111 (2012), p. 054510. eprint: 1108.3117.
- [6] S. Probst et al. “Efficient and robust analysis of complex scattering data under noise in microwave resonators”. In: *Review of Scientific Instruments* 86 (2015), p. 024706. eprint: 1410.3365.
- [7] Alice & Bob EDA Research Team. *Optimal method for resonator scattering parameter fit*. Internal technical note. 2026.
- [8] P. G. Baity et al. “Circle fit optimization for resonator quality factor measurements: point redistribution for maximal accuracy”. In: *Physical Review Research* 6 (2024). eprint: 2301.06364.
- [9] Joshua Combes, Jason Kerckhoff, and Mohan Sarovar. “The SLH framework for modeling quantum input-output networks”. In: *Advances in Physics: X* 2.3 (2017), pp. 784–888. eprint: 1611.00375.
- [10] William P. Livingston. “Continuous Feedback on Quantum Superconducting Circuits”. PhD thesis. University of California, Berkeley, 2021.
- [11] S. E. Nigg et al. “Black-Box Superconducting Circuit Quantization”. In: *Physical Review Letters* 108 (2012), p. 240502. eprint: 1204.0587.
- [12] Z. K. Mineev et al. “Energy-participation quantization of Josephson circuits”. In: *npj Quantum Information* 7 (2021), p. 131.